Vector potentials in three-dimensional non-smooth domains: detailed proofs

Supplement to the Lean formalization of C. Amrouche, C. Bernardi, M. Dauge and V. Girault (1998)

Abstract

This manuscript states the principal numbered results of Amrouche–Bernardi–Dauge–Girault (1998) precisely and supplies detailed mathematical proofs: canonical vector potentials, harmonic kernels and estimates, compactness and regularity, the full Dirichlet potential, finite-element approximation, and both Stokes formulations including pressure recovery. The main potential construction uses orthogonal projection and Riesz duality; the discrete coercivity proof uses a stable scalar correction and does not depend on the polyhedral shift theorem. The required analytic, geometric, and polynomial ingredients are developed explicitly. Several literal source assertions are false; separate counterexamples identify the necessary higher-order and mixed-boundary corrections.

Scope and proof status. The coverage table in Section 1.5 identifies every principal numbered theorem of the paper and its proof here. It also distinguishes source statements requiring correction from theorems that hold as written. The cited sharp refinements in source Remarks 2.16, 2.19, and 3.8 are identified separately; they are not silently included among the proved results. Main Theorems A and B (Theorems 3.12 and 3.17 of the paper) are formalized in Lean in this folder: they are the two theorems of `Showcase.lean`, proved in the library `LeanCode`. The proof route of the formalization is described in the blueprint (`blueprint/Blueprint.pdf`) and may differ from the arguments given here. All other results are proved in this manuscript only.

Foundations. We use Lebesgue integration, Fubini and dominated convergence, convolution approximation, elementary smooth calculus and one-dimensional absolute continuity, Plancherel, and Hilbert-space completeness, orthogonal projection, and Riesz representation. Domain-specific Sobolev, trace, extension, compactness, boundary regularity, cut, singular-kernel, and finite-element statements are proved in the body rather than inserted as background PDE assumptions. For conventional geometric surface-integral interpretations we use the surface-area formula in a Lipschitz graph; the scalar weak traces and fluxes used in the main statements are defined by volume identities.

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1 Precise public statements

1.1 Geometry, functions, and equality

All vector fields are real-valued. Write $H = L^2(\Omega;\mathbb{R}^3)$, with inner product $(v,w) = \int_\Omega v \cdot w$. Elements of all function spaces are equivalence classes modulo equality almost everywhere. Thus every uniqueness assertion in this manuscript is uniqueness in the indicated function space, never uniqueness of pointwise representatives.

Definition 1.1 (Admissible geometric data)**.** The set $\Omega \subset \mathbb{R}^3$ is nonempty, bounded, open, connected, and Lipschitz: locally at every boundary point, after a rigid change of coordinates, it lies on one side of the graph of a Lipschitz function. Write its connected boundary components as $\Gamma_0, \dots, \Gamma_I$, with Γ_0 bordering the unbounded complementary component (the outer component in the paper). An admissible cutting system is a finite family of connected, oriented, smooth open surfaces $\Sigma_1, \dots, \Sigma_J$ in Ω, whose boundaries lie in $\partial\Omega$, whose closures are pairwise disjoint, and for which

$$D = \Omega \setminus \bigcup_{j=1}^{J} \Sigma_j$$

is open, simply connected, and pseudo-Lipschitz. Here pseudo-Lipschitz means that at each boundary point, every sufficiently small ball meets D in one or two connected Lipschitz open pieces, as in [1, Definition 3.1]. Each cut has the two local sides through its closure asserted just after [1, Hypothesis 3.3], and the chosen unit normal of [1, Notation 3.9]; denote the side from which the normal points by minus and the side to which it points by plus.

The case $J = 0$ is included. In the paper J is also identified with the sum of the genera of the boundary components. The arguments below use the specified cutting system and its cardinality directly; they do not require a classification theorem for compact surfaces. Finiteness of the boundary components follows from the finite local graph cover, as established below.

For an open set U, $H^1(U)$ is the space of L^2 functions whose three distributional first partial derivatives belong to $L^2(U)$. The subspace $H^1_0(U)$ is the closure of $C_c^\infty(U)$ in $H^1(U)$. For a vector field $v \in L^2(U)^3$, the equations $\operatorname{div} v = d$ and $\operatorname{curl} v = f$ mean, respectively,

$$\int_U v \cdot \nabla\phi = -\int_U d\phi \qquad (\phi \in C_c^\infty(U)), \tag{1.1}$$

$$\int_U v \cdot \operatorname{curl}\varphi = \int_U f \cdot \varphi \qquad (\varphi \in C_c^\infty(U)^3). \tag{1.2}$$

The right-hand sides, when represented by L^2 functions, are unique by density. Set

$$H(\operatorname{div},U) = \{v \in L^2(U)^3 : \operatorname{div} v \in L^2(U)\}, \qquad H(\operatorname{curl},U) = \{v \in L^2(U)^3 : \operatorname{curl} v \in L^2(U)^3\},$$

and

$$X(U) = H(\operatorname{curl},U) \cap H(\operatorname{div},U), \qquad \|v\|_X^2 = \|v\|_2^2 + \|\operatorname{curl} v\|_2^2 + \|\operatorname{div} v\|_2^2.$$

1.2 Boundary conditions and the finitely many fluxes

Definition 1.2 (Zero normal and tangential traces)**.** For $v \in H(\operatorname{div},\Omega)$, the assertion $v \cdot n = 0$ means

$$\int_\Omega v \cdot \nabla q + \int_\Omega (\operatorname{div} v) q = 0 \quad (q \in H^1(\Omega)). \tag{1.3}$$

For $v \in H(\operatorname{curl}, \Omega)$, the assertion $v \times n = 0$ means

$$\int_\Omega v \cdot \operatorname{curl} w - \int_\Omega \operatorname{curl} v \cdot w = 0 \quad (w \in H^1(\Omega)^3). \tag{1.4}$$

Write $X_T = \{v \in X : v \cdot n = 0\}$ and $X_N = \{v \in X : v \times n = 0\}$.

For a bounded Lipschitz open set U write

$$H_0(\operatorname{curl}, U) = \{v \in H(\operatorname{curl}, U) : v \times n = 0\}, \qquad H_0(\operatorname{div}, U) = \{v \in H(\operatorname{div}, U) : v \cdot n = 0\}.$$

These are definitions by the weak Green identities above. Proposition 2.6 proves that they are also the closures of $C_c^\infty(U)^3$ in their respective graph norms; that closure characterization is not being assumed here.

These are genuine weak boundary conditions. For smooth fields the formulas are Green's identities and give precisely the indicated classical conditions. Their equivalence to zero boundary traces for rough fields, and the replacement of H^1 tests in (1.4) by all $H(\operatorname{curl})$ tests, are proved below. The subscripts describe the direction of the field: T means tangential to the boundary, not zero tangential component.

Choose smooth scalar cutoffs η_i equal to one near Γ_i and zero near every other boundary component. Their restrictions to Ω are in $H^1(\Omega)$, and $\nabla\eta_i \in C_c^\infty(\Omega)^3$. For $v \in H(\operatorname{div}, \Omega)$, define

$$F_i(v) = \int_\Omega v \cdot \nabla\eta_i + (\operatorname{div} v)\eta_i. \tag{1.5}$$

The value is independent of the cutoff: the difference of two such cutoffs has zero boundary trace and lies in $H_0^1(\Omega)$. In particular $F_i(v) = \langle v \cdot n, 1\rangle_{\Gamma_i}$, and

$$\sum_{i=0}^{I} F_i(v) = \int_\Omega \operatorname{div} v. \tag{1.6}$$

For $q \in H^1(D)$ let $[q]_j = q_- - q_+$ be the difference of its two traces on Σ_j. Local trace theory on the Lipschitz sides defines these traces. Write $\nabla_b q$ for the gradient on D, regarded as an $L^2(\Omega)^3$ function; the cuts have volume zero. Define

$$\Theta = \{q \in H^1(D) : [q]_j \text{ is constant for every } j\}.$$

We prove that there exist $r_j \in \Theta$ satisfying $[r_j]_k = \delta_{jk}$. If $v \in H(\operatorname{div}, \Omega)$ has zero normal trace, its cut flux is

$$C_j(v) = \int_D v \cdot \nabla r_j + (\operatorname{div} v) r_j. \tag{1.7}$$

This is independent of the lift because a function with all jumps zero glues to $H^1(\Omega)$, to which (1.3) applies. For smooth fields it is the integral of $v \cdot n_j$ over the cut. Thus this definition gives exactly the scalar pairing $\langle v \cdot n_j, 1\rangle_{\Sigma_j}$ appearing in the paper; we do not need to construct the entire fractional trace space on a cut.

1.3 The two potential theorems and the harmonic kernels

Main Theorem A (Tangential potential; paper Theorem 3.12)**.** *Under Definition 1.1, for $u \in L^2(\Omega)^3$ the following assertions are equivalent:*

(i) $\operatorname{div} u = 0$ *distributionally and* $F_i(u) = 0$ *for* $0 \le i \le I$.

(ii) There exists a unique $\psi \in X(\Omega)$ *such that*

$$\operatorname{curl} \psi = u, \quad \operatorname{div} \psi = 0, \quad \psi \cdot n = 0, \quad C_j(\psi) = 0 \quad (1 \le j \le J). \tag{1.8}$$

The solution depends linearly on u, and there is a constant depending only on Ω such that $\|\psi\|_X \le C_\Omega \|u\|_2$.

Main Theorem B (Normal potential; paper Theorem 3.17)**.** *Under Definition 1.1, for $u \in L^2(\Omega)^3$ the following assertions are equivalent:*

(i) $\operatorname{div} u = 0$, $u \cdot n = 0$, *and* $C_j(u) = 0$ *for* $1 \le j \le J$.

(ii) There exists a unique $\psi \in X(\Omega)$ *such that*

$$\operatorname{curl} \psi = u, \quad \operatorname{div} \psi = 0, \quad \psi \times n = 0, \quad F_i(\psi) = 0 \quad (0 \le i \le I). \tag{1.9}$$

The solution depends linearly on u, and $\|\psi\|_X \le C_\Omega \|u\|_2$.

The weak divergence conditions imply the requisite $H(\operatorname{div})$-membership, so these formulations have exactly the same data as the source statements. A requirement $\psi \in H^1(\Omega)^3$ would in general strengthen these theorems incorrectly. The source paper's Lemma 3.5 does give an H^1 potential, but without the prescribed boundary condition.

Main Theorem C (Harmonic kernels and flux estimates)**.** *Put*

$$K_T = \{v \in X_T : \operatorname{curl} v = 0,\ \operatorname{div} v = 0\}, \qquad K_N = \{v \in X_N : \operatorname{curl} v = 0,\ \operatorname{div} v = 0\}.$$

The cut-flux map $v \mapsto (C_1(v), \dots, C_J(v))$ is an isomorphism from K_T to $\mathbb{R}^J$, and the inner-boundary-flux map $v \mapsto (F_1(v), \dots, F_I(v))$ is an isomorphism from K_N to $\mathbb{R}^I$. In particular $\dim K_T = J$ *and* $\dim K_N = I$. *Moreover*

$$\|v\|_X \le C\Big(\|\operatorname{curl} v\|_2 + \|\operatorname{div} v\|_2 + \sum_{j=1}^{J} |C_j(v)|\Big) \qquad (v \in X_T), \tag{1.10}$$

$$\|v\|_X \le C\Big(\|\operatorname{curl} v\|_2 + \|\operatorname{div} v\|_2 + \sum_{i=1}^{I} |F_i(v)|\Big) \qquad (v \in X_N). \tag{1.11}$$

The reverse inequalities with another constant hold as well. These are the content of paper Propositions 3.14, 3.18 and Corollaries 3.16, 3.19.

Why topology occurs. For divergence-free fields the total boundary flux is automatically zero, but the flux through an individual component need not vanish. Thus the outer flux condition is redundant once all the inner ones vanish. A spherical shell has one inner boundary component and no handle; a solid torus has a handle and no inner boundary component. The two flux systems measure different obstructions. Main Theorem C identifies the exact ambiguity removed by each normalization.

1.4 The remaining principal statements of the original paper

For an integer $m \geq 1$, $W^{m,2}$ denotes functions with weak derivatives through order m in L^2. For real $s \geq 0$, $H^s(\Omega)$ is the restriction of the Fourier-defined Bessel-potential space $H^s(\mathbb{R}^3)$, with the infimum norm over extensions. The extension and Fourier arguments below identify H^1 and H^2 with their weak-derivative definitions on Lipschitz domains, and H^m with $W^{m,2}$ on $C^{m,1}$ domains. Higher integer-order data in the polyhedral finite-element results retain the restriction-space meaning; no unproved all-order Lipschitz extension theorem is used. Boundary spaces on a $C^{m,1}$ boundary are defined by finite charts and a partition of unity; their local norm is the corresponding Fourier Sobolev norm. A $C^{m,1}$ domain has boundary graphs with m continuous derivatives and Lipschitz derivatives of order m. A polyhedron below is a bounded Lipschitz domain with finitely many planar polygonal faces. All continuous-embedding constants may depend on the fixed domain.

Main Theorem D (Regularity and compactness; paper Section 2)**.** *For every bounded connected Lipschitz domain, without a cut hypothesis,*

$$X_T \cap X_N = H_0^1(\Omega)^3, \qquad \|\nabla v\|_2^2 = \|\operatorname{curl} v\|_2^2 + \|\operatorname{div} v\|_2^2 \quad (v \in H_0^1(\Omega)^3).$$

The inclusion $X \hookrightarrow L^2(\Omega)^3$ is not compact, whereas the inclusions $X_T \hookrightarrow L^2(\Omega)^3$ and $X_N \hookrightarrow L^2(\Omega)^3$ are compact. If the domain is $C^{1,1}$ or convex, both X_T and X_N embed continuously into $H^1(\Omega)^3$. More generally, on a $C^{m,1}$ domain,

$$v \in L^2(\Omega)^3, \quad \operatorname{curl} v \in H^{m-1}(\Omega)^3, \quad \operatorname{div} v \in H^{m-1}(\Omega),$$

together with either $v \cdot n \in H^{m-1/2}(\partial\Omega)$ or $v \times n \in H^{m-1/2}(\partial\Omega)^3$, implies $v \in H^m(\Omega)^3$, with a norm estimate in terms of these four data. These are paper Theorems 2.5, 2.8, 2.9, 2.12, 2.17, Proposition 2.7, and Corollary 2.15.

The distinct polyhedral assertion. Paper Proposition 3.7 asserts that for every Lipschitz polyhedron there is $s > 1/2$, depending on the polyhedron, for which $X_T, X_N \hookrightarrow H^s(\Omega)^3$ continuously. A proof is supplied in Theorem 8.32 in the regularity chapter. Neither the $C^{1,1}$ result nor convex regularity may be used as a substitute for this assertion on a general nonconvex polyhedron.

For a harmonic scalar $b \in L^2(\Omega)$ define its total weak normal derivative on a boundary component by

$$N_i(b) = -\int_\Omega b\Delta\eta_i,$$

where the smooth cutoff η_i is one near Γ_i and zero near the other components. The definition is independent of the cutoff and requires no unproved assumption that $\nabla b \in L^2$.

Main Theorem E (Full Dirichlet potential; paper Theorem 3.20)**.** *On a $C^{1,1}$ domain with the admissible cuts, the conditions*

$$\operatorname{div} u = 0, \qquad u \cdot n = 0, \qquad C_j(u) = 0 \quad (1 \leq j \leq J)$$

on $u \in L^2(\Omega)^3$ are equivalent to the existence of a unique $\psi \in H_0^1(\Omega)^3$ such that

$$\operatorname{curl} \psi = u, \qquad \Delta(\operatorname{div} \psi) = 0, \qquad N_i(\operatorname{div} \psi) = 0 \quad (0 \leq i \leq I).$$

The solution map is bounded and linear into H^1. The corresponding homogeneous kernel before the N_i constraints has dimension I, as in paper Proposition 3.21.

Finite-element data. Let $\mathcal{T}_h$ be a conforming tetrahedral triangulation of a Lipschitz polyhedron, $h = \max_K \operatorname{diam} K$, in a family with uniformly bounded diameter-to-inradius ratio. Require a mesh-aligned admissible cut system as in paper Hypothesis 4.1. For $k \geq 1$ put

$$\mathcal{N}_k(K) = P_{k-1}(K)^3 + \{b \in \widetilde{P}_k(K)^3 : b(x) \cdot x = 0\}, \qquad Y_h = \{v \in H(\operatorname{curl}, \Omega) : v|_K \in \mathcal{N}_k(K)\},$$

where P_r denotes polynomials of degree at most r and $\widetilde{P}_r$ homogeneous polynomials of degree r. Let Y_h^0 impose zero tangential trace. Put $\Theta_h = \{q \in \Theta : q|_K \in P_k(K)\}$ and $\Theta_h^0 = \{q \in \Theta_0 : q|_K \in P_k(K)\}$. For the two cases take $(X_h, Q_h, G) = (Y_h, \Theta_h, \nabla_b)$ or $(Y_h^0, \Theta_h^0, \nabla)$, and set $V_h = \{v_h \in X_h : (v_h, Gq_h) = 0 \ \forall q_h \in Q_h\}$.

Main Theorem F (Finite-element solvability and error)**.** *For every $u \in L^2(\Omega)^3$ there is a unique $\psi_h \in V_h$ with*

$$(\operatorname{curl} \psi_h, \operatorname{curl} \varphi_h) = (u, \operatorname{curl} \varphi_h) \qquad (\varphi_h \in X_h).$$

The associated mixed scalar multiplier is zero (modulo constants in the first case). Uniformly over the mesh family, $\|v_h\|_2 \leq C\|\operatorname{curl} v_h\|_2$ for $v_h \in V_h$. For $0 < \sigma \leq k$ and $u \in H^\sigma(\Omega)^3$ satisfying the compatibility conditions of Main Theorem A in the first case, or Main Theorem B in the second case,

$$\|u - \operatorname{curl} \psi_h\|_2 \leq Ch^\sigma \|u\|_{H^\sigma(\Omega)}.$$

The constant is independent of u and h and may depend on the fixed domain, k, σ, and the shape-regularity bound. These are Propositions 4.5, 4.6, 4.11, 4.12 and Theorems 4.8, 4.13.

Stokes spaces. Let $V = \{v \in H_0^1(\Omega)^3 : \operatorname{div} v = 0\}$. The space W_T consists of $\psi \in X_T$ with $\Delta\psi, \operatorname{curl}\operatorname{curl}\psi \in L^2$, $C_j(\psi) = 0$, and $\operatorname{curl}\psi \in H_0^1(\Omega)^3$. The space W_N consists of $\psi \in X_N$ with the same two second-order L^2 conditions, $F_i(\psi) = 0$ for every i, and a field $H\psi \in K_T$ such that $\operatorname{curl}\psi + H\psi \in H_0^1$. This harmonic field, when it exists, is unique. These definitions are equivalent to the trace definitions of the spaces in paper Section 4.2.

Main Theorem G (Both Stokes formulations; paper Theorems 4.15–4.22)**.** *Let $\mu > 0$ and $f \in L^2(\Omega)^3$. Each of the following problems has a unique solution in its indicated space:*

$$\begin{aligned} \mu(\Delta\psi_T, \Delta\varphi) &= (f, \operatorname{curl}\varphi) && (\varphi \in W_T), && \psi_T \in W_T, \\ \mu(\Delta\psi_N, \Delta\varphi) &= (f, \operatorname{curl}\varphi + H\varphi) && (\varphi \in W_N), && \psi_N \in W_N. \end{aligned}$$

Both solutions have zero weak divergence. The reconstructed velocities

$$u = \operatorname{curl}\psi_T = \operatorname{curl}\psi_N + H\psi_N$$

coincide with the unique variational Stokes velocity in V. There is a unique mean-zero $p \in L^2(\Omega)$ such that $-\mu\Delta u + \nabla p = f$ distributionally. Conversely the potentials of a primitive-variable Stokes solution solve these two problems. The Laplacian L^2 norm is equivalent to the natural five-term graph norm on each W_T, W_N.

The three Helmholtz decompositions, the cylindrical potential of Remark 3.23, and the source statements requiring correction are also treated in later chapters. The higher-order Remark 3.22 and the unnumbered mixed-boundary claims are not silently assumed: explicit counterexamples to their literal formulations are given in Section 18.

1.5 Coverage of the original paper

The following table checks statements, not just theorem numbers. The domain hypotheses, weak traces, topological fluxes, uniqueness gauges, mesh uniformity, fractional exponent range, and harmonic correction in the normal Stokes formulation are retained. "Proved" in this table means that a mathematical proof is supplied in this manuscript; of these results, Theorems 3.12 and 3.17 are also formalized in Lean.

Original result	Content checked	Proof or correction here
Theorem 2.5; Remark 2.6	Both zero traces give H_0^1; exact div–curl energy identity	Theorem 8.1
Proposition 2.7	Failure of compactness without boundary conditions	Proposition 8.2
Theorem 2.8	Compactness of both one-trace Maxwell spaces	Theorem 8.4
Theorems 2.9, 2.12	$C^{1,1}$ regularity for both boundary conditions	Theorem 8.7
Corollary 2.15; Remark 2.14	Higher-order estimates including nonzero boundary traces	Theorem 8.12
Theorem 2.17	H^1 regularity on nonsmooth convex domains	Theorem 8.15
Lemma 3.5	Regular potential without imposed boundary values	Theorem 3.4
Proposition 3.7	A positive margin above $H^{1/2}$ on general Lipschitz polyhedra	Theorem 8.32
Theorems 3.12, 3.17	Both canonical vector potentials, iff conditions and uniqueness	Main Theorems A, B
Propositions 3.14, 3.18; Corollaries 3.16, 3.19	Harmonic bases, dimensions, and graph-norm estimates	Proposition 5.7; Corollary 7.1
Section 3.5, (3.30)–(3.37)	Three orthogonal Helmholtz decompositions and correction formulas	Theorem 9.1
Theorem 3.20; Proposition 3.21	Full Dirichlet potential, different gauge, and kernel	Theorem 10.5; Proposition 10.4
Remark 3.23	Cylindrical potential with the averaged weak-flux interpretation	Section 15
Propositions 4.5, 4.11	Both mixed discrete problems and their zero multipliers	Theorem 16.4
Propositions 4.6, 4.12	Mesh-uniform coercivity in both discrete constrained spaces	Theorems 16.9, 16.17
Lemma 4.7	Weak edge moments, continuity, and commuting interpolation	Lemma 16.6
Theorems 4.8, 4.13	Optimal curl error for every $0 < \sigma \leq k$	Theorem 16.18
Theorems 4.15, 4.17; Lemma 4.16	Tangential Stokes formulation and coercivity	Theorem 11.4
Theorems 4.19, 4.21; Lemma 4.20	Normal Stokes formulation, with its harmonic correction	Theorem 12.3
Corollaries 4.18, 4.22	Recovery of the full Stokes pair including L^2 pressure	Corollary 14.5
Remark 3.22 as printed	Missing necessary higher regularity and boundary conditions	Counterexample 18.1

Original result	Content checked	Proof or correction here
Section 3.5, (3.39)–(3.42) as printed	Insufficient relative-topology conditions	Counterexamples 18.3, 18.4

The last two rows are mathematical corrections, not proof omissions: their literal assertions are false, and they are nowhere used as hypotheses. The additional general-Lipschitz and sharp regularity statements attributed to external work in Remarks 2.16, 2.19, and 3.8 are not asserted as newly proved theorems here; the numbered polyhedral result actually used in the paper is Proposition 3.7. These cited refinements remain outside the coverage of this manuscript.

2 Analytic foundations on Lipschitz domains

We use real-valued functions except when taking Fourier transforms. The foundational results about Euclidean measure and Hilbert spaces used below are Fubini's theorem, dominated and monotone convergence, completeness of L^2, the representation theorem for continuous linear functionals on a Hilbert space, and the Plancherel theorem. We also use the elementary one-dimensional facts that a Lipschitz function is absolutely continuous and that its derivative exists almost everywhere and is bounded by its Lipschitz constant. Smooth bump functions, convolution, and smooth partitions of unity are used in their usual Euclidean meanings. The approximation and domain-dependent facts required in the proof are established here.

For an open set $D \subset \mathbb{R}^3$, $H^1(D)$ means the space of L^2 functions whose three distributional first derivatives belong to $L^2(D)$, with norm $\|v\|^2_{H^1(D)} = \|v\|^2_{L^2(D)} + \|\nabla v\|^2_{L^2(D)}$. It is complete: if a sequence and its derivatives converge in L^2, testing against a compactly supported smooth function identifies the limits as the weak derivatives. We write $H^1_0(D)$ for the closure of $C^\infty_c(D)$ in this norm. Vector versions are componentwise.

2.1 The local calculus used in flattening a graph

Lemma 2.1 (Interior approximation and slices)**.** *On every open set D, $C^\infty(D) \cap H^1(D)$ is dense in $H^1(D)$. On a rectangular box, an H^1 function has, on almost every line parallel to each coordinate axis, an absolutely continuous representative whose one-dimensional derivative is the corresponding weak derivative. For $f \in H^1(0,h)$ this representative satisfies*

$$|f(0)|^2 \le \frac{2}{h}\int_0^h |f(t)|^2\,dt + 2h\int_0^h |f'(t)|^2\,dt. \tag{2.1}$$

Proof. We give the approximation argument, in particular to avoid assuming boundary regularity at this stage. Choose a locally finite cover of D by balls whose closures lie in D, and a subordinate locally finite family of smooth functions χ_k with $\sum_k \chi_k = 1$ on D. One explicit construction starts with relatively compact sets exhausting D, covers each compact layer by finitely many sufficiently small balls, and uses nonnegative bump functions positive on smaller balls; dividing each bump by their locally finite positive sum gives the required family. The balls and slightly enlarged balls can be chosen locally finite. The product rule in distributions gives $\partial_j(\chi_k v) = \chi_k \partial_j v + (\partial_j \chi_k)v$. Extend each $\chi_k v$ by zero outside its compact support in D and convolve it with a smooth approximate identity of radius ϵ_k. Choose ϵ_k small enough to retain support in the enlarged ball and to make the $H^1(\mathbb{R}^3)$ error less than $\epsilon 2^{-k}$. The resulting locally finite sum is smooth in D and differs from v in $H^1(D)$ by at most ϵ. Here convergence of convolution in L^2 follows first for continuous compactly supported functions by uniform continuity, and then for all L^2 functions

by their elementary approximation by step functions and the inequality $\|\rho_\epsilon * f\|_2 \le \|\rho_\epsilon\|_1 \|f\|_2$. Convolution commutes with weak derivatives. This also proves the corresponding approximation assertion on $\mathbb{R}^3$; multiplication by cutoffs tending to one makes the approximants compactly supported if needed.

For the slicing assertion, apply Fubini to the weak-derivative identity, first with product test functions and then with a countable dense set of one-dimensional test functions. For almost every transverse coordinate the slice has weak derivative equal to the slice of the designated derivative. If f and g satisfy the one-dimensional weak-derivative identity, subtract $\int_{t_0}^t g(s)\,ds$ from f. The difference has zero distributional derivative and is constant: every compactly supported smooth test function of integral zero is the derivative of another such test function. This proves absolute continuity and the asserted derivative. Finally, $f(0) = f(t) - \int_0^t f'(s)\,ds$, the inequality $|a+b|^2 \le 2|a|^2 + 2|b|^2$, and Cauchy–Schwarz give (2.1) after averaging over t. □

Lemma 2.2 (Flattening, reflection, and flat traces)**.** *Let $Q \subset \mathbb{R}^2$ be a rectangle, let $h > 0$, and let $\varphi : Q \to \mathbb{R}$ be Lipschitz. Put*

$$F(y,t) = (y, \varphi(y) + t), \qquad C = F(Q \times (-h,h)), \qquad C_+ = F(Q \times (0,h)).$$

Composition with F and with F^{-1} gives bounded mutually inverse maps between the respective H^1 spaces. If $w \in H^1(Q \times (0,h))$, its even reflection $w(y,|t|)$ belongs to $H^1(Q \times (-h,h))$. There is a bounded trace map $w \mapsto w(\cdot,0)$ into $L^2(Q)$. Two H^1 functions on the upper and lower half boxes glue to an H^1 function on the full box if and only if their traces agree.

Proof. Fubini and a translation in the vertical variable show directly that F preserves volume. For smooth v on C_+, the one-dimensional chain rule on coordinate lines gives, almost everywhere,

$$\partial_{y_j}(v \circ F) = (\partial_j v) \circ F + (\partial_3 v) \circ F\, \partial_j \varphi, \qquad \partial_t (v \circ F) = (\partial_3 v) \circ F.$$

The weak partial derivatives of φ are bounded by its Lipschitz constant; this follows from its absolute continuity on coordinate lines and Fubini. Thus the displayed identities give the required norm bound. Approximate an arbitrary $v \in H^1(C_+)$ using Lemma 2.1. The norm bound makes the compositions Cauchy in H^1, and volume preservation identifies their L^2 limit as $v \circ F$. This proves the formulas for v. Applying the same argument to $F^{-1}(x) = (x', x_3 - \varphi(x'))$ proves the reverse bound. The full-box and lower-half versions are identical. Rigid changes of coordinates do not affect this argument.

The slice representative and (2.1), integrated over y, define the trace and prove its boundedness. Moreover, for almost every $t > 0$,

$$\|w(\cdot,t) - w(\cdot,0)\|_{L^2(Q)}^2 \le t \int_{Q \times (0,t)} |\partial_t w|^2. \tag{2.2}$$

Even reflection has vertical weak derivative $\operatorname{sgn}(t)\partial_t w(y,|t|)$ and tangential derivatives $\partial_{y_j} w(y,|t|)$. For completeness, the vertical identity follows by integration by parts on the absolutely continuous slices: the two terms at zero cancel. For a tangential derivative, test on the upper half against the sum of the test function at t and $-t$; a cutoff that vanishes for $0 < t < \delta$ makes it an admissible test, and letting $\delta \downarrow 0$ introduces no differentiated cutoff because the derivative is tangential. The H^1 norm squared of the reflection is exactly twice that of w.

The same argument proves gluing when the two traces agree: the vertical boundary terms cancel and the tangential calculation is unchanged. If the piecewise function is in H^1 of the full box, its almost every vertical slice is absolutely continuous through zero. Its two one-sided traces therefore agree. This proves necessity as well. □

2.2 Extension, traces, and zero boundary values

Proposition 2.3 (Lipschitz extension and traces)**.** *Let D be a bounded open set with Lipschitz boundary, with the convention that D lies on one side of its boundary in every boundary chart. It need not be connected. There is a bounded linear operator*

$$E_D : H^1(D) \longrightarrow H^1(\mathbb{R}^3), \qquad (E_D v)|_D = v,$$

whose outputs are supported in one fixed bounded set. Restrictions of $C_c^\infty(\mathbb{R}^3)$ are dense in $H^1(D)$. The local graph traces define a continuous boundary trace. Its kernel is exactly $H_0^1(D)$. Across a Lipschitz interface two H^1 functions glue to an H^1 function if and only if their traces agree.

Proof. Compactness of ∂D gives finitely many smaller graph cylinders contained in larger cylinders of the form in Lemma 2.2. Together with an open set compactly contained in D, these smaller cylinders cover $\overline{D}$. Choose a finite smooth partition $\chi_0, \dots, \chi_N$ equal to one in sum on a neighborhood of $\overline{D}$, with χ_0 compactly supported in D and each other χ_j compactly supported in its larger cylinder. Such a partition follows from the bump-function construction already described, using an additional cutoff equal to one near $\overline{D}$. Flatten $\chi_j v$, reflect it evenly, and undo the flattening. Its support stays away from the lateral and top and bottom edges of the larger cylinder, so its extension by zero outside that cylinder belongs to $H^1(\mathbb{R}^3)$. Indeed a function zero in a neighborhood of those edges can be tested distributionally without a boundary term. Extend $\chi_0 v$ by zero and sum the finitely many pieces. The product rule and the bounds in Lemma 2.2 prove boundedness, linearity, and a common bounded support for this extension. Convolving $E_D v$ gives the stated smooth density.

The trace in a boundary chart is the trace of $v \circ F$ at $t = 0$. One can formulate everything below using these finitely many $L^2(Q)$ traces, without introducing fractional boundary spaces or boundary integrals. To verify consistency on overlaps, approximate v by ambient smooth functions as just proved. The traces of these approximants are their ordinary restrictions and are consistent. Coordinate transitions between two graph parametrizations are locally bi-Lipschitz. Such maps preserve sets of measure zero and bound the measure of images by a constant times the original measure: cover a set by small squares and use that the image of a square of side r is contained in a square of side at most Cr, then take the infimum over coverings. Consequently the L^2 limits of the smooth traces agree on overlapping relatively compact chart pieces. This verifies that equality or vanishing of traces has an intrinsic meaning. If a single boundary norm is desired, take the sum of the finitely many chart L^2 norms after a subordinate partition; all statements made here are independent of this choice.

Every compactly supported smooth function in D has zero trace, so $H_0^1(D)$ is contained in the trace kernel. Conversely suppose that v has zero trace. In a flattened boundary chart extend $\chi_j v$ by zero to the lower half box. The gluing assertion of Lemma 2.2, with the lower function equal to zero, shows that this extension belongs to H^1; thus the global extension $\overline{\chi_j v}$ by zero belongs to $H^1(\mathbb{R}^3)$. In the physical coordinates of that chart translate this function a distance ϵ in the inward vertical direction. Its support is then compactly contained in D: its original support was away from the edges of the chart and its new vertical height above the graph is at least ϵ. Quantitatively its distance from the graph is at least $\epsilon/\sqrt{1 + L_j^2}$, where L_j is the Lipschitz constant. For sufficiently small ϵ the other chart margins persist. Translations converge in $H^1(\mathbb{R}^3)$, since they do so for smooth functions and are isometries. Mollify the translated function at a radius less than half the distance of its support from D^c. This gives elements of $C_c^\infty(D)$ converging to $\chi_j v$ in $H^1(D)$. The interior piece is treated by direct mollification. Adding these finitely many approximations proves $v \in H_0^1(D)$. Finally, flattening an interface reduces the last assertion of the proposition to the gluing assertion of Lemma 2.2; a partition of unity makes the local assertion global. □

Remark 2.4. There is a convenient precise meaning of "normal trace" requiring no fractional-space theorem. Give $\mathcal{T}(D) = H^1(D)/H^1_0(D)$ its quotient Hilbert norm. For $u \in H(\operatorname{div}, D)$, where both u and $\operatorname{div} u$ belong to L^2, the formula

$$\langle \gamma_n u, [v] \rangle = \int_D u \cdot \nabla v + \int_D (\operatorname{div} u) v$$

defines a bounded functional on $\mathcal{T}(D)$. The expression vanishes on $H^1_0(D)$ by approximation with compactly supported tests, so it is well-defined; its bound follows from Cauchy–Schwarz and the infimum defining the quotient norm. Proposition 2.3 identifies the equivalence class $[v]$ with its graph trace. Thus saying $u \cdot n = 0$ means exactly that the displayed expression is zero for every $v \in H^1(D)$. This definition also fixes the signs used below. The corresponding vector trace quotient is $\mathcal{T}_{\mathrm{vec}}(D) = H^1(D)^3/H^1_0(D)^3$. For $v \in H(\operatorname{curl}, D)$ set

$$\langle \gamma_\tau v, [w] \rangle = \int_D v \cdot \operatorname{curl} w - \int_D \operatorname{curl} v \cdot w.$$

It vanishes on $H^1_0(D)^3$ by compactly supported approximation and the weak curl identity. Cauchy–Schwarz followed by the quotient infimum proves its continuity. For smooth fields this functional represents $v \times n$, with exactly the sign in (1.4).

Proposition 2.5 (Density in the curl and divergence graph norms)**.** *On a bounded Lipschitz open set D, restrictions of $C^\infty_c(\mathbb{R}^3)^3$ are dense in each of*

$$H(\operatorname{curl}, D) = \{v \in L^2(D)^3 : \operatorname{curl} v \in L^2(D)^3\}, \qquad H(\operatorname{div}, D) = \{v \in L^2(D)^3 : \operatorname{div} v \in L^2(D)\}.$$

They are also dense in the intersection equipped with the sum of the three squared L^2 norms.

Proof. Use the finite partition from Proposition 2.3. Localization is continuous in either graph norm because, in distributions,

$$\operatorname{curl}(\chi v) = \chi \operatorname{curl} v + \nabla\chi \times v, \qquad \operatorname{div}(\chi v) = \chi \operatorname{div} v + \nabla\chi \cdot v.$$

An interior piece can be extended by zero and mollified. Consider a boundary piece $a = \chi_j v$, supported in a compact subset of a graph cylinder, with D above its graph. Let A be its extension by zero to $\mathbb{R}^3$ and let B be the extension by zero of $\operatorname{curl} a$ (or of $\operatorname{div} a$). We do not assert that the distributional curl or divergence of A is B across the boundary. For $\epsilon > 0$ and $0 < \delta < \epsilon/[2(1 + L_j)]$, define the ambient smooth field

$$a_{\epsilon,\delta}(x) = (\rho_\delta * A)(x + \epsilon e_3),$$

where the coordinate e_3 points into D. All supports remain in a fixed compact set. If $x \in D$ and the convolution can meet the support of A, sufficiently small ϵ, δ keep the whole relevant ball inside the larger chart. For $|z| < \delta$ its points satisfy

$$x_3 + \epsilon - z_3 - \varphi(x' - z') > \epsilon - (1 + L_j)\delta > 0.$$

Thus the convolution kernel, wherever it can meet the localized field, is an admissible interior test function. Differentiating the convolution in the weak identity gives, on D,

$$\operatorname{curl} a_{\epsilon,\delta}(x) = (\rho_\delta * B)(x + \epsilon e_3)$$

in the curl case, and exactly the corresponding divergence identity in the divergence case. The support of a distributional derivative is contained in the support of the original distribution, so the support of B is contained in the localized support as well. Where the relevant ball misses this support the identities both read zero; the compact chart margins justify the stated alternatives. Translation continuity and approximate identity convergence in $L^2(\mathbb{R}^3)$ show that these fields and their designated derivatives converge on D to a and its derivatives as $\epsilon \downarrow 0$, for example with $\delta = \epsilon/[4(1 + L_j)]$. Sum the finitely many pieces. The same choices work simultaneously for curl and divergence when both are in L^2. □

Proposition 2.6 (Zero-extension characterization and interior density)**.** *Let D be bounded and Lipschitz and let Z_D denote extension by zero from D to $\mathbb{R}^3$. A field $v \in H(\mathrm{div}, D)$ has zero normal trace in the sense of Remark 2.4 if and only if*

$$\mathrm{div}(Z_D v) = Z_D(\mathrm{div}\, v) \quad \textit{in distributions on } \mathbb{R}^3.$$

These conditions are also equivalent to v being in the closure of $C_c^\infty(D)^3$ in the divergence graph norm. Likewise, for $v \in H(\mathrm{curl}, D)$, the identity

$$\mathrm{curl}(Z_D v) = Z_D(\mathrm{curl}\, v) \quad \textit{in distributions on } \mathbb{R}^3$$

is equivalent to membership in the closure of $C_c^\infty(D)^3$ in the curl graph norm.

Proof. Testing the divergence identity with $\phi \in C_c^\infty(\mathbb{R}^3)$ gives exactly $\int_D v \cdot \nabla\phi + \int_D (\mathrm{div}\, v)\phi = 0$. The density in Proposition 2.3 extends this identity from ambient smooth restrictions to every $\phi \in H^1(D)$, proving the first equivalence. For curl, testing its zero-extension identity with ambient smooth vector fields gives the zero tangential Green identity on their restrictions. The H^1 density just proved extends it to all $H^1(D)^3$ tests, and the reverse implication follows by those same ambient tests. Thus the curl zero-extension condition is exactly the definition of $H_0(\mathrm{curl}, D)$. Compactly supported smooth fields plainly satisfy either zero-extension identity; convergence in the appropriate graph norm preserves the identity distributionally.

For the converse use the finite partition of Proposition 2.3. Multiplication by a smooth cutoff preserves the zero-extension identity, by the distributional product formulas and $Z_D(\chi v) = \chi Z_D v$. A localized boundary piece therefore has a zero extension A in the corresponding whole-space graph space, with derivative equal to the zero extension of the derivative of that piece. In a chart where D lies above the graph, translate A inward, using $A_\epsilon(x) = A(x - \epsilon e_3)$. Its support is compactly contained in D for all sufficiently small $\epsilon > 0$, by precisely the support and distance argument in the zero-trace proof of Proposition 2.3. Translations commute with constant-coefficient distributional derivatives and converge in L^2 both for A and for its designated derivative. Mollify A_ϵ at a radius smaller than half its distance from D^c and tending to zero. Convolution commutes with curl and divergence and converges in both required L^2 norms. The result lies in $C_c^\infty(D)^3$ and approximates the localized piece in its graph norm. Mollify the interior piece directly and sum the finitely many approximations. □

2.3 Compactness, Poincare inequalities, and the Neumann problem

Lemma 2.7 (Compactness and Poincare)**.** *For bounded Lipschitz D, the inclusion $H^1(D) \hookrightarrow L^2(D)$ is compact. If D is connected and $v_D = |D|^{-1} \int_D v$, then*

$$\|v - v_D\|_{L^2(D)} \le C_D \|\nabla v\|_{L^2(D)}. \tag{2.3}$$

The gradient norm is therefore a complete Hilbert norm on $H^1_*(D) = \{v \in H^1(D) : \int_D v = 0\}$.

Proof. Extend a bounded sequence in $H^1(D)$ by the fixed operator in Proposition 2.3. Its extensions are bounded in $H^1(\mathbb{R}^3)$ and supported in one fixed cube. On a cube Q of side δ, successive one-dimensional differences and Cauchy–Schwarz give

$$\int_Q |f - f_Q|^2 \le 3\delta^2 \int_Q |\nabla f|^2. \tag{2.4}$$

Here are details of the estimate. Jensen bounds its left side by $|Q|^{-1} \int_{Q\times Q} |f(x) - f(y)|^2\, dx\, dy$. Insert the three points obtained by replacing coordinates of x by those of y, one at a time. The square of the sum of the three differences is at most three times the sum of their squares. Each

coordinate difference is an integral of the corresponding partial derivative over an interval of length at most δ; integrating the resulting bound over the six coordinates and dividing by $|Q|$ gives at most $\delta^2 \int_Q |\partial_j f|^2$ for that difference. The slicing lemma justifies this for H^1 functions.

Partition a large cube containing all supports into finitely many cubes of side δ and replace each extension by its average on each cube. Equation (2.4) bounds the L^2 error uniformly by $\sqrt{3}\delta$ times the common gradient bound. The averages form a bounded subset of a finite-dimensional space and hence have finite L^2 nets at every positive accuracy. Letting $\delta \downarrow 0$ shows that the original sequence is totally bounded in L^2 and has a convergent subsequence. Restriction proves compactness on D.

If (2.3) failed, subtracting means and normalizing would give v_k with mean zero, $\|v_k\|_2 = 1$, and $\|\nabla v_k\|_2 \to 0$. Compactness gives a limit v of norm one and mean zero with distributional gradient zero. A zero-gradient function is constant on every ball compactly contained in D: convolve on smaller balls, observe that the smooth convolution has zero gradient, and pass to the L^2 limit. Constants agree on overlapping balls. Any two points of a connected open set are joined by a finite chain of overlapping interior balls: the points reachable from one ball form a relatively open set whose complement is also relatively open. Hence v is constant on D, a contradiction. Finally, mean zero defines a closed subspace of $H^1(D)$ and (2.3) makes its gradient norm equivalent to its complete H^1 norm. □

Proposition 2.8 (Finite Lipschitz covers and cut domains)**.** *Suppose a bounded connected open set D is the union of finitely many connected Lipschitz open subsets $D_1, \dots, D_m$. Then $H^1(D) \hookrightarrow L^2(D)$ is compact and (2.3) holds on D. In particular these assertions hold on a bounded connected pseudo-Lipschitz domain in the sense that, in a sufficiently small ball about each boundary point, the domain has one or two connected components, each a Lipschitz open set.*

Proof. A sequence bounded in $H^1(D)$ is bounded on every D_j. Finitely many successive applications of Lemma 2.7 give one subsequence Cauchy in every $L^2(D_j)$. Since $\|f\|^2_{L^2(D)} \le \sum_j \|f\|^2_{L^2(D_j)}$, it is Cauchy in $L^2(D)$. The contradiction argument proving (2.3) then applies verbatim, since zero-gradient rigidity used only connectedness and interior balls. Equivalently, the overlap graph of the cover is connected, and every nonempty overlap is open and has positive volume, so constants on the pieces must agree. For a bounded pseudo-Lipschitz domain take finitely many of the indicated boundary balls covering its compact boundary. The portion of its closure outside their union is compactly contained in D and is covered by finitely many interior balls. The one or two local pieces in each boundary ball, together with these interior balls, supply the required finite Lipschitz cover. □

Proposition 2.9 (Variational Neumann problem)**.** *Let D be bounded, connected, and Lipschitz, and let ℓ be a bounded linear functional on $H^1(D)$ such that $\ell(1) = 0$. There is a unique $s \in H^1_*(D)$ satisfying*

$$\int_D \nabla s \cdot \nabla q = \ell(q) \quad \text{for every } q \in H^1(D). \tag{2.5}$$

It depends linearly on ℓ and obeys $\|s\|_{H^1(D)} \le C_D \|\ell\|_{(H^1(D))'}$.

Proof. On the Hilbert space $H^1_*(D)$ with gradient inner product, Poincare's inequality makes ℓ continuous. Hilbert-space representation gives exactly one s representing it and bounds $\|\nabla s\|_2$ by $C_D\|\ell\|$. The representation identity extends to all q by subtracting q_D and using $\ell(1) = 0$. Poincare bounds s itself, and uniqueness gives linearity. □

3 A bounded divergence-free vector-potential operator

3.1 Boundary components and volume fluxes

Lemma 3.1 (The complementary pieces)**.** *Let $\Omega \subset \mathbb{R}^3$ be bounded, connected, and Lipschitz, and choose a ball B with $\overline{\Omega} \subset B$. The boundary $\partial\Omega$ has finitely many connected components $\Gamma_1, \dots, \Gamma_N$. The open set $B \setminus \overline{\Omega}$ has finitely many connected components $D_1, \dots, D_M$, each a bounded connected Lipschitz open set. Each Γ_i is a boundary component of exactly one of the D_a in the sense that it is wholly contained in its boundary and disjoint from the boundaries of the other pieces. The rest of ∂D_a, if nonempty, is ∂B. Thus $\partial D_a \cap \partial\Omega$ is a union of entire Γ_i. Here the indexing is local to this analytic construction: the components $\Gamma_1, \dots, \Gamma_N$ are the components usually denoted $\Gamma_0, \dots, \Gamma_I$, relabeled with $N = I + 1$.*

Proof. Shrink each boundary chart so that its graph patch is connected. Each such relatively open connected patch lies in a single connected component of $\partial\Omega$. A finite cover by these patches shows that there are finitely many boundary components; each is compact, and distinct components have positive distance.

The side of a chart outside $\overline{\Omega}$ is connected, so lies in one component of $B \setminus \overline{\Omega}$. This assignment is locally constant as the boundary point moves in the graph patch. It is therefore constant on each connected Γ_i. Every point of that Γ_i is in the boundary of the assigned component, and no other component can approach it, by the same local description. Near ∂B there is a full annulus disjoint from $\overline{\Omega}$; it is connected, so all of ∂B is assigned to one component. A component D_a has a nonempty boundary, and its boundary lies in $\partial\Omega \cup \partial B$. To justify the latter assertion, a point of the open set $B \setminus \overline{\Omega}$ has an interior ball contained in its own component and hence cannot be a boundary point of another component. There are only finitely many graph patches and one sphere, so only finitely many components can occur. At every point of ∂D_a the local set is precisely the designated graph side or the interior side of the sphere. This proves its Lipschitz property and all remaining assertions. No classification of surfaces or separation theorem for embedded surfaces is used here. □

Choose $\eta_i \in C^\infty(\mathbb{R}^3)$ equal to one in a neighborhood of Γ_i and zero in neighborhoods of every other Γ_j. These cutoffs exist because finitely many disjoint compact sets have positive mutual distance: choose disjoint neighborhoods, convolve their smaller-neighborhood indicators with sufficiently small mollifiers, and if necessary multiply by an exterior compact cutoff. For a divergence-free field $u \in L^2(\Omega)^3$ define

$$\mathcal{F}_i(u) = \int_\Omega u \cdot \nabla\eta_i. \tag{3.1}$$

This value is independent of the cutoff: the difference of two admissible cutoffs, restricted to Ω, has compact support in Ω, and the distributional equation $\operatorname{div} u = 0$ annihilates its gradient. It is the normal-trace pairing of Remark 2.4 with the boundary function that is one on Γ_i and zero elsewhere. For smooth fields this is the usual outward flux, but no classical surface integral is needed in the definition. Also $\sum_i \mathcal{F}_i(u) = 0$, since $1 - \sum_i \eta_i$ is compactly supported in Ω after restriction. Each $\mathcal{F}_i$ is bounded on the divergence-free L^2 subspace.

Here is a direct verification of the agreement with classical flux. In a boundary chart with the domain above $x_3 = \varphi(x')$, let w be a smooth vector field supported away from the chart edges. Fubini, the fundamental theorem of calculus in x_3, and the one-dimensional chain rule for the Lipschitz function φ give

$$\int_{x_3 > \varphi(x')} \operatorname{div} w \, dx = \int_{\mathbb{R}^2} \big(w_1\partial_1\varphi + w_2\partial_2\varphi - w_3\big)(x', \varphi(x'))\, dx'.$$

For example, differentiating $\int_{\varphi(x')}^{\infty} w_j(x',t)\,dt$ in x_j and integrating that compactly supported derivative over x' gives the jth horizontal term; vertical integration gives the last term. The right side is exactly the graph surface integral of $w\cdot n$, since the outward graph normal times its surface element is $(\partial_1\varphi,\partial_2\varphi,-1)\,dx'$. An interior compactly supported field has divergence integral zero. Applying the finite partition of unity and summing proves the divergence formula for smooth fields on the whole domain, because the terms involving derivatives of the partition sum to zero. Apply it to $w=\eta_i u$ to obtain the claimed identification of (3.1) for smooth divergence-free u.

Lemma 3.2 (Every curl has zero component flux)**.** *If $v\in H(\operatorname{curl},\Omega)$ and $u=\operatorname{curl} v$, then $\operatorname{div} u=0$ in distributions and $\mathcal{F}_i(u)=0$ for every i.*

Proof. Commutation of distributional derivatives gives $\operatorname{div}\operatorname{curl} v=0$. Because η_i is constant near every boundary component, $\nabla\eta_i|_\Omega$ belongs to $C_c^\infty(\Omega)^3$. The distributional definition of curl therefore gives

$$\mathcal{F}_i(u)=\int_\Omega \operatorname{curl} v\cdot\nabla\eta_i=\int_\Omega v\cdot\operatorname{curl}\nabla\eta_i=0.$$

□

3.2 Solenoidal extension through the complementary pieces

Proposition 3.3 (Compactly supported solenoidal extension)**.** *Define the closed subspace*

$$\mathcal{E}(\Omega)=\{u\in L^2(\Omega)^3:\operatorname{div} u=0,\ \mathcal{F}_i(u)=0\ (1\le i\le N)\}.$$

There is a bounded linear map $S:\mathcal{E}(\Omega)\to L^2(\mathbb{R}^3)^3$ such that

$$(Su)|_\Omega=u,\qquad \operatorname{supp}(Su)\subset\overline{B},\qquad \operatorname{div}(Su)=0\quad \text{in } \mathbb{R}^3.$$

Proof. Closedness follows by testing the distributional divergence and by continuity of the finitely many fluxes. For each complementary piece D_a let I_a denote the indices of its interfaces Γ_i with Ω. Fix a smooth function θ_a equal to one near $\bigcup_{i\in I_a}\Gamma_i$ and zero near all other boundary components of Ω. If I_a is empty, take $\theta_a=0$. Use the extension operator E_{D_a} from Proposition 2.3 to set

$$L_a q=(\theta_a E_{D_a} q)|_\Omega\quad (q\in H^1(D_a)),\qquad \ell_a(q)=-\int_\Omega u\cdot\nabla L_a q.$$

This is a bounded linear functional of q, of norm at most $C_a\|u\|_{L^2(\Omega)}$. The trace of $L_a q$ agrees with the trace of q on its interfaces and is zero on every other Γ_i. The agreement across an interface follows either from the local gluing lemma applied to the ambient function $E_{D_a}q$, or from ambient smooth approximation of that function. Any two lifts with these traces differ by $H_0^1(\Omega)$, so the value of ℓ_a is independent of the chosen lift.

In particular, $L_a 1-\theta_a\in H_0^1(\Omega)$ and hence

$$\ell_a(1)=-\int_\Omega u\cdot\nabla\theta_a=-\sum_{i\in I_a}\mathcal{F}_i(u)=0.$$

Proposition 2.9 supplies $s_a\in H^1_*(D_a)$ such that

$$\int_{D_a}\nabla s_a\cdot\nabla q=-\int_\Omega u\cdot\nabla L_a q,\qquad q\in H^1(D_a),\qquad \|\nabla s_a\|_{L^2(D_a)}\le C_a\|u\|_2. \tag{3.2}$$

Define $U=Su$ to equal u on Ω, ∇s_a on D_a, and zero on $\mathbb{R}^3\setminus\overline{B}$. Lipschitz graphs have three-dimensional measure zero by Fubini, so this specifies an L^2 field up to the usual equivalence, and the displayed estimates give $\|U\|_2\le C\|u\|_2$.

For $p \in C_c^\infty(\mathbb{R}^3)$ equation (3.2) gives

$$\int_{\mathbb{R}^3} U \cdot \nabla p = \int_\Omega u \cdot \nabla \left(p|_\Omega - \sum_{a=1}^{M} L_a(p|_{D_a}) \right) = 0.$$

Indeed every boundary component of Ω is assigned to exactly one D_a, so the expression in parentheses has zero trace and belongs to $H_0^1(\Omega)$. The divergence equation for u extends from smooth compactly supported tests to that space by its defining density. This proves the global divergence equation, including at ∂B.

The minus sign in (3.2) is essential: the outward normal of a complementary piece is the negative of the outward normal of Ω at their interface. In classical notation that equation prescribes $\partial_{n_{D_a}} s_a = -u \cdot n_\Omega$ there and zero outward derivative on ∂B. No normal derivative regularity was assumed in the construction. Every operation used was linear in u, which also proves linearity of S. □

3.3 The whole-space Fourier construction

Theorem 3.4 (Bounded H^1 vector-potential right inverse)**.** *Let $\Omega \subset \mathbb{R}^3$ be bounded, connected, and Lipschitz. There is a bounded linear map*

$$R : \mathcal{E}(\Omega) \longrightarrow H^1(\Omega)^3$$

such that, for every $u \in \mathcal{E}(\Omega)$,

$$\operatorname{curl} Ru = u, \qquad \operatorname{div} Ru = 0, \qquad \|Ru\|_{H^1(\Omega)} \le C_\Omega \|u\|_{L^2(\Omega)}. \tag{3.3}$$

Consequently a divergence-free L^2 field has an H^1 vector potential if and only if its flux through every connected component of the boundary is zero. Necessity holds even for potentials in $H(\operatorname{curl}, \Omega)$.

Proof. Let $U = Su$ be the extension of Proposition 3.3. Use the unitary Fourier convention

$$\widehat{f}(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\, dx.$$

Since U has bounded support, $U \in L^1 \cap L^2$ and

$$|\widehat{U}(\xi)| \le (2\pi)^{-3/2} \|U\|_1 \le (2\pi)^{-3/2} |B|^{1/2} \|U\|_2. \tag{3.4}$$

The Fourier transform of the distributional divergence equation is $\xi \cdot \widehat{U}(\xi) = 0$ almost everywhere. This follows directly by testing against inverse transforms of Schwartz functions; the Fourier weak-derivative identity follows by integration by parts for smooth functions and distributional approximation. For $\xi \neq 0$ define

$$\widehat{\Psi}(\xi) = \frac{i\,\xi \times \widehat{U}(\xi)}{|\xi|^2}, \qquad \widehat{\Psi}(0) = 0. \tag{3.5}$$

The value at zero is immaterial. Pointwise, $|\widehat{\Psi}| \le |\widehat{U}|/|\xi|$ and $|\xi|^2 |\widehat{\Psi}|^2 \le |\widehat{U}|^2$. At high frequencies this bounds the L^2 norm by that of $\widehat{U}$. At low frequencies, (3.4) and the explicit radial integral give

$$\int_{|\xi|<1} |\widehat{\Psi}(\xi)|^2\, d\xi \le (2\pi)^{-3} |B| \|U\|_2^2 \int_{|\xi|<1} |\xi|^{-2}\, d\xi = 4\pi (2\pi)^{-3} |B| \|U\|_2^2.$$

Thus $\widehat{\Psi}$ and each $\xi_j\widehat{\Psi}$ are in L^2. Plancherel supplies $\Psi \in H^1(\mathbb{R}^3)^3$ and

$$\|\Psi\|^2_{H^1(\mathbb{R}^3)} \le \big(2 + 4\pi(2\pi)^{-3}|B|\big)\|U\|_2^2. \tag{3.6}$$

The identification with the distributional H^1 space follows by testing the Fourier inverse against smooth compactly supported functions: multiplication by $i\xi_j$ is precisely weak differentiation. Since U is real, its Fourier transform has conjugate symmetry; formula (3.5) preserves this symmetry, so Ψ is real as well.

Finally the elementary triple-product identity gives, almost everywhere away from zero,

$$i\xi \times \widehat{\Psi} = -\frac{\xi \times (\xi \times \widehat{U})}{|\xi|^2} = \widehat{U}, \qquad i\xi \cdot \widehat{\Psi} = 0.$$

Fourier inversion proves $\operatorname{curl}\Psi = U$ and $\operatorname{div}\Psi = 0$ in $\mathbb{R}^3$. Set $Ru = \Psi|_\Omega$. Linearity, the bound, and (3.3) follow from the construction. The converse is Lemma 3.2. □

Remark 3.5 (The low-frequency point). The construction gives a potential in the full space $H^1(\mathbb{R}^3)^3$, not merely in H^1_{loc}. The crucial additional input is that the extended field has bounded support, hence a bounded Fourier transform. An L^2 input alone would not justify dividing its Fourier transform by $|\xi|$ near zero. In three dimensions boundedness suffices because $|\xi|^{-1} \in L^2(\{\xi : |\xi| < 1\})$.

4 A constructive Hilbert-space range argument

This section isolates the functional-analysis step. Its proof is included so that no general unbounded closed-range theorem is needed.

Lemma 4.1 (Normalized inverse and adjoint inverse)**.** *Let H be a real Hilbert space and $A : \mathcal{D}(A) \subset H \to H$ a densely defined closed linear operator. Suppose its range E is closed and there is $M < \infty$ such that each $f \in E$ admits an $x \in \mathcal{D}(A)$ with*

$$Ax = f, \qquad \|x\| \le M\|f\|.$$

Write $Z = \ker A$. Then:

(a) There is a bounded linear map $R : E \to Z^\perp$, with $Rf \in \mathcal{D}(A)$, $ARf = f$, and $\|Rf\| \le M\|f\|$. It is the unique solution in $Z^\perp$.

(b) The adjoint, defined by

$$z \in \mathcal{D}(A^*),\ A^*z = y \quad\Longleftrightarrow\quad (z, Ax) = (y, x) \quad \text{for every } x \in \mathcal{D}(A),$$

has kernel $E^\perp$ and range $Z^\perp$.

(c) For every $y \in Z^\perp$ there is a unique $Sy \in \mathcal{D}(A^*) \cap E$ with $A^*Sy = y$. The map $S : Z^\perp \to E$ is linear and $\|Sy\| \le M\|y\|$.

Proof. Closedness of A makes Z a closed subspace of H: if $z_n \to z$ and $Az_n = 0$, graph closedness gives $z \in \mathcal{D}(A)$ and $Az = 0$. Let P_Z be its orthogonal projection. For a solution x of $Ax = f$, set $Rf = x - P_Z x$. This lies in $\mathcal{D}(A)$, since $P_Z x \in Z \subset \mathcal{D}(A)$, and has image f. Two choices of x differ by an element of Z, so the definition is independent of choice. It is linear by this uniqueness. Orthogonal projection is contractive, hence choosing the solution in the hypothesis gives $\|Rf\| \le M\|f\|$. If another solution is in $Z^\perp$, their difference lies in $Z \cap Z^\perp = \{0\}$.

Density of $\mathcal{D}(A)$ ensures that the y in the definition of A^* is unique. Directly from that definition, $\ker A^* = E^\perp$, and every A^*z is orthogonal to Z. To prove the converse range inclusion, fix $y \in Z^\perp$. On the Hilbert space E, consider the continuous linear functional

$$\ell_y(f) = (y, Rf).$$

Its norm is at most $M\|y\|$. Riesz representation supplies a unique $z \in E$ with $(z, f) = \ell_y(f)$ for every $f \in E$, and $\|z\| \le M\|y\|$. For $x \in \mathcal{D}(A)$, part (a) gives $RAx = x - P_Z x$; therefore

$$(z, Ax) = (y, RAx) = (y, x - P_Z x) = (y, x).$$

This proves $z \in \mathcal{D}(A^*)$ and $A^*z = y$. Set $Sy = z$. Riesz uniqueness gives linearity; if two such solutions lie in E, their difference belongs to $E \cap \ker A^* = E \cap E^\perp = \{0\}$. All assertions follow. □

Remark 4.2 (No completeness assumption is hidden in the range). In the application below, the range E is identified explicitly by zero distributional divergence and finitely many continuous volume fluxes. These are closed conditions in L^2. The bound M comes from the Neumann-extension/Fourier construction, not from an assumed Maxwell estimate or compactness theorem.

4.1 The two realizations of curl

Define $C_{\max}$ on H by

$$\mathcal{D}(C_{\max}) = H(\operatorname{curl}, \Omega), \qquad C_{\max} v = \operatorname{curl} v.$$

It is densely defined because its domain contains $C_c^\infty(\Omega)^3$. It is closed: if $v_n \to v$ and $\operatorname{curl} v_n \to f$ in L^2, taking limits in (1.2) shows $\operatorname{curl} v = f$.

Lemma 4.3 (Adjoint boundary condition)**.** *The adjoint $C_{\max}^*$ is precisely curl with zero tangential trace:*

$$\mathcal{D}(C_{\max}^*) = \{v \in H(\operatorname{curl}, \Omega) : v \times n = 0\}, \qquad C_{\max}^* v = \operatorname{curl} v.$$

We denote this operator by $C_{\min}$.

Proof. If $v \in \mathcal{D}(C_{\max}^*)$, test the adjoint identity with $w \in C_c^\infty(\Omega)^3$. It is exactly the distributional definition of $\operatorname{curl} v = C_{\max}^* v$, so $v \in H(\operatorname{curl})$. Testing next with $w \in H^1(\Omega)^3$ gives (1.4). Conversely, suppose (1.4) holds. Ambient smooth restrictions, and hence $H^1(\Omega)^3$, are dense in $H(\operatorname{curl}, \Omega)$ for its graph norm, by the local translation and mollification proof in Proposition 2.5. For $w_m \to w$ in that graph norm, both terms in (1.4) converge by Cauchy–Schwarz. The identity therefore holds for every $w \in \mathcal{D}(C_{\max})$, which is the definition of the adjoint with value $\operatorname{curl} v$. □

Lemma 4.4 (The maximal-curl range)**.** *Set*

$$E = \{u \in H : \operatorname{div} u = 0,\ F_i(u) = 0\ (0 \le i \le I)\}.$$

Then E is closed in H and $\operatorname{ran} C_{\max} = E$. For every $u \in E$ there exists $a \in H^1(\Omega)^3$ with

$$\operatorname{curl} a = u, \qquad \operatorname{div} a = 0, \qquad \|a\|_{H^1(\Omega)} \le M_\Omega \|u\|_2.$$

Proof. Distributional divergence zero is the intersection of the kernels of the continuous functionals $u \mapsto \int u \cdot \nabla\phi$. On that subspace, $F_i(u) = \int u \cdot \nabla\eta_i$, also a continuous L^2 functional. Thus E is closed. If $u = \operatorname{curl} v$, testing against gradients of compactly supported scalar functions proves $\operatorname{div} u = 0$. Moreover $\nabla\eta_i \in C_c^\infty(\Omega)^3$, so

$$F_i(u) = \int_\Omega \operatorname{curl} v \cdot \nabla\eta_i = \int_\Omega v \cdot \operatorname{curl} \nabla\eta_i = 0.$$

This proves necessity without a regularity or density argument. The converse and the estimate are exactly Theorem 3.4. □

5 Scalar primitives and the topology carried by the cuts

In this section Ω° is an alias for the cut domain:

$$\Omega^\circ := D = \Omega \setminus \bigcup_{j=1}^{J} \Sigma_j.$$

A function on D is also regarded as an almost-everywhere defined function on Ω, since the finitely many smooth cuts have three-dimensional measure zero. Its *broken gradient* $\nabla_b q$ is its weak gradient on D, regarded in the same way as a field on Ω. Throughout this section the normals and signs are fixed as follows: n_j points from the minus side of Σ_j to its plus side, and

$$[q]_j = q^- - q^+.$$

This convention makes the cut contribution in Green's formula positive. Only the vanishing of the fluxes occurs in the two main theorems, so changing a cut's orientation has no effect on their statements.

5.1 A weak Poincaré lemma with its global step proved

Lemma 5.1 (Weak scalar Poincaré lemma)**.** *Let $G \subset \mathbb{R}^3$ be open, connected, and simply connected. If $v \in L^2_{\mathrm{loc}}(G)^3$ and $\operatorname{curl} v = 0$ in $\mathcal{D}'(G)$, there is $q \in H^1_{\mathrm{loc}}(G)$ with $\nabla q = v$. The primitive is unique up to an additive constant. If, in addition, G is bounded, admits a finite cover by connected Lipschitz subdomains, and $v \in L^2(G)^3$, then q can be chosen in $H^1(G)$ and*

$$\int_G q = 0, \qquad \|q\|_{H^1(G)} \le C_G \|v\|_{L^2(G)}.$$

In particular this conclusion applies to the cut domain D.

Proof. First work on a ball $B = B(a, r)$ whose closure is contained in G. Mollify v in a slightly larger neighborhood of $\overline{B}$. For sufficiently small ε, the field v_ε is smooth and curl-free there, and $v_\varepsilon \to v$ in $L^2(B)^3$. The formula

$$p_\varepsilon(x) = \int_0^1 v_\varepsilon(a + t(x-a)) \cdot (x-a)\, dt$$

gives $\nabla p_\varepsilon = v_\varepsilon$ on B. Indeed, writing the kth derivative and using $\partial_k (v_\varepsilon)_\ell = \partial_\ell (v_\varepsilon)_k$ turns the integrand into $\frac{d}{dt}\{t (v_\varepsilon)_k(a + t(x-a))\}$. Subtract the mean of p_ε on B. The usual Poincaré inequality on a ball then gives

$$\|p_\varepsilon - p_\delta\|_{H^1(B)} \le C_B \|v_\varepsilon - v_\delta\|_{L^2(B)}.$$

Thus the normalized primitives converge in $H^1(B)$ to a primitive p_B. On the intersection of two balls the difference of the primitives has zero gradient. Since a nonempty intersection of balls is connected, that difference is a constant.

Here is the global gluing argument, including the precise use of simple connectivity. Fix a primitive germ at a base point. Along a path in G, choose a finite chain of balls covering successive closed subintervals of the path. At each transition adjust the additive constant of the new primitive so that it agrees with the preceding one on the overlap. Refining the subdivision or inserting another ball does not alter the continued germ: on a neighborhood of a transition point all the relevant primitives differ by constants, and those constants add transitively. For a homotopy of two paths with fixed endpoints, the compact image of the

homotopy square has a finite ball cover. Uniform continuity and the Lebesgue-number property permit a rectangular subdivision fine enough that the image of each small rectangle lies in a single ball. Continuation around such a rectangle changes no additive constant, because one primitive is available throughout that ball. Cancelling the oppositely traversed interior edges of the rectangles shows that continuation along the two boundary paths agrees; the fixed-endpoint edges contribute zero. Simple connectivity therefore makes the continued germ independent of the path. The primitives used here are equivalence classes on neighborhoods, not prescribed point values of H^1 functions. To obtain a single measurable function, choose a countable cover of G by balls with closures in G; rational balls provide such a cover. On each ball choose its primitive with the additive constant determined by continuation from the fixed initial germ. Path independence makes these primitives equal almost everywhere on every overlap. Choose measurable representatives and remove the union of their pairwise exceptional null sets. This union is null because the cover is countable. At each remaining point take the value from the first ball containing it, and define any value on the exceptional set. The resulting function q is measurable and agrees almost everywhere with the chosen H^1 primitive on every ball. It is therefore in $H^1_{\mathrm{loc}}(G)$ and has gradient v. A locally H^1 function with zero gradient is constant on balls and hence, by connectedness, on G; this proves uniqueness.

It remains to justify global integrability rather than assume it. The finite-cover Poincaré inequality of Lemma 2.8 implies, for any fixed ball $B_0 \Subset G$,

$$\|w - w_{B_0}\|_{L^2(G)} \le C\|\nabla w\|_{L^2(G)}, \qquad w \in H^1(G). \tag{5.1}$$

To see this, subtract the mean on G and estimate its difference from the mean on B_0 by Cauchy–Schwarz on B_0. For the truncation $T_m(q) = \max(-m, \min(q, m))$, use the absolutely continuous representatives on almost every coordinate line from Lemma 2.1. Composition of an absolutely continuous scalar function with this 1-Lipschitz truncation is absolutely continuous; its derivative has magnitude at most the original derivative almost everywhere. Fubini and one-dimensional integration by parts therefore give $T_m(q) \in H^1_{\mathrm{loc}}(G)$ and $|\partial_k T_m(q)| \le |\partial_k q|$ almost everywhere for every k. The truncation is bounded and G has finite measure, so it belongs to $H^1(G)$: its locally identified weak derivatives are globally in L^2, and each compactly supported test function checks the weak identity. Apply (5.1) to $T_m(q)$. Its mean on B_0 is bounded, because $q \in L^2(B_0)$, and its gradient norm is at most $\|v\|_2$. Fatou's lemma now gives $q \in L^2(G)$. Thus $q \in H^1(G)$; subtracting its global mean and applying Poincaré proves the estimate. □

5.2 Constant-jump functions and scalar cut fluxes

Define

$$\Theta = \{q \in H^1(D) : [q]_j \text{ is a constant on } \Sigma_j,\ 1 \le j \le J\}.$$

Traces in this definition may be taken first on relatively compact smooth patches of a cut. The two adjacent sides are Lipschitz locally, so their H^1 traces exist; agreement on overlapping patches makes the definition unambiguous. No trace condition is imposed at the rim of a cut. The following elementary details are useful because only *constant* jumps, rather than an arbitrary jump trace theorem, are needed here.

Lemma 5.2 (Gluing, closedness, and unit-jump lifts)**.** *The space Θ is closed in $H^1(D)$. A function $q \in H^1(D)$ belongs to $H^1(\Omega)$ if and only if every jump is zero. For each j there is $r_j \in \Theta$ with*

$$[r_j]_k = \delta_{jk} \qquad (1 \le k \le J). \tag{5.2}$$

For $q \in \Theta$ one has

$$|[q]_j| \le C_j \|q\|_{H^1(D)}. \tag{5.3}$$

Consequently $\Theta/\mathbb{R}$, or equivalently its mean-zero representatives, is a Hilbert space for the gradient inner product.

Proof. The local Sobolev gluing lemma across a smooth interior interface says that the piecewise function is H^1 across that interface exactly when the two traces agree. For completeness, integration by parts on its two sides shows that the distributional derivative is the broken derivative minus $[q]n_j$ times surface measure. Thus equal traces give an L^2 weak derivative. Conversely an H^1 function has a single trace across the interface, by local approximation by smooth functions. Localizing inside Ω proves the assertion globally: the rims of the cuts lie on $\partial\Omega$ and therefore never meet a compactly supported test function in Ω.

Choose a fixed smooth disk compactly contained in each connected cut. The ordinary trace inequality on the two sides of that disk, and its positive area, give (5.3) for a constant jump. If $q_m \to q$ in $H^1(D)$ and each q_m has constant jumps, the constants are Cauchy by this estimate. Local trace continuity on every other patch identifies the limit jump with the same constant. This proves closedness.

We describe the lifts, including what happens where a cut meets the boundary. In the oriented two-sided cut geometry of Hypothesis 3.3 and Notation 3.9, every point of $\overline{\Sigma_j}$ has a neighborhood whose intersection with D consists of its two local Lipschitz sides. The local labels minus and plus are coherent on overlapping neighborhoods and agree with the fixed normal on the interior of the cut. Here we use precisely the operational cut convention of the source: its chosen normals and two-sided traces in Notation 3.9, together with its assertion after Hypothesis 3.3 that the local side count is two on the cut closures. We are not inferring orientability for an arbitrary embedded smooth surface. The neighborhoods can avoid every other cut closure because the finitely many cut closures are disjoint and compact. Choose finitely many such neighborhoods U_α covering $\overline{\Sigma_j}$ and smooth functions χ_α supported in U_α whose sum is one on a neighborhood of $\overline{\Sigma_j}$. On $D \cap U_\alpha$ let b_α be one on the minus side and zero on the plus side. Set

$$r_j = \sum_\alpha \chi_\alpha b_\alpha \quad \text{on } D,$$

where each summand is zero outside U_α. For every α, the indicator b_α is locally constant on $D \cap U_\alpha$, since its two side pieces are disjoint open components. Thus $\chi_\alpha b_\alpha$ is locally smooth there and its weak gradient is $b_\alpha \nabla \chi_\alpha$. The compact support of χ_α in U_α leaves a neighborhood of its artificial boundary where the product is zero. Extending it by zero outside U_α hence produces a locally smooth function on all of D, with no artificial interface term. Its value and gradient are bounded and D has finite volume, so both belong to $L^2(D)$. This proves the asserted $H^1(D)$ membership directly, including at chart overlaps near the cut rim.

Its jump on Σ_j is χ_α, with the same sign on overlaps. The sum has jump one on Σ_j and zero on all the other cuts. This proves (5.2). In particular the construction does not require arbitrary $H^{1/2}$ jump data to be lifted. Finally, closedness and the Poincaré inequality on D identify the gradient norm with the H^1 norm on the mean-zero subspace of Θ, which is complete. □

For $v \in H(\mathrm{div}; \Omega)$ with zero normal trace, define its scalar flux through the jth cut by

$$C_j(v) = \int_D v \cdot \nabla_b r_j + \int_D (\mathrm{div}\, v) r_j. \tag{5.4}$$

Here and below a zero normal trace means that the usual Green functional vanishes on every $H^1(\Omega)$ test function. Formula (5.4) gives a continuous linear functional on this closed subspace of $H(\mathrm{div}; \Omega)$.

Lemma 5.3 (Cut Green identity)**.** *The definition of C_j is independent of the choice of unit-jump lifts. For every $q \in \Theta$ and every $v \in H(\mathrm{div}; \Omega)$ with zero normal trace,*

$$\int_D v \cdot \nabla_b q + \int_D (\mathrm{div}\, v) q = \sum_{j=1}^{J} C_j(v)[q]_j. \tag{5.5}$$

For a smooth compactly supported field, $C_j(v)$ is the classical integral $\int_{\Sigma_j} v \cdot n_j$. Hence it is precisely the weak scalar cut flux used in the main theorems.

Proof. The difference of two lifts with identical jumps belongs to $H^1(\Omega)$ by Lemma 5.2. The zero-normal Green identity makes the corresponding difference of the right side of (5.4) zero. More generally,

$$q_0 = q - \sum_j [q]_j r_j \in H^1(\Omega).$$

Apply the zero-normal Green identity to q_0 and rearrange to obtain (5.5). For smooth compactly supported v, integration by parts locally on the two sides of a cut gives the interface term $(r_j^- - r_j^+)v \cdot n_j$. A partition of unity and the fact that v vanishes near the cut rims give exactly $\int_{\Sigma_j} v \cdot n_j$. Lemma 2.6 gives density of compactly supported smooth fields in the zero-normal $H(\mathrm{div})$ space, so this continuous extension is unique. □

5.3 The two curl kernels

We use the maximal distributional curl $C_{\max}$ and its minimal realization $C_{\min}$ defined in the operator section. Thus $\mathcal{D}(C_{\max}) = H(\mathrm{curl};\Omega)$ and $\mathcal{D}(C_{\min})$ consists of the fields in this space with zero tangential trace. Put $Z = \ker C_{\max}$ and $Z_0 = \ker C_{\min}$.

Lemma 5.4 (The maximal curl kernel)**.** *For $q \in H^1(D)$, the field $\nabla_b q$ has zero distributional curl on Ω if and only if $q \in \Theta$. Consequently*

$$Z = \{\nabla_b q : q \in \Theta\}. \tag{5.6}$$

The primitive on D is unique up to a constant, and its mean-zero representative satisfies $\|q\|_{H^1(D)} \le C\|\nabla_b q\|_{L^2(\Omega)}$.

Proof. We give a local argument, so no regularity assertion about surface derivatives of jump traces is needed. Suppose first that $\mathrm{curl}(\nabla_b q) = 0$ in Ω. Around an interior point of a cut choose a sufficiently small ball $B \Subset \Omega$ that meets no other cut and in which the cut separates B into its two connected sides. The local part of Lemma 5.1 gives $p \in H^1(B)$ with $\nabla p = \nabla_b q$ there. On each side of the cut, $q - p$ has zero gradient, hence is a constant. Since p has matching traces, $[q]$ is the difference of these two constants. The jump is therefore locally constant on the cut. Its constants agree on overlapping patches, so connectedness of each Σ_j makes the jump constant on all of Σ_j.

Conversely, suppose the jump on this patch is the constant c. Replace q by $q - c$ on its minus side and leave it unchanged on its plus side. The new piecewise function has equal traces, so the local H^1 gluing lemma makes it an $H^1(B)$ function p. Its gradient is $\nabla_b q$ on both sides, hence $\mathrm{curl}(\nabla_b q) = \mathrm{curl}\,\nabla p = 0$ on B. Away from the cuts the same identity holds directly. These neighborhoods cover Ω: the rims lie on $\partial\Omega$ and do not need to be covered. Distributional identities are local, so the curl vanishes on all of Ω. This proves the first assertion.

If $v \in Z$, its restriction to the simply connected cut domain is curl-free. Lemma 5.1 gives $v = \nabla_b q$ with $q \in H^1(D)$; the assertion just proved forces $q \in \Theta$. The converse and the estimate follow from that assertion and the Poincaré inequality on D. □

Let the boundary components be $\Gamma_0, \dots, \Gamma_I$, where Γ_0 borders the unbounded complementary component, and define

$$\Theta_0 = \{q \in H^1(\Omega) : q|_{\Gamma_0} = 0, \quad q|_{\Gamma_i} = c_i(q) \text{ constant}, \ 1 \le i \le I\}.$$

All these equalities are trace equalities. Because finitely many compact boundary components are disjoint, there are smooth functions η_i equal to one near Γ_i and zero near every other component. For $i \geq 1$ their restrictions belong to Θ_0 and $c_k(\eta_i) = \delta_{ik}$. Write

$$F_i(v) = \langle v \cdot n, 1 \rangle_{\Gamma_i} = \int_\Omega v \cdot \nabla \eta_i + (\operatorname{div} v)\eta_i$$

for the boundary-component flux; the same definition applies to $i = 0$. Its independence of the cutoff follows from the zero-trace Green identity. The divergence theorem gives $\sum_{i=0}^{I} F_i(v) = \int_\Omega \operatorname{div} v$.

Lemma 5.5 (The minimal curl kernel)**.** *One has*

$$Z_0 = \{\nabla q : q \in \Theta_0\}. \tag{5.7}$$

The primitive in Θ_0 is unique. Moreover

$$\|q\|_{H^1(\Omega)} + \sum_{i=1}^{I} |c_i(q)| \leq C \|\nabla q\|_{L^2(\Omega)}, \qquad q \in \Theta_0. \tag{5.8}$$

Proof. Let $v \in Z_0$, and extend it by zero to all of $\mathbb{R}^3$, obtaining $\widetilde{v}$. The defining zero-tangential Green identity implies

$$\int_{\mathbb{R}^3} \widetilde{v} \cdot \operatorname{curl} \varphi = \int_\Omega v \cdot \operatorname{curl}(\varphi|_\Omega) = 0, \qquad \varphi \in C_c^\infty(\mathbb{R}^3)^3.$$

Thus $\widetilde{v}$ is curl-free on the simply connected space $\mathbb{R}^3$. The first, global H^1_{loc} assertion of Lemma 5.1 provides $Q \in H^1_{\mathrm{loc}}(\mathbb{R}^3)$ with $\nabla Q = \widetilde{v}$. Since $\overline{\Omega}$ lies in a ball, $Q|_\Omega \in H^1(\Omega)$. On each connected component of $\mathbb{R}^3 \setminus \overline{\Omega}$, Q is constant because its gradient vanishes. Each connected boundary component borders a single complementary component: the local exterior side in a Lipschitz chart is connected, so the identity of that component is locally constant along the boundary, and hence constant on a connected Γ_i. No bijection of the two lists of components is needed. Subtract its constant on the unbounded component. Across a Lipschitz boundary a single ambient H^1 function has matching traces from both sides. It follows that $q = Q|_\Omega$ has zero trace on Γ_0 and a constant trace on each Γ_i; hence $q \in \Theta_0$ and $v = \nabla q$.

Conversely, let $q \in \Theta_0$ and write

$$q_0 = q - \sum_{i=1}^{I} c_i(q)\eta_i.$$

This function has zero trace on the entire boundary, hence belongs to $H^1_0(\Omega)$. Choose $\phi_m \in C_c^\infty(\Omega)$ converging to q_0 in H^1. Their gradients converge to ∇q_0 in $H(\operatorname{curl}; \Omega)$, since every such gradient has zero curl. Every $\nabla \phi_m$ has zero tangential trace, and continuity of the Green trace functional therefore gives the same condition for ∇q_0. Also $\nabla \eta_i$ is smooth with compact support inside Ω: the cutoff is constant near every boundary component. Thus each $\nabla \eta_i$ has zero tangential trace and zero curl. Their linear combination ∇q belongs to Z_0. This argument requires no identification between boundary components and connected components of the complement.

For the estimate, write $q = q_\Omega + (q - q_\Omega)$ and use Poincaré. Trace continuity on Γ_0 and its positive area, together with $q|_{\Gamma_0} = 0$, bound the constant q_Ω by $C\|\nabla q\|_2$. This gives the H^1 bound. Trace continuity on each other component then bounds $|c_i(q)|$. The same estimate proves uniqueness and completeness of Θ_0 for the gradient norm. □

5.4 Orthogonality is exactly the stated compatibility

Proposition 5.6 (Concrete annihilators)**.** *For $u \in L^2(\Omega)^3$ the following identities hold:*

$$Z_0^\perp = \{u \in H(\mathrm{div};\Omega) : \mathrm{div}\, u = 0,\ F_i(u) = 0\ (0 \le i \le I)\}, \tag{5.9}$$
$$Z^\perp = \{u \in H(\mathrm{div};\Omega) : \mathrm{div}\, u = 0,\ u \cdot n = 0,\ C_j(u) = 0\ (1 \le j \le J)\}. \tag{5.10}$$

The statements include $I = 0$ and $J = 0$, with empty lists of additional conditions interpreted as vacuous.

Proof. Suppose $u \perp Z_0$. Since $C_c^\infty(\Omega) \subset \Theta_0$, Lemma 5.5 gives $\int_\Omega u \cdot \nabla\phi = 0$ for every scalar test function. Thus $\mathrm{div}\, u = 0$ distributionally, which is an L^2 function, so $u \in H(\mathrm{div};\Omega)$. Testing with η_i, $1 \le i \le I$, gives $F_i(u) = 0$. The divergence theorem then gives $F_0(u) = 0$. Conversely, if these divergence and flux conditions hold, the trace Green formula for $q \in \Theta_0$ gives

$$(u, \nabla q)_{L^2} = \sum_{i=1}^{I} c_i(q) F_i(u) = 0.$$

Lemma 5.5 proves (5.9).

Next suppose $u \perp Z$. The space Z contains every gradient of an $H^1(\Omega)$ function. The same compactly supported test first gives $\mathrm{div}\, u = 0$; the identity against all $H^1(\Omega)$ gradients then says that the entire normal trace is zero. Finally $\nabla_b r_j \in Z$, and (5.4) gives $C_j(u) = (u, \nabla_b r_j) = 0$. Conversely, the conditions on the right of (5.10), together with the cut Green identity, give

$$(u, \nabla_b q) = \sum_j C_j(u)[q]_j = 0 \qquad (q \in \Theta).$$

Lemma 5.4 completes the proof. □

5.5 Harmonic fields and flux-normalized bases

Define the two finite-dimensional obstructions by

$$K_T = \{h \in H(\mathrm{curl};\Omega) \cap H(\mathrm{div};\Omega) : \mathrm{curl}\, h = 0,\ \mathrm{div}\, h = 0,\ h \cdot n = 0\},$$
$$K_N = \{h \in H(\mathrm{curl};\Omega) \cap H(\mathrm{div};\Omega) : \mathrm{curl}\, h = 0,\ \mathrm{div}\, h = 0,\ h \times n = 0\}.$$

The subscripts follow the paper: a tangential field has zero normal component, and a normal field has zero tangential component.

Proposition 5.7 (Kernel dimensions and normalized bases)**.** *There are unique mean-zero functions $q_j^T \in \Theta$, $1 \le j \le J$, satisfying*

$$\int_D \nabla_b q_j^T \cdot \nabla_b r = [r]_j \qquad (r \in \Theta). \tag{5.11}$$

The fields $h_j^T = \nabla_b q_j^T$ form a basis of K_T and satisfy $C_k(h_j^T) = \delta_{jk}$. Thus $\dim K_T = J$. There are unique $q_i^N \in \Theta_0$, $1 \le i \le I$, satisfying

$$\int_\Omega \nabla q_i^N \cdot \nabla r = c_i(r) \qquad (r \in \Theta_0). \tag{5.12}$$

The fields $h_i^N = \nabla q_i^N$ form a basis of K_N and satisfy $F_k(h_i^N) = \delta_{ik}$ for $1 \le k \le I$, and $F_0(h_i^N) = -1$. Thus $\dim K_N = I$.

Proof. The jump functional is continuous on the Hilbert space $\Theta/\mathbb{R}$, by (5.3) and Poincaré. The Riesz representation theorem therefore gives exactly one mean-zero solution of (5.11). For $r \in H^1(\Omega)$ the right side is zero, so this variational identity says that h_j^T has zero divergence and zero normal trace. Its curl is zero by Lemma 5.4. Taking $r = r_k$ gives $C_k(h_j^T) = \delta_{jk}$. These flux identities give linear independence. For any $h \in K_T$, write $h = \nabla_b q$ with $q \in \Theta$ and set $h' = h - \sum_j C_j(h) h_j^T$. Then $h' \in K_T$, every cut flux of h' is zero, and $h' = \nabla_b q'$ for some $q' \in \Theta$. The cut Green identity gives

$$\|h'\|_2^2 = \sum_j C_j(h')[q']_j = 0.$$

This proves spanning without assuming finite dimensionality beforehand.

For the other kernel, (5.8) makes c_i a continuous functional on the Hilbert space Θ_0 with its gradient inner product. Riesz gives (5.12). Testing with $H_0^1(\Omega)$ functions gives zero divergence, while Lemma 5.5 gives zero curl and zero tangential trace. Testing with η_k gives $F_k(h_i^N) = \delta_{ik}$, and total flux zero gives $F_0(h_i^N) = -1$. These identities imply independence. For $h \in K_N$, subtract $\sum_i F_i(h) h_i^N$ and represent the remainder as $\nabla q'$ with $q' \in \Theta_0$. The ordinary Green formula gives its squared norm as $\sum_i c_i(q') F_i(h') = 0$, proving spanning. □

6 Proofs of the two potential theorems

We now assemble the preceding results without an additional PDE theorem. Write

$$Z = \ker C_{\max}, \qquad Z_0 = \ker C_{\min}.$$

The kernel identifications and Green identities above give

$$Z = \{\nabla_b q : q \in \Theta\}, \tag{6.1}$$

$$Z_0 = \{\nabla q : q \in \Theta_0\}, \tag{6.2}$$

$$Z^\perp = \{v \in H : \operatorname{div} v = 0,\ v \cdot n = 0,\ C_j(v) = 0 \text{ for all } j\}, \tag{6.3}$$

$$Z_0^\perp = \{v \in H : \operatorname{div} v = 0,\ F_i(v) = 0 \text{ for all } i\} = E. \tag{6.4}$$

Here Θ_0 consists of the $H^1(\Omega)$ functions with zero trace on Γ_0 and constant trace on each inner component. The equalities involving orthogonal complements are equalities of subspaces of actual L^2 fields, not abstract assumptions on the geometry.

For the two main existence results, only the maximal-kernel identity (6.1) and its annihilator (6.3) are needed from the cut analysis. Indeed, (6.4) also follows directly from $\operatorname{ran} C_{\max} = E$ and $\ker C_{\min} = E^\perp$ in Lemma 4.1. The explicit scalar description of Z_0 is useful for the harmonic kernel and norm-equivalence results, but is not a prerequisite of the normal-potential construction.

6.1 Tangential potential

Proof of Main Theorem A. Suppose first that the data satisfy (i), so $u \in E$. Choose the regular potential a supplied by Lemma 4.4, and set

$$\psi = R_T u := a - P_Z a. \tag{6.5}$$

The definition is independent of the particular potential a, since any two potentials differ by an element of Z. Lemma 4.1(a), applied to $C_{\max}$, proves that $\psi \in H(\operatorname{curl}, \Omega)$, $\operatorname{curl} \psi = u$, $\psi \in Z^\perp$, and $\|\psi\|_2 \le M_\Omega \|u\|_2$. Equation (6.3) supplies simultaneously zero weak divergence, zero normal trace, and all the zero cut fluxes. In particular $\psi \in X$. Its graph norm satisfies

$$\|\psi\|_X^2 = \|\psi\|_2^2 + \|u\|_2^2 \le (M_\Omega^2 + 1)\|u\|_2^2.$$

If ψ_1, ψ_2 satisfy all the conditions in (ii), their difference has zero curl and hence belongs to Z. By (6.3), each potential belongs to $Z^\perp$; their difference is consequently zero. This proves uniqueness in L^2, and therefore also in the graph space X.

Conversely, the existence of a potential in (ii) implies $u \in \operatorname{ran} C_{\max} = E$ by Lemma 4.4, proving (i). Finally R_T is linear by Lemma 4.1, or directly by uniqueness after taking linear combinations of the defining identities. This proves the asserted stability and linearity. □

6.2 Normal potential

Proof of Main Theorem B. The conditions in (i) are exactly $u \in Z^\perp$, by (6.3). Lemma 4.1(c) gives a unique $\psi = R_N u \in \mathcal{D}(C_{\min}) \cap E$ satisfying $C_{\min}\psi = u$, with $\|\psi\|_2 \le M_\Omega \|u\|_2$. For clarity, the construction is characterized by the volume identity

$$(R_N u, f) = (u, R_T f) \qquad (f \in E). \tag{6.6}$$

It is a Riesz representation on the closed Hilbert subspace E, whose boundedness follows from the estimate for R_T.

Lemma 4.3 says that ψ has zero tangential trace and $\operatorname{curl}\psi = u$. Membership in E says exactly that $\operatorname{div}\psi = 0$ and $F_i(\psi) = 0$ for all i. These are all the conditions in (ii), and $\|\psi\|_X \le \sqrt{M_\Omega^2 + 1}\,\|u\|_2$.

If two potentials satisfy (ii), their difference lies in $\ker C_{\min} = Z_0$ and, by (6.4), in $Z_0^\perp$. It is zero. Conversely, a potential in (ii) lies in $\mathcal{D}(C_{\min})$; Lemma 4.1(b) puts its curl in $Z^\perp$, which gives (i). Linearity follows from the Riesz construction, or again from uniqueness. □

Corollary 6.1 (Independence of the admissible cuts)**.** *If two admissible cutting systems are chosen for the same domain, the tangential potential in Main Theorem A is the same L^2 field for both systems. The data space in Main Theorem B is also independent of that choice.*

Proof. The maximal-curl kernel Z is defined by a distributional equation on Ω, independently of cuts. In both systems the tangential potential is the unique preimage of u in $Z^\perp$, and the normal data space is $Z^\perp$. Apply the two identifications for the two systems. No comparison of individual flux coordinates is necessary. □

Remark 6.2 (Minimum-norm characterization)*.* The tangential potential uniquely minimizes the L^2 norm among all $H(\operatorname{curl})$ potentials of its data. The normal potential uniquely minimizes that norm among all zero-tangential-trace potentials of its data. Indeed, every other potential is the normalized one plus an element of the corresponding kernel, and the two summands are orthogonal. Since their curls are fixed and the normalized divergence is zero, this also directly explains the stability bound.

7 Graph-norm inequalities and stability

The next argument uses the two main theorems only through their bounded linear solution operators. Denote them by R_T and R_N: $R_T u$ is the divergence-free potential with zero normal trace and zero cut fluxes, and $R_N u$ is the divergence-free potential with zero tangential trace and zero boundary-component fluxes. Suppose, as proved in the operator argument, that

$$\|R_T u\|_2 \le M_T \|u\|_2, \qquad \|R_N u\|_2 \le M_N \|u\|_2.$$

Corollary 7.1 (The estimates of Corollaries 3.16 and 3.19)**.** *There are constants depending only on Ω and its chosen cuts such that*

$$\|w\|_{L^2(\Omega)} \le C \left(\|\operatorname{curl} w\|_2 + \|\operatorname{div} w\|_2 + \sum_{j=1}^{J} |C_j(w)| \right), \qquad w \in X_T, \tag{7.1}$$

$$\|w\|_{L^2(\Omega)} \le C \left(\|\operatorname{curl} w\|_2 + \|\operatorname{div} w\|_2 + \sum_{i=1}^{I} |F_i(w)| \right), \qquad w \in X_N. \tag{7.2}$$

Here $X_T = X \cap \{w \cdot n = 0\}$ and $X_N = X \cap \{w \times n = 0\}$, with $X = H(\operatorname{curl};\Omega) \cap H(\operatorname{div};\Omega)$. Thus each right-hand expression is a norm equivalent to the graph norm on its indicated space. Including F_0 in the second sum gives an equivalent norm as well.

Proof. For $w \in X_T$, put $a = R_T(\operatorname{curl} w)$. This is defined because the curl of an $H(\operatorname{curl})$ field satisfies the first theorem's compatibility conditions, as established in the operator argument. The remainder $z = w - a$ has zero curl, so $z = \nabla_b q$ for a mean-zero $q \in \Theta$. Both z and w have zero normal trace, and

$$\operatorname{div} z = \operatorname{div} w, \qquad C_j(z) = C_j(w).$$

The cut Green identity therefore gives

$$\|z\|_2^2 = -\int_D (\operatorname{div} w) q + \sum_j C_j(w)[q]_j.$$

Poincaré and (5.3) bound the absolute value of the right side by $C(\|\operatorname{div} w\|_2 + \sum_j |C_j(w)|)\|z\|_2$. Divide by $\|z\|_2$ if it is nonzero; if it is zero the same conclusion is automatic. Adding the estimate for a proves (7.1).

For $w \in X_N$, put $a = R_N(\operatorname{curl} w)$. Its compatibility follows either from the second main theorem's necessity, already proved, or directly from $(C_{\min} w, z) = (w, C_{\max} z) = 0$ for $z \in Z$ and Proposition 5.6. Now $w - a \in Z_0$, so $w - a = \nabla q$ for $q \in \Theta_0$. The ordinary Green identity gives

$$\|w - a\|_2^2 = -\int_\Omega (\operatorname{div} w) q + \sum_{i=1}^{I} c_i(q) F_i(w).$$

Use (5.8), divide as before, and add the bound for a to obtain (7.2). The reverse norm bounds follow from the continuity of every boundary flux on $H(\operatorname{div})$, and of every cut flux on its zero-normal subspace. Finally $F_0(w) = \int_\Omega \operatorname{div} w - \sum_{i=1}^{I} F_i(w)$, so its inclusion is immaterial to norm equivalence. □

8 Compactness and regularity of the Maxwell graph spaces

Write $X = H(\operatorname{curl},\Omega) \cap H(\operatorname{div},\Omega)$, with the sum of the three squared L^2 norms, and retain the paper's notation: X_T imposes zero *normal* trace and X_N imposes zero *tangential* trace. The subscripts describe the direction of the field, not the trace being set equal to zero. Put $X_0 = X_T \cap X_N$. The boundary conditions below always have the weak meanings already defined. In particular zero tangential trace means the zero-extension identity of Proposition 2.6, equivalently

$$(\operatorname{curl} v, z)_\Omega = (v, \operatorname{curl} z)_\Omega \quad (z \in H(\operatorname{curl},\Omega)).$$

8.1 Both traces, no trace, and one trace

Theorem 8.1 (Both traces; original Theorem 2.5)**.** *On every bounded Lipschitz domain, $X_0 = H^1_0(\Omega)^3$, and for v in this space*

$$\|\nabla v\|_2^2 = \|\operatorname{curl} v\|_2^2 + \|\operatorname{div} v\|_2^2. \tag{8.1}$$

Every $v \in X$ belongs to $H^1_{\mathrm{loc}}(\Omega)^3$.

Proof. If $v \in X_0$, its extension V by zero has both curl and divergence equal to the corresponding zero-extended derivatives. For a complex vector a and a real vector ξ, the elementary identity

$$|\xi|^2|a|^2 = |\xi \times a|^2 + |\xi \cdot a|^2$$

follows by rotating ξ to the first coordinate axis, or by expanding both sides. Apply it to $a = \widehat{V}(\xi)$ and integrate. Plancherel implies $V \in H^1(\mathbb{R}^3)^3$ and gives (8.1). Since V is zero on the exterior side of each boundary chart, the trace-gluing result makes its interior trace zero. Proposition 2.3 gives $v \in H^1_0(\Omega)^3$. Conversely approximating a member of $H^1_0(\Omega)^3$ by compactly supported smooth fields gives both zero-extension identities and the asserted inclusion. For the local assertion take a smooth cutoff χ supported in Ω. The zero extension of χv has L^2 curl and divergence by the product formulas, and the same Fourier identity puts it in $H^1(\mathbb{R}^3)^3$. □

Theorem 8.2 (No boundary condition; original Proposition 2.7)**.** *The inclusion $X \hookrightarrow L^2(\Omega)^3$ is not compact for any nonempty bounded open set Ω.*

Proof. For $k \geq 1$ take the real harmonic polynomial $p_k(x) = \operatorname{Re}(x_1 + ix_2)^k$. The fields ∇p_k are linearly independent on Ω: a linear relation would make the gradient of their linear combination vanish on an interior ball; a polynomial vanishing on a ball has every coefficient zero, and the different homogeneous degrees then force the relation to be trivial. The polynomial assertion follows by repeatedly viewing it as a one-variable polynomial on coordinate line intervals. Each gradient has zero curl and divergence. Gram–Schmidt in L^2 therefore produces an infinite orthonormal sequence of finite linear combinations of these gradients. Its X norms are one and the L^2 distance between distinct members is $\sqrt{2}$. □

Lemma 8.3 (Two regular decompositions)**.** *There is a finite cover of $\overline{\Omega}$ by boundary graph cylinders and interior balls, a subordinate smooth partition χ_j, and connected simply connected Lipschitz pieces $U_j \subset \Omega$ containing the support of $\chi_j|_\Omega$ away from the artificial edges, with the following properties. For $v \in H(\operatorname{curl}, \Omega)$ there are $w \in H^1(\Omega)^3$ and $q_j \in H^1(U_j)$ such that*

$$v = w + \nabla q_j \text{ on } U_j, \qquad \|w\|_{H^1(\Omega)} + \sum_j \|q_j\|_{H^1(U_j)} \leq C\|v\|_{H(\operatorname{curl},\Omega)}. \tag{8.2}$$

The choices can be linear. If v has zero tangential trace, one can instead choose a fixed ball B with $\overline{\Omega} \subset B$ and

$$w \in H^1(\mathbb{R}^3)^3, \quad q \in H^1(B), \quad Z_\Omega v = w + \nabla q \text{ on } B, \qquad \|w\|_{H^1(\mathbb{R}^3)} + \|q\|_{H^1(B)} \leq C\|v\|_{H(\operatorname{curl},\Omega)}. \tag{8.3}$$

In both constructions $\operatorname{div} w = 0$.

Proof. Take $w = R(\operatorname{curl} v)$ from Theorem 3.4; Lemma 3.2 verifies its hypotheses. The difference $v - w$ is curl-free. Each graph-side cylinder is homeomorphic to a half box by the explicit flattening map and hence simply connected. Apply Lemma 5.1 on each such piece and on each interior ball, normalizing q_j to have mean zero. Poincare's inequality bounds q_j by its

gradient. This proves (8.2). The finite partition and the artificial-edge margins are exactly those in Proposition 2.3.

In the second case $V = Z_\Omega v$ belongs to $H(\operatorname{curl}, \mathbb{R}^3)$ and $f = \operatorname{curl} V$ is supported in $\overline{\Omega}$. It is divergence-free. Apply directly the Fourier formula (3.5) to f, without a complementary-domain extension, to obtain $w \in H^1(\mathbb{R}^3)^3$ with $\operatorname{curl} w = f$, $\operatorname{div} w = 0$, and $\|w\|_{H^1} \le C\|f\|_2$. The curl-free field $V - w$ has an $H^1(B)$ scalar potential by Lemma 5.1; choose its mean on B to be zero. Poincare gives the last estimate. Uniqueness of the normalized scalar potentials proves linearity in both cases. □

Theorem 8.4 (One homogeneous trace; original Theorem 2.8)**.** *For every bounded Lipschitz domain the inclusions $X_T \hookrightarrow L^2(\Omega)^3$ and $X_N \hookrightarrow L^2(\Omega)^3$ are compact. No cutting hypothesis is required for this assertion.*

Proof. Let (v_k) be bounded in X_T and use (8.2). Rellich compactness and finitely many subsequences make w_k Cauchy in $L^2(\Omega)$ and every $q_{j,k}$ Cauchy in $L^2(U_j)$. Write $a = v_k - v_l$, $b = w_k - w_l$, and $r_j = q_{j,k} - q_{j,l}$. Since $\sum_j \chi_j = 1$,

$$\|a\|_2^2 = (a, b)_\Omega + \sum_j \int_{U_j} \chi_j a \cdot \nabla r_j.$$

The function $\chi_j r_j$ extends by zero past the artificial boundary of U_j to an $H^1(\Omega)$ function; its support is away from that boundary. Zero normal trace and the product rule give

$$\int_{U_j} \chi_j a \cdot \nabla r_j = -\int_{U_j} \chi_j (\operatorname{div} a) r_j - \int_{U_j} r_j a \cdot \nabla \chi_j.$$

Every right-hand term tends to zero as $k, l \to \infty$, since a and its divergence are uniformly bounded and the other indicated factors converge strongly. Thus (v_k) is L^2 Cauchy.

Now let (v_k) be bounded in X_N and use (8.3). After taking a subsequence, w_k is Cauchy in $L^2(B)$ and q_k is Cauchy in $L^2(B)$. On every complementary component D_a of $B \setminus \overline{\Omega}$ we have $\nabla q_k = -w_k$. Consequently $q_k|_{D_a}$ is Cauchy in $H^1(D_a)$. Use the fixed interface lifts L_a from the proof of Proposition 3.3 and put

$$Q_k = \sum_a L_a(q_k|_{D_a}), \qquad r_k = q_k|_\Omega - Q_k.$$

Then Q_k is Cauchy in $H^1(\Omega)$, $r_k \in H_0^1(\Omega)$, and r_k is Cauchy in $L^2(\Omega)$. For differences of two indices the decomposition gives

$$\|v_k - v_l\|_2^2 = (v_k - v_l, w_k - w_l)_\Omega + (v_k - v_l, \nabla(Q_k - Q_l))_\Omega - (\operatorname{div}(v_k - v_l), r_k - r_l)_\Omega.$$

All three terms tend to zero. This proves the second compactness assertion using only scalar extension and Rellich compactness. □

8.2 The scalar boundary estimate needed for smooth domains

Lemma 8.5 (Tangential difference quotients)**.** *Let $Q_+ = (-2, 2)^2 \times (0, 2)$, let A be a symmetric matrix of Lipschitz functions on its closure, and assume $\lambda|\zeta|^2 \le \zeta^T A(x)\zeta \le \Lambda|\zeta|^2$. Suppose $q \in H^1(Q_+)$, $G \in H^1(Q_+)^3$, and $F \in L^2(Q_+)$ satisfy*

$$\int_{Q_+} A\nabla q \cdot \nabla \phi = \int_{Q_+} G \cdot \nabla \phi + \int_{Q_+} F\phi. \tag{8.4}$$

There are two permitted cases. In the Neumann case all H^1 tests supported away from the lateral and upper faces are permitted. In the Dirichlet case q has zero trace on the bottom face and the tests also have zero trace there. Then $q \in H^2((-1,1)^2 \times (0,1))$ and

$$\|q\|_{H^2((-1,1)^2\times(0,1))} \le C\big(\|q\|_{H^1(Q_+)} + \|G\|_{H^1(Q_+)} + \|F\|_2\big), \tag{8.5}$$

where C depends only on the box margins, ellipticity constants, and the Lipschitz bound of A.

Proof. Fix a smooth cutoff η equal to one on the smaller half box and supported away from the lateral and upper faces of Q_+. For $\alpha = 1, 2$ and small $h \neq 0$, write $D_h f(x) = (f(x+he_\alpha) - f(x))/h$. Translation and the one-dimensional fundamental theorem give

$$\|D_h f\|_{L^2(V)} \le \|\partial_\alpha f\|_{L^2(V_h)}, \qquad \int f D_{-h} g = -\int (D_h f) g,$$

whenever the translated supports stay in the larger set V_h. Use $\phi = -D_{-h}(\eta^2 D_h q)$ in (8.4). This is an admissible H^1 test for each fixed h. In the Dirichlet case its trace is zero because tangential translations and multiplication preserve zero trace. Moving the difference quotient by the second displayed identity turns the left-hand side into

$$\int \big(A(x+he_\alpha)\nabla D_h q + (D_h A)\nabla q\big) \cdot \big(\eta^2 \nabla D_h q + 2\eta \nabla \eta D_h q\big).$$

The principal term is at least $\lambda \|\eta \nabla D_h q\|_2^2$. All remaining terms on this side are bounded by

$$C\big(\|q\|_{H^1}\|\eta \nabla D_h q\|_2 + \|q\|_{H^1}^2\big),$$

because $\|D_h A\|_\infty \le \mathrm{Lip}(A)$ and $\|D_h q\|_2 \le \|\partial_\alpha q\|_2$ on the slightly larger support. The G term is estimated by the same expression with one $\|q\|_{H^1}$ factor replaced by $\|G\|_{H^1}$. For the F term, do not differentiate F: instead use

$$\|D_{-h}(\eta^2 D_h q)\|_2 \le \|\partial_\alpha(\eta^2 D_h q)\|_2 \le C\big(\|\eta \nabla D_h q\|_2 + \|q\|_{H^1}\big).$$

The inequality $ab \le \varepsilon a^2 + (4\varepsilon)^{-1} b^2$ absorbs every occurrence of the unknown gradient into half of the principal term. We obtain uniformly for small h

$$\|\eta \nabla D_h q\|_2 \le C(\|q\|_{H^1} + \|G\|_{H^1} + \|F\|_2).$$

A bounded sequence in a Hilbert space has a weakly convergent subsequence; this elementary fact follows by extracting convergent coordinates on a countable orthonormal basis of the closed span of the sequence and using its uniform norm bound. Testing the difference quotients against smooth functions identifies any weak limit with $\partial_\alpha \nabla q$. Hence all second derivatives having at least one tangential index belong to L^2 on the smaller half box, with the asserted bound.

For interior tests equation (8.4) says $-\operatorname{div}(A\nabla q) = -\operatorname{div} G + F$. Expand it distributionally and solve for the remaining derivative:

$$A_{33}\partial_{33} q = \operatorname{div} G - F - \sum_{(i,j)\neq(3,3)} A_{ij}\partial_{ij} q - \sum_{i,j} (\partial_i A_{ij})\partial_j q.$$

All right-hand terms belong to L^2 and $A_{33} \ge \lambda$. Multiplication by its Lipschitz reciprocal identifies $\partial_{33} q$ as an L^2 distribution and proves the estimate. At this step the identities may first be read in H^{-1}: multiplication by a Lipschitz function is bounded on H_0^1 by the product rule, and hence on its dual, so the multiplication and cancellation are legitimate before the L^2 conclusion. □

Lemma 8.6 (H^2 changes of coordinates and reflection)**.** *If a graph function φ is $C^{1,1}$, its flattening map and inverse act boundedly on H^2 in smaller graph cylinders. An H^2 function on one side of that graph has a bounded linear H^2 extension to a smaller cylinder on both sides.*

Proof. The first-derivative formulas in Lemma 2.2, differentiated once more in distributions, contain only the first and second weak derivatives of the function and the bounded first and second derivatives of the coordinate map. The product rule for Lipschitz coefficients follows by mollifying the coefficient and passing to the weak-derivative identity. Volume preservation and the already proved H^1 composition rule therefore bound all terms. This proves the H^2 composition assertion directly, without assuming boundary-smooth density in H^2.

In flat coordinates suppose f is given for $t < 0$. For small $t > 0$ set

$$Ef(y,t) = 3f(y,-t) - 2f(y,-2t), \qquad Ef(y,t) = f(y,t)\ (t<0).$$

At zero the values match because $3-2=1$, the normal first derivatives match because $-3+4=1$, and the tangential first derivatives match for the first reason. The traces of the first derivatives are available from the H^1 trace theorem. Apply the H^1 gluing theorem first to the function and then to its three first derivatives. It follows that $Ef \in H^2$, with a bound obtained by changing the vertical variable by factors one and two. Local cutoffs and the H^2 coordinate change prove the graph assertion. □

Theorem 8.7 ($C^{1,1}$ regularity; original Theorems 2.9 and 2.12)**.** *If Ω is bounded and $C^{1,1}$, then $X_T \cup X_N \subset H^1(\Omega)^3$ and, for either boundary condition,*

$$\|v\|_{H^1(\Omega)} \le C_\Omega(\|v\|_2 + \|\operatorname{curl} v\|_2 + \|\operatorname{div} v\|_2). \tag{8.6}$$

Proof. Interior regularity was proved in Theorem 8.1, so only smaller boundary cylinders must be considered. For $v \in X_T$ use the local decomposition $v = w + \nabla q$ of Lemma 8.3. For every test supported away from the artificial chart edges, zero normal trace gives

$$\int \nabla q \cdot \nabla \phi = -\int w \cdot \nabla \phi - \int (\operatorname{div} v)\phi.$$

Under the graph map F this is (8.4) with $A = (DF)^{-1}(DF)^{-T}$, $G = -(DF)^{-1}(w \circ F)$, and $F_{\mathrm{rhs}} = -(\operatorname{div} v) \circ F$. The coefficient is uniformly elliptic and Lipschitz, and $G \in H^1$ because DF is Lipschitz and $w \in H^1$. Lemma 8.5, followed by the coordinate-change part of Lemma 8.6, yields $q \in H^2$ on the smaller chart, with a bound by $\|v\|_X$. Hence $v \in H^1$ there.

For $v \in X_N$ use (8.3). On the exterior side of each chart $\nabla q = -w$. Thus q belongs to H^2 on that side, and its H^2 norm is bounded by $\|q\|_{H^1(B)} + \|w\|_{H^1(B)}$. Lemma 8.6 extends that exterior function to an H^2 function Q across a smaller chart. The original $q \in H^1(B)$ has equal traces from the two sides; therefore $r = q - Q$ has zero interior trace on the boundary graph. Inside Ω one has $\Delta r = \operatorname{div} v - \Delta Q \in L^2$, since $\operatorname{div} w = 0$. Its flattened weak Dirichlet equation has $G = 0$ and L^2 right side. Lemma 8.5 gives $r \in H^2$ on a still smaller chart, hence $q \in H^2$ and $v = w + \nabla q \in H^1$. The finitely many local bounds, the interior bounds, and the decomposition estimate give (8.6). □

8.3 Fractional boundary data and an explicit lifting

For $s \ge 0$ define $H^s(\mathbb{R}^d)$ by

$$\|f\|^2_{H^s(\mathbb{R}^d)} = \int_{\mathbb{R}^d} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 \, d\xi.$$

For an open subset, the restriction norm is the infimum of the whole-space norms of its extensions. For integer orders on the $C^{m,1}$ graph sets used below, the weak-derivative norm is equivalent to this norm: one direction follows by restriction and Plancherel, and the reverse follows from local reflection extension. For order m, the flat reflection is $\sum_{j=1}^{m} c_j f(y, -jt)$ on the missing side, where

$$\sum_{j=1}^{m} c_j (-j)^r = 1 \qquad (0 \le r \le m-1). \tag{8.7}$$

The Vandermonde matrix is invertible because the numbers $-j$ are distinct; its determinant is the product of their pairwise differences, as follows by successively factoring the polynomial determinant at equal arguments. Matching the traces of derivatives through order $m-1$ and repeatedly applying the H^1 gluing theorem proves this extension is H^m. Chain and product rules show that $C^{m,1}$ graph changes preserve H^m, and a finite partition gives a global extension. This is the higher-order version of Lemma 8.6, with exactly the same support argument.

Lemma 8.8 (Fractional multiplication and plane trace)**.** *If $0 < \theta < 1$, then on $\mathbb{R}^d$*

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \frac{|f(x+h) - f(x)|^2}{|h|^{d+2\theta}}\, dx\, dh = c_{d,\theta} \int_{\mathbb{R}^d} |\xi|^{2\theta} |\widehat{f}(\xi)|^2\, d\xi, \tag{8.8}$$

where $0 < c_{d,\theta} < \infty$. Multiplication by a bounded Lipschitz function is bounded on H^θ; multiplication by a function in $W^{k+1,\infty}$ is bounded on $H^{k+\theta}$. For $s > 1/2$, restriction to a coordinate hyperplane is a bounded map $H^s(\mathbb{R}^d) \to H^{s-1/2}(\mathbb{R}^{d-1})$.

Proof. Plancherel applied to the difference and Tonelli's theorem give (8.8): the inner integral of $|e^{ih\cdot\xi} - 1|^2 |h|^{-d-2\theta}$ scales as $|\xi|^{2\theta}$ and is rotation invariant. It is finite and positive because the numerator is bounded by $\min\{|h|^2|\xi|^2, 4\}$ and is not identically zero. For a multiplier a, split

$$a(x+h)f(x+h) - a(x)f(x) = a(x+h)(f(x+h) - f(x)) + (a(x+h) - a(x))f(x).$$

The second term is bounded using $|a(x+h) - a(x)| \le \min\{\mathrm{Lip}(a)|h|, 2\|a\|_\infty\}$; the squared minimum is integrable with the weight in (8.8). The first term uses $\|a\|_\infty$ and the fractional seminorm of f. Apply this estimate to every derivative of order at most k and use Leibniz's rule. Plancherel identifies the sum of their H^θ norms with an equivalent $H^{k+\theta}$ norm.

For a Schwartz function the Fourier transform of its hyperplane restriction is $(2\pi)^{-1/2} \int \widehat{f}(\xi', t)\, dt$. Cauchy–Schwarz and the substitution $t = (1 + |\xi'|^2)^{1/2}\tau$ give

$$\left| \int \widehat{f}(\xi', t)\, dt \right|^2 \le C_s (1 + |\xi'|^2)^{1/2-s} \int (1 + |\xi'|^2 + t^2)^s |\widehat{f}(\xi', t)|^2\, dt.$$

Multiply by $(1 + |\xi'|^2)^{s-1/2}$ and integrate. Truncating and smoothing in Fourier space proves density of Schwartz functions in H^s, so the estimate defines the trace. The same argument, with $e^{iht} - 1$ in the integral, and dominated convergence show continuity of restrictions as a parallel plane approaches the given plane. Thus the trace depends only on the function on either adjacent open half-space: an extension zero on that half-space has zero restrictions at almost all parallel planes and hence zero trace at its boundary. This also proves that the estimate gives a well-defined trace for restriction Sobolev spaces on half-spaces and on interiors of planar faces. □

On a $C^{m,1}$ boundary, $H^{m-1/2}(\partial\Omega)$ means that the data, multiplied by a finite subordinate smooth partition and written in graph coordinates, belong to $H^{m-1/2}(\mathbb{R}^2)$ after extension by zero outside their coordinate supports. This definition is invariant up to equivalent norms. Indeed the chain rule reduces integer derivatives of a change of coordinates to products of derivatives of

the function and bounded Lipschitz coefficients. For the remaining half derivative use (8.8): a bi-Lipschitz change bounds the denominator above and below by fixed multiples and bounds the two Jacobians, while the coefficient products are bounded by Lemma 8.8. Finite chart covers and partitions then give both norm comparisons.

Lemma 8.9 (Boundary-value lifting)**.** *On a bounded $C^{m,1}$ domain, $m \geq 1$, every $g \in H^{m-1/2}(\partial\Omega)$ has a linear extension $Eg \in H^m(\Omega)$ with trace g and $\|Eg\|_{H^m} \leq C\|g\|_{H^{m-1/2}(\partial\Omega)}$. The assertion holds componentwise for vector data.*

Proof. In flat coordinates put $\lambda(\xi) = (1 + |\xi|^2)^{1/2}$ and define, for $t > 0$,

$$\widehat{Eg}(\xi, t) = e^{-t\lambda(\xi)}\widehat{g}(\xi),$$

where the transform here is in the two tangential variables only. For $|\alpha| + j \leq m$, tangential Plancherel gives

$$\int_0^\infty \|\partial_y^\alpha \partial_t^j Eg(\cdot, t)\|_2^2 \, dt \leq \frac{1}{2} \int \lambda(\xi)^{2m-1} |\widehat{g}(\xi)|^2 \, d\xi.$$

The weak derivatives are identified first for smooth data and then by the completeness of the derivative norm. Dominated convergence shows $Eg(\cdot, t) \to g$ in $H^{m-1/2}$ as $t \downarrow 0$. For a chart-localized datum multiply the extension by a fixed cutoff supported inside the chart and equal to one on the support of its boundary datum. Its trace is unchanged and its H^m norm remains bounded by the product rule. Undo the $C^{m,1}$ flattening, extend by zero past the artificial chart edges, and add the finitely many chart contributions. All choices are fixed and linear, so the stated map is linear and bounded. □

Lemma 8.10 (A fractional embedding needed for face moments)**.** *If $\sigma > 0$ and $2 < p < \infty$ satisfy $d(1/2 - 1/p) < \sigma$, then $H^\sigma(\mathbb{R}^d) \hookrightarrow L^p(\mathbb{R}^d)$ continuously. The same holds on an open set with the restriction norm. In particular a planar face trace of an $H^{1/2+\delta}(K)$ function on a tetrahedron belongs to some L^p of that face with $p > 2$.*

Proof. Split Fourier space into the unit ball and the disjoint annuli $2^j \leq |\xi| < 2^{j+1}$, and let f_j be the inverse transforms of the corresponding pieces. Cauchy–Schwarz on an annulus and Fourier inversion give $\|f_j\|_\infty \leq C2^{jd/2}\|f_j\|_2$. The elementary inequality $\|f_j\|_p \leq \|f_j\|_2^{2/p}\|f_j\|_\infty^{1-2/p}$ then gives

$$\|f_j\|_p \leq C2^{jd(1/2-1/p)}\|f_j\|_2.$$

The sum of these bounds is finite by Cauchy–Schwarz with weights $2^{j\sigma}$, because the remaining geometric series has strictly negative exponent. The low-frequency piece has the same bound without the dyadic factor. The series therefore converges in L^p, and its L^2 limit identifies it with f. For an open set apply the bound to an extension and take the infimum. The last assertion follows by the plane trace estimate and the $d = 2$, $\sigma = \delta$ instance of this lemma. Taking extensions arbitrarily close to the restriction norm gives the claimed bounds on each open planar face. □

8.4 Higher regularity, including nonhomogeneous traces

Lemma 8.11 (Represented boundary traces and tangentiality)**.** *On a bounded $C^{1,1}$ domain, the assertion that a weak normal trace is represented by $g \in L^2(\partial\Omega)$ means*

$$\int_\Omega v \cdot \nabla\phi + (\operatorname{div} v)\phi = \int_{\partial\Omega} g\,\gamma\phi \qquad (\phi \in H^1(\Omega)).$$

The assertion that the weak trace $v \times n$ is represented by $g \in L^2(\partial\Omega)^3$ means

$$\int_\Omega v \cdot \operatorname{curl} \Phi - \operatorname{curl} v \cdot \Phi = \int_{\partial\Omega} g \cdot \gamma\Phi \qquad (\Phi \in H^1(\Omega)^3).$$

These formulas suffice for the nonhomogeneous boundary conditions in Theorem 8.12; no additional fractional-dual trace construction is required. In the second formula necessarily $g \cdot n = 0$ almost everywhere. For an H^1 field the represented traces are its ordinary Sobolev traces dotted or crossed with n.

Proof. The products in the boundary integrals are integrable by the L^2 trace bound. Let η be a smooth scalar test function supported in a boundary coordinate patch. The boundary vector $n\eta$ belongs to $H^{1/2}(\partial\Omega)^3$: n is Lipschitz in these coordinates, and the fractional multiplier estimate applies. The $m = 1$ case of Lemma 8.9 supplies $\Phi \in H^1(\Omega)^3$ with $\gamma\Phi = n\eta$. Choose ambient smooth restrictions v_j converging to v in the $H(\operatorname{curl})$ graph norm. For each of them the Green formula, first for smooth tests and then by H^1 density, gives

$$\int_\Omega v_j \cdot \operatorname{curl} \Phi - \operatorname{curl} v_j \cdot \Phi = \int_{\partial\Omega} (v_j \times n) \cdot n\eta = 0.$$

Pass to the graph-norm limit and use the represented-trace identity. This yields $\int_{\partial\Omega}(g \cdot n)\eta = 0$ for every such local test. Hence the L^2 function $g \cdot n$ is zero.

For an H^1 field, ambient smooth restrictions converge in H^1, so their curls, divergences, and ordinary traces converge in the respective L^2 spaces. Pass to the limit in both classical Green identities. The boundary products converge because the unit normal is bounded. This proves agreement with ordinary Sobolev traces. In particular, if an H^1 lifting has boundary vector $b = gn$ or $b = n \times g$, its represented designated trace is exactly g: $b \cdot n = g$ in the first case and $(n \times g) \times n = g$ in the second, using $g \cdot n = 0$. Subtracting the lifting therefore produces precisely the homogeneous weak boundary condition used in the rest of the proof. □

Theorem 8.12 (Full higher-order statement; original Corollary 2.15)**.** *Let $m \geq 1$ and let Ω be bounded and $C^{m,1}$. If $v \in L^2(\Omega)^3$ has $\operatorname{curl} v \in H^{m-1}(\Omega)^3$ and $\operatorname{div} v \in H^{m-1}(\Omega)$, and either*

$$g = v \cdot n \in H^{m-1/2}(\partial\Omega) \quad \text{or} \quad g = v \times n \in H^{m-1/2}(\partial\Omega)^3,$$

then $v \in H^m(\Omega)^3$ and

$$\|v\|_{H^m} \leq C\big(\|v\|_2 + \|\operatorname{curl} v\|_{H^{m-1}} + \|\operatorname{div} v\|_{H^{m-1}} + \|g\|_{H^{m-1/2}(\partial\Omega)}\big). \tag{8.9}$$

For $m = 1$ this includes the two inhomogeneous assertions of the paper's Remark 2.14.

Proof. First reduce to homogeneous data. In the normal case prescribe the boundary vector $b = gn$; in the tangential case prescribe $b = n \times g$. Since Lemma 8.11 gives $g \cdot n = 0$ for an actual tangential trace, the identity $(n \times g) \times n = g$ verifies the second choice. In $C^{m,1}$ coordinates the normal has bounded weak derivatives through order m. The multiplier estimate in Lemma 8.8 therefore bounds b in the same boundary space. Lemma 8.9 extends it to an H^m vector field. Subtract this lift from v. Its weak trace is the prescribed represented trace by Lemma 8.11, so the difference has zero designated trace. The curls and divergences of the lift have H^{m-1} bounds. It remains to prove (8.9) with $g = 0$.

We use induction on m; Theorem 8.7 is the base case. The interior induction follows immediately by applying the whole-space Fourier identity of Theorem 8.1 to a cutoff, multiplying it by $(1+|\xi|^2)^{m-1}$, and using the already established H^{m-1} regularity to bound the cutoff commutators. We detail the boundary induction.

In graph coordinates choose the invertible matrix $M(y)$ whose rows are the two graph tangent vectors $(1,0,\partial_1\varphi)$ and $(0,1,\partial_2\varphi)$ and the unit normal. Let $a = M(v\circ F)$. Zero tangential trace says $a_1 = a_2 = 0$ on $t = 0$; zero normal trace says $a_3 = 0$ there. The matrices $M, M^{-1}, DF, (DF)^{-1}$ have bounded derivatives through order m. Writing the four components consisting of curl and divergence in these coordinates gives a first-order operator

$$Pa = \sum_{j=1}^{3} A_j(y)\partial_j a + B(y)a, \tag{8.10}$$

where $A_j \in W^{m,\infty}$, $B \in W^{m-1,\infty}$, and $Pa = ((\operatorname{curl} v)\circ F, (\operatorname{div} v)\circ F)$. These assertions follow by substituting $v \circ F = M^{-1}a$ in the four first-derivative definitions: the leading coefficients contain only DF^{-1} and M^{-1}, and the lower coefficients contain one derivative of M^{-1}. All coefficients depend only on y.

For fields supported away from the artificial chart edges and with the designated components zero at $t = 0$, the base estimate, applied after undoing the coordinate and matrix changes, gives

$$\|a\|_{H^1} \le C(\|Pa\|_2 + \|a\|_2). \tag{8.11}$$

A cutoff reduces the induction to fields with this support: $P(\eta a) = \eta Pa + \sum_j A_j(\partial_j\eta)a$, whose extra term is H^{m-1} by the induction hypothesis.

Let α be a tangential multiindex with $|\alpha| = m-1$, and apply (8.11) to its iterated finite difference $D_h^\alpha a$. This still has the required zero components on the flat boundary. Iterating the elementary identity $D_h(fg) = fD_hg + (D_hf)g(\cdot+h)$ shows that $P(D_h^\alpha a) - D_h^\alpha(Pa)$ is a finite sum of products of the following kinds, with harmless translates of the factors:

$$(D_h^\beta A_j)\partial_j D_h^{\alpha-\beta}a, \qquad (D_h^\beta B)D_h^{\alpha-\beta}a, \qquad 0 < \beta \le \alpha.$$

Every first factor is bounded in L^∞ uniformly in h; every second factor requires at most $m-1$ derivatives of a. The translation estimate used in Lemma 8.5 therefore bounds these commutators by $C\|a\|_{H^{m-1}}$. It also bounds $D_h^\alpha(Pa)$ in L^2 by $\|Pa\|_{H^{m-1}}$ and $D_h^\alpha a$ in L^2 by $\|a\|_{H^{m-1}}$. Consequently

$$\|D_h^\alpha a\|_{H^1} \le C(\|Pa\|_{H^{m-1}} + \|a\|_{H^{m-1}}).$$

Weak limits identify $\partial_y^\alpha a \in H^1$ as in Lemma 8.5. Thus all derivatives of total order m containing at most one normal derivative are in L^2.

The principal matrix A_3 has a bounded left inverse $L(y)$. Indeed, put $k = (DF)^{-T}e_3 \neq 0$. Its action on a vector b is the pair $(k \times M^{-1}b, k \cdot M^{-1}b)$. The identity $|k \times z|^2 + |k \cdot z|^2 = |k|^2|z|^2$ gives a left inverse explicitly: apply the transpose of this cross/dot map, divide by $|k|^2$, and multiply by M. Its derivatives through order m are bounded. Hence

$$\partial_3 a = L\big(Pa - A_1\partial_1 a - A_2\partial_2 a - Ba\big). \tag{8.12}$$

Differentiate this identity $m-1$ times distributionally. Induct on the number of normal derivatives in the desired derivative of order m. On the right the derivatives of Pa have order at most $m-1$, the derivatives of a in the two leading tangential terms have one fewer normal derivative, and every coefficient-derivative or lower-order term uses at most $m-1$ derivatives of a. All are already controlled. This proves every remaining derivative is L^2, with the same bound. Coordinate changes, the finite cover, and the induction hypothesis finish the homogeneous estimate. Adding back the boundary lift proves the complete statement. □

8.5 Convex domains without smooth boundary

Lemma 8.13 (Smooth convex exhaustion; original Lemma 2.18)**.** *Every bounded convex open set Ω has smooth convex open subsets D_k such that $\overline{D}_k \subset D_{k+1}$, $\overline{D}_k \subset \Omega$, and $\bigcup_k D_k = \Omega$.*

Proof. Translate an interior point to zero and choose $B_r \subset \Omega \subset B_R$. The gauge $p(x) = \inf\{t > 0 : x \in t\Omega\}$ is convex, positively homogeneous, and satisfies $|x|/R \le p(x) \le |x|/r$. Convexity follows by writing x and y as positive multiples of points in Ω and taking their convex combination. It implies subadditivity, and then $|p(x) - p(y)| \le |x - y|/r$ by applying it in both directions. Also $\Omega = \{p < 1\}$: one inclusion uses openness along the ray through a point of Ω, and the other uses convexity with the interior point zero.

For integers $k \ge 1$, let $a_k = 2^{-k}$ and choose positive δ_k, ϵ_k tending to zero so fast that the smooth strongly convex function

$$f_k = p * \rho_{\delta_k} + \epsilon_k |x|^2$$

satisfies $|f_k - p| \le a_k/8$ on B_{2R+1} and $\delta_k/r < 1/4$. This is possible since convolution changes a Lipschitz function uniformly by at most $C\delta_k/r$. Convolution preserves convexity by integrating the defining convexity inequality, and adding $\epsilon_k|x|^2$ makes the Hessian positive definite. Set $D_k = \{f_k < 1 - a_k\}$. The lower bound $f_k \ge p - C\delta_k/r$ ensures every such set is in B_{2R+1} after decreasing the parameters further. The preceding uniform estimate then gives

$$\overline{D}_k \subset \{p \le 1 - 7a_k/8\} \subset \Omega, \qquad f_{k+1} \le 1 - 7a_k/8 + a_k/16 < 1 - a_{k+1} \text{ on } \overline{D}_k.$$

Thus the sets are nested with the asserted closure inclusion. For the finitely many initial choices decrease parameters so that $f_k(0) < 1 - a_k$; all D_k are then nonempty. Strong convexity implies that a point with $\nabla f_k = 0$ is its strict global minimum, and hence is not on this level. The implicit function theorem makes ∂D_k smooth. Finally $p(x) < 1$ implies $f_k(x) < 1 - a_k$ for all sufficiently large k, proving exhaustion. □

Lemma 8.14 (The convex boundary identity)**.** *If D is smooth and convex and $z \in H^1(D)^3$ has either zero normal or zero tangential trace, then*

$$\|\nabla z\|^2_{L^2(D)} \le \|\operatorname{curl} z\|^2_{L^2(D)} + \|\operatorname{div} z\|^2_{L^2(D)}. \tag{8.13}$$

Proof. For a smooth field integration of the coordinate identity yields

$$\int_D \big(|\operatorname{curl} z|^2 + |\operatorname{div} z|^2 - |\nabla z|^2\big) = \int_{\partial D} \big((z \cdot n) \operatorname{div} z - ((z \cdot \nabla)z) \cdot n\big). \tag{8.14}$$

To check it, expand $|\operatorname{curl} z|^2$ as $\sum_{i,j}(\partial_i z_j)^2 - \sum_{i,j} \partial_i z_j \partial_j z_i$ and integrate the two cross terms once. Their interior second-derivative terms cancel because derivatives commute. The remaining boundary terms are exactly those displayed.

Write $\mathrm{II}(t,t) = t \cdot (Dn)t$ for tangent vectors and $H = \operatorname{div} n$ on the boundary, using the unit normal extended constantly along normal lines. If $z \cdot n = 0$, differentiating that identity along the tangent vector z gives $((z \cdot \nabla)z) \cdot n = -\mathrm{II}(z,z)$. If $z \times n = 0$, write $z = bn$ at the boundary; the two normal derivatives in (8.14) cancel, leaving Hb^2. Thus the right side is respectively $\int \mathrm{II}(z,z)$ or $\int H|z|^2$. Convexity implies $\mathrm{II} \ge 0$: at a boundary point the domain lies on the interior side of its supporting tangent plane, so the second tangential derivative of a local defining function with outward gradient is nonnegative. Equivalently, for the convex level sets in Lemma 8.13 it is $t^T D^2 f_k t/|\nabla f_k| \ge 0$. Only these level sets are needed below.

We justify passage to H^1 fields with the boundary condition. Flatten a smooth boundary patch and use the smooth invertible matrix M from (8.10). The components required to vanish on the boundary have odd H^1 extensions across the flat plane; the other components have even

extensions. The odd extension is H^1 because its two zero traces match. Convolution with a smooth kernel even in the normal variable preserves oddness and evenness, hence preserves the required zero boundary components. Local cutoffs, with support inside the chart, preserve those zero components as well. Undo the smooth coordinate and matrix changes and add a finite partition. This constructs smooth fields obeying the same boundary condition and converging in H^1 to z. The H^1 trace estimate gives convergence of their traces in $L^2(\partial D)$. Since the smooth boundary curvatures are bounded, the boundary quadratic forms converge. Formula (8.14) therefore passes to H^1, and its nonnegative right side proves the assertion. □

Theorem 8.15 (Convex regularity; original Theorem 2.17)**.** *If the standing bounded Lipschitz domain Ω is convex, then $X_T \cup X_N \subset H^1(\Omega)^3$, and $\|v\|_{H^1} \le C\|v\|_X$ for either boundary condition.*

Proof. We use the smooth exhaustion D_k of Lemma 8.13. All norms in the estimates below are controlled by norms on Ω, independently of the curvature and chart constants of D_k.

First let $v \in X_T$. On $H^1(D_k)$ use the full H^1 inner product to solve

$$\int_{D_k} (s_k q + \nabla s_k \cdot \nabla q) = \int_{D_k} \big((\operatorname{div} v) q + v \cdot \nabla q\big) \quad (q \in H^1(D_k)). \tag{8.15}$$

Hilbert-space representation gives $\|s_k\|_{H^1(D_k)} \le (\|v\|_2^2 + \|\operatorname{div} v\|_2^2)^{1/2}$. Interior tests give $\Delta s_k = s_k$. Put $z_k = v|_{D_k} - \nabla s_k$. Equation (8.15) says exactly that z_k has zero normal trace on D_k, and

$$\operatorname{curl} z_k = \operatorname{curl} v, \qquad \operatorname{div} z_k = \operatorname{div} v - s_k.$$

Hence $z_k \in X_T(D_k)$. Theorem 8.7 puts it in $H^1(D_k)^3$, and the curvature-free estimate (8.13) bounds its gradient uniformly. Thus z_k, ∇z_k, s_k, and ∇s_k, each extended by zero solely as an L^2 array to Ω, have weakly convergent subsequences. Testing on compact subsets eventually contained in D_k identifies the weak limits of the gradient arrays as the distributional gradients of the limits. Call the limits $z \in H^1(\Omega)^3$ and $s \in H^1(\Omega)$.

For any fixed $q \in H^1(\Omega)$ insert its restriction in (8.15). Weak convergence of the zero-extended arrays and dominated convergence of the fixed data give

$$\int_\Omega (sq + \nabla s \cdot \nabla q) = \int_\Omega \big((\operatorname{div} v) q + v \cdot \nabla q\big) = 0.$$

The last equality is the original zero normal trace. Taking $q = s$ gives $s = 0$, so $z = v$. Weak lower semicontinuity of the norm supplies the required estimate.

Now let $v \in X_N$. We use a correction on the entire $H(\operatorname{curl}, D_k)$ space, which avoids any divergence constraint on auxiliary test fields. Define $\xi_k \in H(\operatorname{curl}, D_k)$ by

$$\int_{D_k} \big(\xi_k \cdot \phi + \operatorname{curl} \xi_k \cdot \operatorname{curl} \phi\big) = \int_{D_k} \big(v \cdot \operatorname{curl} \phi - \operatorname{curl} v \cdot \phi\big) \quad (\phi \in H(\operatorname{curl}, D_k)). \tag{8.16}$$

The graph space is Hilbert by passing to limits in its distributional definition, so representation gives $\|\xi_k\|_{H(\operatorname{curl}, D_k)} \le \|v\|_{H(\operatorname{curl}, \Omega)}$. For compactly supported smooth tests the right side vanishes; hence $\operatorname{curl}\operatorname{curl} \xi_k = -\xi_k$ in D_k. Let $z_k = v|_{D_k} - \operatorname{curl} \xi_k$. Then

$$\operatorname{curl} z_k = \operatorname{curl} v + \xi_k, \qquad \operatorname{div} z_k = \operatorname{div} v.$$

Subtracting the two sides of (8.16) shows $(\operatorname{curl} z_k, \phi) = (z_k, \operatorname{curl} \phi)$ for every $\phi \in H(\operatorname{curl}, D_k)$. Thus $z_k \in X_N(D_k)$, and again Theorem 8.7 followed by (8.13) bounds z_k in $H^1(D_k)$ uniformly in k.

Extract weak limits of the zero-extended L^2 arrays $z_k, \nabla z_k, \xi_k, \operatorname{curl} \xi_k$. Interior tests identify a limit $z \in H^1(\Omega)^3$ and $\xi \in H(\operatorname{curl}, \Omega)$, with $z = v - \operatorname{curl} \xi$. For any fixed $\phi \in H(\operatorname{curl}, \Omega)$ test

(8.16) with its restriction to D_k and pass to the limit. Its right side tends to zero by the zero tangential trace of v. Hence

$$(\xi, \phi)_\Omega + (\operatorname{curl} \xi, \operatorname{curl} \phi)_\Omega = 0 \quad (\phi \in H(\operatorname{curl}, \Omega)).$$

Taking $\phi = \xi$ proves $\xi = 0$, so $z = v$ and weak lower semicontinuity yields the estimate. At no point was a zero extension asserted to preserve an H^1 norm on D_k; only the component arrays were extended in L^2, and the derivative identity was recovered by interior tests. □

8.6 A fractional local estimate for finite-element use

Lemma 8.16 (Intrinsic fractional extension and scaling)**.** *Let D be bounded and Lipschitz and let $0 < t < 1$. Define*

$$|f|^2_{H^t(D),\mathrm{int}} = \int_D \int_D \frac{|f(x) - f(y)|^2}{|x - y|^{3+2t}}\, dx\, dy.$$

There is a linear extension E to $\mathbb{R}^3$ such that

$$\|Ef\|_{H^t(\mathbb{R}^3)} \le C\big(\|f\|_{L^2(D)} + |f|_{H^t(D),\mathrm{int}}\big).$$

For a shape-regular family of tetrahedra K of diameter h_K, and $p > 2$ with $3(1/2 - 1/p) < t$, this gives the uniform bound

$$h_K^{3/2-3/p} \|f\|_{L^p(K)} \le C\big(\|f\|_{L^2(K)} + h_K^t |f|_{H^t(K),\mathrm{int}}\big). \tag{8.17}$$

Proof. Use the same finite cutoffs, graph flattening, and even reflection as in Proposition 2.3. Multiplication by a bounded Lipschitz cutoff is bounded for the intrinsic norm by the exact difference decomposition used in Lemma 8.8; its error term is bounded by $\|f\|_2^2$ times the finite integral of $\min\{L|h|, 2A\}^2 |h|^{-3-2t}$. The graph change of variables preserves volume and changes distances by at most fixed multiplicative factors, so it is bounded for the intrinsic double integral in both directions.

For a flat half box, the same-side contributions of even reflection are copies of the original double integral. For a cross-side pair with heights $s > 0$ and $-r < 0$, the values are $f(x', s)$ and $f(y', r)$, while

$$|(x', s) - (y', -r)|^2 = |x' - y'|^2 + (s + r)^2 \ge |x' - y'|^2 + (s - r)^2.$$

Thus its double integral is bounded by the corresponding same-side integral. The reflected localized function has support a fixed positive distance d from the artificial edges of the full box. Its extension by zero past those edges adds at most $Cd^{-2t}\|f\|_2^2$, because for any point of its support the integral of $|x - y|^{-3-2t}$ over the outside is bounded by the integral over $|x - y| \ge d$. Sum the finitely many reflected pieces; the intrinsic seminorm of the sum is bounded by the sum of their seminorms times the number of pieces. Equation (8.8) identifies the resulting whole-space norm with H^t.

Apply Lemma 8.10 on a fixed reference tetrahedron. An affine map to K has singular values bounded above and below by fixed multiples of h_K under shape regularity. A single change of variables scales the squared L^2 norm by h_K^3, the L^p norm by $h_K^{3/p}$, and the squared intrinsic seminorm by h_K^{3-2t}, up to those fixed constants. This gives (8.17) with a uniform C. □

8.7 Re-entrant edges and the polyhedral shift problem

Proposition 8.17 (The loss of H^1 on a re-entrant edge)**.** *The H^1 conclusions cannot be extended to arbitrary Lipschitz polyhedra. In particular there are fields in X_T and in X_N which do not belong to H^1 on a polyhedron with a re-entrant dihedral edge.*

Proof. Consider a polyhedron which near an interior segment of an edge is the wedge $0 < r < r_0$, $0 < \theta < \omega$, $|z| < z_0$, with $\pi < \omega < 2\pi$. Put $\alpha = \pi/\omega \in (1/2, 1)$. Choose smooth cutoffs $\chi(r)$ and $\zeta(z)$, supported inside the displayed radial and longitudinal ranges and equal to one on smaller ranges. Set

$$q_N = \chi(r)\zeta(z)r^\alpha \sin(\alpha\theta), \qquad q_T = \chi(r)\zeta(z)r^\alpha \cos(\alpha\theta).$$

The uncut wedge functions are harmonic in the two transverse variables. The first derivatives behave as $r^{\alpha-1}$, whose squares are integrable with the volume factor $r\,dr$. Terms in the Laplacian involving the radial cutoff are away from $r = 0$; the longitudinal cutoff contributes only a multiple of r^α. Thus $q_N, q_T \in H^1$ and their Laplacians are in L^2. These distributional assertions have no source on the edge: performing integration by parts outside $r = \epsilon$ produces inner-cylinder terms of order ϵ^α against a smooth test and its gradient, which tend to zero.

The function q_N has zero trace on both wedge faces and is zero near every remaining boundary face. Consequently $q_N \in H^1_0$, and ∇q_N has zero tangential trace, as follows by differentiating compactly supported H^1 approximations in the curl graph norm. For q_T the angular normal derivative is zero on both wedge faces. Integration by parts as above, now with arbitrary smooth boundary tests, gives zero normal trace of ∇q_T; ambient H^1 density extends that identity to all weak normal-trace tests. Both gradients have curl zero and divergence in L^2. On the region where the cutoffs are one, their transverse Hessians have squared angular integral equal to a positive constant times $r^{2\alpha-4}$. This follows by differentiating the non-affine homogeneous harmonic functions twice; their Hessians cannot vanish identically since $\alpha \neq 1$. The radial integral $\int_0^\epsilon r^{2\alpha-3}\,dr$ diverges. Hence neither gradient is in H^1. □

Remark 8.18 (The additional ingredients for original Proposition 3.7). The source Proposition 3.7 asserts an exponent strictly greater than $1/2$ on every Lipschitz polyhedron. Compactness, $C^{1,1}$ regularity, and convex regularity alone do not prove this: a polyhedral graph has slope jumps at its edges, so the smooth coordinate argument cannot simply be reused.

The following subsections establish the extra ingredients. For X_N the regular decomposition produces a scalar potential which is H^2 on the exterior side. An explicit H^2 extension on a general Lipschitz graph reduces it to a homogeneous Dirichlet problem. For X_T the local scalar potential has nonhomogeneous Neumann data generated by an H^1 vector field. We retain those data in the variational term $\int G \cdot \nabla\phi$ and prove a weighted polyhedral estimate with this term present. Edge and vertex angular gaps, radial Green kernels, and dyadic localization then give the exponent $s > 1/2$ in Theorem 8.32. The homogeneous scalar theorem by itself is not being substituted for either boundary-data step.

Lemma 8.19 (H^2 extension on an arbitrary Lipschitz domain)**.** *For every bounded Lipschitz domain D there is a bounded linear extension $H^2(D) \to H^2(\mathbb{R}^3)$, with outputs supported in a fixed bounded set. Here $H^2(D)$ is defined by its weak derivatives, without assuming an extension in its definition.*

Proof. We give the construction that is needed beyond Lemma 8.6. A Lipschitz graph function on a chart can be extended to all of $\mathbb{R}^2$ with the same Lipschitz constant L by $\inf_z\{\varphi(z) + L|y - z|\}$, the infimum being over its original domain. The triangle inequality proves the Lipschitz bound and the original Lipschitz inequality proves agreement on that domain. Write $\varphi_t = \rho_{\epsilon t} * \varphi$, with a fixed smooth unit-integral kernel supported in the unit disk and with $\epsilon > 0$ small depending only on L. Cancellation of the derivatives of that kernel, followed by $|\varphi(y - h) - \varphi(y)| \le L|h|$, proves

$$|\varphi_t - \varphi| \le C\epsilon L t, \quad |\partial_t\varphi_t| \le C\epsilon L, \quad |\nabla_y\varphi_t| \le L, \quad |D^2_{y,t}\varphi_t| \le C_{L,\epsilon}/t.$$

For example, differentiating a convolution kernel twice produces a kernel of integral zero and size bounded by a constant times its scale to the power -4; subtracting $\varphi(y)$ in its integral gives a factor bounded by L times that scale. Differentiation in t is the same calculation with the

additional factor ϵ from the scale. These observations prove all four bounds. Choose ϵ so the first two constants are at most $1/2$ in their displayed relative bounds.

For $x_3 < \varphi(y)$ define $\delta(y, x_3) > 0$ implicitly by

$$\delta = \varphi_\delta(y) - x_3.$$

The function $t - \varphi_t(y) + x_3$ is strictly increasing with derivative between $1/2$ and $3/2$, starts negative at zero, and tends to infinity. It has exactly one zero. The implicit function theorem and the preceding estimates give

$$\tfrac{2}{3}(\varphi(y) - x_3) \le \delta \le 2(\varphi(y) - x_3), \quad -2 \le \partial_3\delta \le -\tfrac{2}{3}, \quad |\nabla\delta| \le C, \quad |D^2\delta| \le C/\delta. \tag{8.18}$$

For the last inequality differentiate the implicit identity twice: its denominator is $1 - \partial_t\varphi_t \ge 1/2$, its first derivatives are bounded, and its second-derivative terms are bounded by C/δ.

Choose a smooth signed function ψ supported in $(3, 4)$ with $\int \psi = 1$ and $\int \lambda\psi(\lambda)\, d\lambda = 0$. For instance combine two unit-integral bumps with distinct first moments $a < b$ using coefficients $b/(b-a)$ and $-a/(b-a)$. If u is given in the epigraph, set below it

$$Eu(y, x_3) = \int_3^4 \psi(\lambda) u(y, x_3 + \lambda\delta(y, x_3))\, d\lambda. \tag{8.19}$$

For every fixed $3 \le \lambda \le 4$ the vertical map $x_3 \mapsto x_3 + \lambda\delta$ maps the lower half-line bijectively onto the upper one. Its derivative lies between $1 - 2\lambda$ and $1 - 2\lambda/3$, hence is negative with absolute value bounded above and below independently of λ. Its endpoint limits follow from the first bound in (8.18). Fubini and this one-dimensional change of variables give

$$\|a(y, x_3 + \lambda\delta)\|_{L^2(x_3 < \varphi(y))} \le C\|a\|_{L^2(x_3 > \varphi(y))}. \tag{8.20}$$

For smooth u, differentiate (8.19). The first derivatives contain only Du composed with these maps and bounded factors $1, \lambda\partial_j\delta$. The second derivatives contain bounded multiples of D^2u composed with the same maps, except for

$$(\partial_{ij}\delta) \int_3^4 \lambda\psi(\lambda) u_3(y, x_3 + \lambda\delta)\, d\lambda.$$

Use the zero first moment to subtract $u_3(y, x_3 + 3\delta)$ inside this integral. The fundamental theorem in the vertical variable gives the remaining integrand as $\delta \int_3^\lambda u_{33}(y, x_3 + s\delta)\, ds$. The factor δ cancels the only unbounded factor in $|\partial_{ij}\delta| \le C/\delta$. Equation (8.20), triangle inequality for integrals, and the fixed finite integration interval therefore bound all second derivatives by $C\|u\|_{H^2}$; the same argument bounds the zeroth and first derivatives.

For an ambient smooth u, the function and first derivatives of Eu have the same boundary traces as u. Indeed all evaluation points tend to the boundary point, $\int \psi = 1$, and every additional first-derivative term is $\partial_j\delta$ times an integral whose constant boundary value cancels by $\int \lambda\psi = 0$. The bounded first derivatives of δ make the remaining difference tend to zero. The H^1 gluing theorem, first for the functions and then for their first derivatives, makes this a global H^2 function across the Lipschitz graph.

Ambient smooth restrictions are dense in the weak H^2 norm on a Lipschitz domain independently of this extension theorem. The proof is the localized inward translation and mollification of Proposition 2.5, applied simultaneously to all weak derivatives of order at most two: at points of the original domain each convolution kernel samples only interior points, so differentiation gives the convolution of the designated zero-extended derivative; translation and convolution converge in each L^2 norm. Smooth localization preserves H^2 by the product rule. Thus the bounded

extension just proved on these smooth restrictions extends by completeness to every H^2 function on the graph side.

Finally localize a function on D by finitely many smooth chart cutoffs. View each localized boundary piece as a function on the full epigraph, extending by zero only past its artificial chart edges, where it vanishes in a neighborhood. Apply the graph construction, then multiply by an ambient cutoff equal to one on the original localized support and supported inside the larger chart. These extended pieces agree with the original localized functions on D and have bounded H^2 norm. An interior piece is extended by zero. Their finite sum is the required extension with fixed bounded support. All choices were fixed and linear. □

8.8 Proved model estimates toward the polyhedral shift

We first prove the separated model estimates and their homogeneous localization. The nonhomogeneous boundary-data steps are supplied afterward; the model estimates alone are not asserted to imply them.

Lemma 8.20 (Angular eigenvalues and the positive geometric gap)**.** *Let S be a connected Lipschitz spherical polygon with nonempty boundary. The weak spherical Laplacian with homogeneous Dirichlet, or homogeneous Neumann, condition has an orthonormal $L^2(S)$ basis Y_j of eigenfunctions with eigenvalues $\mu_j \geq 0$ tending to infinity, counted with multiplicity. In the Dirichlet case the first eigenvalue is positive. In the Neumann case the zero eigenspace consists exactly of constants and the next eigenvalue is positive. The positive conical exponents are*

$$\alpha_j = \sqrt{\mu_j + 1/4} - 1/2 > 0. \tag{8.21}$$

For a dihedral angle $0 < \omega < 2\pi$, the nonconstant edge exponents for either homogeneous condition are $k\pi/\omega$, $k \geq 1$, so the first exceeds $1/2$. For any fixed polyhedron there is therefore a number $0 < \delta < 1/2$ smaller than every positive first corner exponent and every first edge exponent minus $1/2$.

Proof. The extension, compactness, and Poincare proofs already given work in dimension two as well: rectangles replace boxes and the constant in the cube-average estimate is two rather than three. Smooth coordinate charts of the sphere change the norms by bounded factors and give a smooth uniformly positive metric. A finite chart partition therefore gives compact $H^1(S) \hookrightarrow L^2(S)$ and the respective Poincare inequality, on mean-zero functions or on $H^1_0(S)$. For the latter, a sequence with gradient tending to zero has a constant limit, and its zero boundary trace forces that constant to be zero.

Let V be $H^1(S)$ or $H^1_0(S)$ and define $Kf \in V$ by

$$\int_S (\nabla_S Kf \cdot \nabla_S q + (Kf)q) = \int_S fq \quad (q \in V).$$

Hilbert representation and the compact inclusion make $K : L^2(S) \to L^2(S)$ compact, positive, self-adjoint, and injective. Here is the needed spectral argument. The supremum of (Kf, f) on the L^2 unit sphere is attained: for a maximizing sequence take a weak subsequence, use compactness to make its images converge strongly, and pass to the pairing. The limiting vector can be normalized, and positivity shows its norm is one. Variation against its orthogonal complement shows it is an eigenvector with this largest eigenvalue. Repeat on the orthogonal complement of each finite collection. The positive eigenvalues must tend to zero, since otherwise their images of orthonormal unit vectors would have no convergent subsequence. The orthogonal complement of all obtained vectors is zero: if nonzero, positivity and injectivity make the same maximization produce another positive eigenvalue there. That eigenvalue would be no larger

than every eigenvalue chosen in the successive maximizations, contradicting their convergence to zero. This proves an orthonormal basis. Writing its eigenvalues as κ_j gives $\mu_j = \kappa_j^{-1} - 1$ and the asserted weak eigen-equation. Zero gradient rigidity and the two Poincare inequalities give the zero-mode statements and a strictly positive first remaining eigenvalue.

For a homogeneous function $r^\alpha Y(\omega)$ in a three-dimensional cone, differentiating in polar coordinates gives

$$\Delta(r^\alpha Y) = r^{\alpha-2}\big(\alpha(\alpha+1)Y + \Delta_S Y\big).$$

The nonnegative root of $\alpha(\alpha+1) = \mu_j$ is (8.21). On an interval of angle length ω, the equation $-Y'' = \mu Y$ with either two Dirichlet or two Neumann conditions has the sine or cosine solutions with $\sqrt{\mu} = k\pi/\omega$; solving the elementary constant-coefficient ODE verifies this list. The transverse two-dimensional radial equation gives $\alpha^2 = \mu$. Finitely many edges and vertices then permit a single strictly positive choice of δ. □

Lemma 8.21 (A quantitative radial Green estimate)**.** *Let $0 < \beta < \nu_*$ and $\nu \geq \nu_*$. For F with $e^{\beta t}F \in L^2(0,\infty)$, the Dirichlet solution of $-v'' + \nu^2 v = F$ on $(0,\infty)$ satisfies*

$$\|e^{\beta t}v''\|_2 + \nu\|e^{\beta t}v'\|_2 + \nu^2\|e^{\beta t}v\|_2 \leq C_{\beta,\nu_*}\|e^{\beta t}F\|_2. \tag{8.22}$$

For $0 < \nu < \beta$ the same estimate holds after subtracting $Ae^{-\nu t}$, where

$$A = \nu^{-1}\int_0^\infty \sinh(\nu s)F(s)\,ds, \tag{8.23}$$

with a constant allowed to depend on ν and β.

Proof. The solution is the integral against

$$G_\nu(t,s) = \frac{e^{-\nu|t-s|} - e^{-\nu(t+s)}}{2\nu}.$$

Direct differentiation on $t < s$ and $t > s$, continuity at $t = s$, and the derivative jump -1 verify its equation and its zero value at $t = 0$. The ordinary exponential convolution bound puts it in $H_0^1(0,\infty)$ for $F \in L^2$; testing the homogeneous difference equation with the difference proves uniqueness. This justifies the formula for the variational solution as well.

For $\beta < \nu$, multiplication by $e^{\beta t}$ replaces the main convolution kernel by $e^{\beta(t-s)}e^{-\nu|t-s|}/(2\nu)$. Its integral over the real difference variable is $(\nu^2 - \beta^2)^{-1}$. The kernel G_ν is nonnegative and bounded by this main kernel before weighting, so the elementary L^1 convolution bound gives $\|e^{\beta t}v\|_2 \leq (\nu^2 - \beta^2)^{-1}\|e^{\beta t}F\|_2$. Also $|\partial_t G_\nu| \leq e^{-\nu|t-s|}$, whose weighted integral is $2\nu/(\nu^2 - \beta^2)$. This bounds v'. The equation bounds v''. The ratios $\nu^2/(\nu^2 - \beta^2)$ are uniformly bounded for $\nu \geq \nu_*$, proving (8.22).

If $\beta > \nu$, Cauchy–Schwarz makes (8.23) convergent, and expansion of the Green formula gives exactly

$$v(t) - Ae^{-\nu t} = \frac{1}{2\nu}\left(-e^{-\nu t}\int_t^\infty e^{\nu s}F(s)\,ds + e^{\nu t}\int_t^\infty e^{-\nu s}F(s)\,ds\right).$$

After weighting, the two one-sided convolution kernels have masses $(\beta - \nu)^{-1}$ and $(\beta + \nu)^{-1}$. Their first derivatives have the same bounds with factors ν; the two endpoint terms $F(t)$ cancel in the first derivative. The equation supplies the second derivative. This proves the second assertion. □

Proposition 8.22 (Radial estimates on a polyhedral cone)**.** *Let $C_S = \{r\omega : 0 < r < 1,\ \omega \in S\}$, and suppose $u \in H^1(C_S)$ vanishes for r near one, satisfies $-\Delta u = f \in L^2(C_S)$, and has homogeneous Dirichlet or Neumann condition on the conical side. Choose $\beta = 1/2 + \delta < 1$*

smaller than $\sqrt{\mu_+ + 1/4}$, where μ_+ is the first positive angular eigenvalue. There is a constant c, zero in the Dirichlet case, for which

$$\begin{aligned}\|r^{-1-\beta}(u-c)\|_{L^2(C_S)} + \|r^{-\beta}\partial_r u\|_{L^2(C_S)} + \|r^{-1-\beta}\nabla_S u\|_{L^2(C_S)} \\ + \|r^{1-\beta}\partial_{rr}u\|_{L^2(C_S)} + \|r^{-1-\beta}\Delta_S u\|_{L^2(C_S)} \le C\|f\|_{L^2(C_S)},\end{aligned} \tag{8.24}$$

and $|c| \le C\|f\|_2$. The angular Laplacian in this statement is the variational angular operator, not an assertion of two ordinary angular weak derivatives at its polygonal corners.

Proof. Expand $u(r,\cdot)$ and $f(r,\cdot)$ in the basis of Lemma 8.20. The form identity $\int_S |\nabla_S q|^2 = \sum_j \mu_j |(q, Y_j)|^2$ holds for q in the form domain: the eigenvectors span that Hilbert space as well, since a vector orthogonal to them in its form inner product is orthogonal in L^2 and hence zero. Thus all changes of sums and radial integrals are justified by nonnegativity and Parseval. Tests of the form $\eta(r)Y_j$ give the radial equation

$$-u_j'' - 2r^{-1}u_j' + \mu_j r^{-2}u_j = f_j.$$

Set $t = -\log r$, $v_j(t) = e^{-t/2}u_j(e^{-t})$, and $F_j(t) = e^{-5t/2}f_j(e^{-t})$. The transformed equation is $-v_j'' + (\mu_j + 1/4)v_j = F_j$, with $v_j(0) = 0$. Its finite-energy meaning follows directly from the radial energy integral. If necessary the elementary Hardy bound

$$\int_0^1 |u_j|^2\,dr \le 4\int_0^1 r^2|u_j'|^2\,dr$$

puts v_j in H^1: integrate $(r|u_j|^2)'$ from ϵ to one, use $u_j(1) = 0$, discard the nonpositive lower-end term, and apply Cauchy–Schwarz, then let $\epsilon \downarrow 0$.

For all positive angular modes, $\nu_j = \sqrt{\mu_j + 1/4} \ge \sqrt{\mu_+ + 1/4} > \beta$. Apply (8.22) and sum its squared estimates. The weighted forcing is bounded because

$$\sum_j \int_0^\infty e^{2\beta t}|F_j(t)|^2\,dt = \int_0^1 r^{4-2\beta}\|f(r,\cdot)\|_{L^2(S)}^2\,dr \le \|f\|_{L^2(C_S)}^2.$$

For the Neumann constant mode $\nu_0 = 1/2 < \beta$, use the second part of Lemma 8.21. The subtracted exponential corresponds to a constant in the original radial coordinate. Its coefficient is bounded by Cauchy–Schwarz in $\int_0^1 r f_0(r)\,dr$ and the radial $L^2(r^2dr)$ norm of f_0, as is seen directly from (8.23). This gives c. Finally substitute $u_j = e^{t/2}v_j$ back into the norms. The first three terms in (8.24) correspond respectively to $e^{\beta t}v_j$, $e^{\beta t}(v_j' + v_j/2)$, and $\sqrt{\mu_j}e^{\beta t}v_j$; the last two correspond to $e^{\beta t}(v_j'' + 2v_j' + 3v_j/4)$ and $\mu_j e^{\beta t}v_j$. The Green estimates control all of these quantities. □

Lemma 8.23 (Passing a Neumann angular mean to the full cylinder)**.** *Let $W_\omega = \{0 < r < 1,\ 0 < \theta < \omega,\ z \in \mathbb{R}\}$, where $0 < \omega < 2\pi$, be the dihedral model and suppose $u \in H^1(W_\omega)$, $G \in L^2(W_\omega)^3$, and $F \in L^2(W_\omega)$ satisfy*

$$\int_{W_\omega} \nabla u \cdot \nabla\phi = \int_{W_\omega} G\cdot\nabla\phi + \int_{W_\omega} F\phi$$

for H^1 tests compactly supported in the radial and axial directions away from $r = 1$, with no zero-trace restriction on the two lateral faces. Suppose there are $0 < \rho < 1$ and $Z < \infty$ such that u, G, F vanish almost everywhere when $r > \rho$ or $|z| > Z$. These are the artificial radial and axial collars used below. Let bars be the averages over $0 < \theta < \omega$ of u, F, G_r, G_z, and define on the full cylinder

$$U(r,\theta,z) = \bar u(r,z), \qquad \mathcal{G} = \overline{G_r}(r,z)e_r + \overline{G_z}(r,z)e_z, \qquad \mathcal{F} = \bar F(r,z).$$

After extension by zero through the artificial collars, $U \in H^1(\mathbb{R}^3)$, $\mathcal{G}, \mathcal{F} \in L^2(\mathbb{R}^3)$, and

$$-\Delta U = \mathcal{F} - \operatorname{div} \mathcal{G} \quad \text{in } \mathcal{D}'(\mathbb{R}^3).$$

Moreover $\|U\|_{H^1} \le C_\omega \|u\|_{H^1}$ and the analogous L^2 bounds hold for the averaged data.

Proof. Away from the axis, polar coordinates are smooth and Fubini in the weak derivative identities shows $\partial_r \bar{u} = \overline{\partial_r u}$ and $\partial_z \bar{u} = \overline{\partial_z u}$. Jensen's inequality, with the ratio $2\pi/\omega$ of the angular measures, gives the displayed bounds for U and its candidate gradient $e_r \partial_r \bar{u} + e_z \partial_z \bar{u}$. It also gives the L^2 data bounds.

For an arbitrary smooth three-dimensional test function supported away from the axis, average it over the full angle $0 < \theta < 2\pi$. The resulting scalar test is independent of angle and is admissible in the original wedge's variational equation. Multiply it by a fixed cutoff equal to one on the union of the data and solution supports and zero near the artificial boundaries; all extra terms occur where the paired fields vanish. All full-cylinder pairings with $U, \mathcal{G}, \mathcal{F}$ equal $2\pi/\omega$ times the corresponding wedge pairings with this averaged test. In particular the angular component of the original G does not appear, because the averaged scalar test has angular derivative zero. This proves the asserted equation on the punctured full space, for arbitrary tests there, not just for axisymmetric tests. The artificial collars make the zero extensions harmless at their other boundaries.

To remove the axis, take a radial cutoff η_ε equal to zero for $r \le \varepsilon^2$, equal to one for $r \ge \varepsilon$, and affine in $\log r$ between those radii. It can be smoothed with the same bounds, and on any fixed bounded axial interval

$$\int |\nabla \eta_\varepsilon|^2 \le C |\log \varepsilon|^{-1} \longrightarrow 0.$$

In the distributional first-derivative identities on the punctured space, test with $\eta_\varepsilon \phi$. The error term is bounded by $\|U\|_2 \|\phi \nabla \eta_\varepsilon\|_2$ and tends to zero. Thus the candidate gradient is the global weak gradient, proving $U \in H^1(\mathbb{R}^3)$. In the punctured weak equation, the corresponding error is bounded by

$$(\|\nabla U\|_2 + \|\mathcal{G}\|_2) \|\phi \nabla \eta_\varepsilon\|_2,$$

which also tends to zero. All remaining terms converge by dominated convergence and Cauchy–Schwarz. This proves the global equation. □

Lemma 8.24 (The homogeneous dihedral estimate)**.** *Let $W_\omega = \{0 < r < 1,\ 0 < \theta < \omega,\ z \in \mathbb{R}\}$, where $0 < \omega < 2\pi$. Suppose $u \in H^1(W_\omega)$ is zero near $r = 1$ and outside a bounded z interval, has homogeneous Dirichlet or Neumann data on its two flat faces, and solves $-\Delta u = f \in L^2(W_\omega)$. If $0 < \beta < \min\{1, \pi/\omega\}$, then*

$$\|r^{-\beta} \nabla u\|_2 + \|r^{1-\beta} D^2 u\|_2 \le C_{\beta,\omega} (\|f\|_2 + \|u\|_2). \tag{8.25}$$

For the nonconstant angular part one also has $\|r^{-1-\beta} u\|_2 \le C \|f\|_2$.

Proof. Expand in the normalized sine functions for Dirichlet data, or cosine functions for Neumann data, and Fourier transform in z. Completeness of these angular systems follows by odd or even extension to the circle: convolution with the nonnegative Fejer kernels approximates continuous periodic functions uniformly, and then all L^2 functions by density; each Fejer convolution is a finite trigonometric sum. The kernel assertion follows directly from $N^{-1} |\sum_{j=0}^{N-1} e^{ij\theta}|^2$, whose integral is 2π and whose mass outside any neighborhood of zero tends to zero. Thus no angular expansion completeness assumption is needed.

For a nonconstant mode $\alpha = k\pi/\omega$ and a longitudinal frequency κ, put $t = -\log r$, $q(t) = u_{k,\kappa}(e^{-t})$, and $F(t) = e^{-2t} f_{k,\kappa}(e^{-t})$. The equation is

$$-q'' + (\alpha^2 + \kappa^2 e^{-2t}) q = F, \qquad q(0) = 0.$$

Let $Q = e^{\beta t}q$, $V = \kappa^2 e^{-2t}$, and $F_\beta = e^{\beta t}F$. Testing the original equation with $e^{2\beta t}q$, after a bounded-weight truncation, gives

$$\|Q'\|_2^2 + (\alpha^2 - \beta^2)\|Q\|_2^2 + \|\sqrt{V}Q\|_2^2 \le \|F_\beta\|_2\|Q\|_2. \tag{8.26}$$

For rigor use the weight $e^{\beta \min\{t,T\}}$, rounded at T with logarithmic derivative at most β; the energy identity contains the subtraction of the square of that logarithmic derivative, so the same lower bound holds. Its estimates are independent of T and monotone convergence gives (8.26). Since $\alpha \ge \pi/\omega > \beta$, it follows that

$$\alpha^2\|Q\|_2 + \alpha\|Q'\|_2 + \alpha\|\sqrt{V}Q\|_2 \le C\|F_\beta\|_2.$$

The transformed equation is $-Q'' + 2\beta Q' + (\alpha^2 - \beta^2 + V)Q = F_\beta$. Test it with VQ and integrate by parts, using $V' = -2V$. This gives

$$\int V|Q'|^2 + \int V^2|Q|^2 + (\alpha^2 - \beta^2 + 2\beta - 2)\int V|Q|^2 = \int F_\beta VQ.$$

The negative part of the parenthesized coefficient is bounded independently of α and κ. The inequalities $V \le \epsilon V^2 + (4\epsilon)^{-1}$ and Cauchy–Schwarz with Young's inequality therefore give

$$\|VQ\|_2 + \|\sqrt{V}Q'\|_2 \le C\|F_\beta\|_2.$$

For each fixed κ, V and V' are bounded, so these tests are justified first by compact cutoffs and then by the H^1 bounds. The displayed bounds, unlike those preliminary cutoff justifications, are uniform in κ. The equation now bounds Q'' as well.

In cylindrical coordinates the Cartesian Hessian is bounded pointwise by a fixed multiple of the sum of u_{rr}, $r^{-1}u_r$, $r^{-1}u_{r\theta}$, $r^{-2}u_\theta$, $r^{-2}u_{\theta\theta}$, u_{rz}, $r^{-1}u_{\theta z}$, and u_{zz}. Under the changes just made, their weighted squared norms are controlled by the squared norms of Q'', Q', $\alpha Q'$, αQ, $\alpha^2 Q$, $\sqrt{V}Q'$, $\alpha\sqrt{V}Q$, and VQ, respectively. The additional harmless β terms are bounded by the same list. The weighted forcing satisfies

$$\int_0^\infty |F_\beta|^2\,dt = \int_0^1 r^{3-2\beta}|f_{k,\kappa}(r)|^2\,dr \le \int_0^1 r|f_{k,\kappa}(r)|^2\,dr.$$

Integrate in κ and sum the nonconstant angular modes by Parseval. This proves their contribution, including the additional weighted zeroth-order bound.

Only the Neumann constant angular mode remains. Apply Lemma 8.23 with $G = 0$ to obtain $U_0 \in H^1(\mathbb{R}^3)$ with $-\Delta U_0 = F_0$ and the stated angular-averaging norm bounds. In particular this is an equation against all three-dimensional tests, including tests meeting the axis.

The whole-space Fourier identity then gives $\|U_0\|_{H^2} \le C(\|F_0\|_2 + \|U_0\|_2)$. For any $h \in H^1(\mathbb{R}^2)$, Holder's inequality and Lemma 8.10 give

$$\|r^{-\beta}h\|_{L^2(r<1)} \le C_\beta\|h\|_{H^1(\mathbb{R}^2)}:$$

choose a finite $p > 2/(1-\beta)$, so $r^{-2\beta} \in L^{p/(p-2)}(r<1)$. Apply this slice by slice to the three components of ∇U_0 and integrate in z. This bounds the weighted gradient; $r^{1-\beta} \le 1$ bounds the weighted Hessian by its unweighted norm. It completes the proof for the zero mode. □

Lemma 8.25 (Weighted derivatives imply fractional regularity)**.** *Let Ω be a fixed Lipschitz polyhedron and $\mathcal{S}$ its finite edge and vertex skeleton. Choose a sufficiently small fixed $d_0 > 0$ and set $d(x) = \min\{d_0, \operatorname{dist}(x,\mathcal{S})\}$. For $0 < \beta < 1$, if $h \in L^2(\Omega)$ is H^1 locally away from $\mathcal{S}$ and*

$$d^{-\beta}h \in L^2(\Omega), \qquad d^{1-\beta}\nabla h \in L^2(\Omega),$$

then $h \in H^\beta(\Omega)$, with norm bounded by the sum of these two weighted norms.

Proof. There is $c > 0$, depending only on the finite face geometry, such that $\Omega \cap B(x, cd(x))$ is convex. Indeed a ball sufficiently small compared with its distance to the skeleton meets at most one face; for two intersecting face planes this follows from the elementary distance bound to their intersection line, with constant depending on their angle. Parallel nonadjacent faces have a positive separation, accommodated by d_0. Thus this intersection is either a ball or a half ball.

In the intrinsic fractional seminorm split pairs into $|x - y| < cd(x)/2$ and its complement. For a close pair the line segment stays in the indicated convex set, and the one-dimensional fundamental theorem and Cauchy–Schwarz give

$$|h(x + z) - h(x)|^2 \le |z|^2 \int_0^1 |\nabla h(x + tz)|^2 \, dt.$$

One may first use smooth approximations on compact subsets of the ball or half ball and then pass to the inequality by H^1 convergence. Since d is Lipschitz, $d(x + tz)$ and $d(x)$ are comparable on this region. Change variables $w = x + tz$ and integrate z up to a fixed multiple of $d(w)$; the radial integral of $|z|^{-1-2\beta}$ with the three-dimensional volume factor after the $|z|^2$ gain is $C_\beta d(w)^{2-2\beta}$. This bounds the close pairs by $C \int d^{2-2\beta} |\nabla h|^2$. For the remaining pairs use $|h(x) - h(y)|^2 \le 2|h(x)|^2 + 2|h(y)|^2$. The first term is bounded by $C \int d(x)^{-2\beta} |h(x)|^2$. For the second note that $d(y) \le d(x) + |x - y| \le (1 + 2/c)|x - y|$, so the same estimate with y applies. The intrinsic seminorm is finite with the asserted bound; Lemma 8.16 converts it to the restriction-space H^β norm. □

Lemma 8.26 (The weighted fractional estimate about a straight axis)**.** *Let $\mathcal{L}$ be a straight line in $\mathbb{R}^3$ and put $d(x) = \min\{1, \operatorname{dist}(x, \mathcal{L})\}$. If $0 < \beta < 1$, $h \in L^2(\mathbb{R}^3)$ is H^1 locally off $\mathcal{L}$, and*

$$d^{-\beta} h \in L^2(\mathbb{R}^3), \qquad d^{1-\beta} \nabla h \in L^2(\mathbb{R}^3),$$

then $h \in H^\beta(\mathbb{R}^3)$, with norm bounded by the sum of these two weighted norms. No boundary extension is involved in this statement; in particular it applies to compactly supported axisymmetric fields extended by zero through collars of their artificial boundaries.

Proof. Split the whole-space fractional double integral into the regions $|x - y| < d(x)/2$ and $|x - y| \ge d(x)/2$. A segment in the first region stays a positive distance from the axis, so the H^1 line estimate, obtained by smooth approximation on that ball, gives

$$|h(x + z) - h(x)|^2 \le |z|^2 \int_0^1 |\nabla h(x + tz)|^2 \, dt.$$

The distance function is 1-Lipschitz. Thus $d(x + tz)$ and $d(x)$ are comparable there. With $w = x + tz$, the allowed z lie in a ball of radius at most a fixed multiple of $d(w)$. Tonelli and translation of the x variable bound the near-pair integral by

$$C \int |\nabla h(w)|^2 \int_{|z| < Cd(w)} |z|^{-1-2\beta} \, dz \, dw = C_\beta \int d(w)^{2-2\beta} |\nabla h(w)|^2 \, dw.$$

For far pairs use $|h(x) - h(y)|^2 \le 2|h(x)|^2 + 2|h(y)|^2$. The first term contributes at most $C \int d(x)^{-2\beta} |h(x)|^2$. For the second term, the distance inequality gives $d(y) \le d(x) + |x - y| \le 3|x - y|$ on this region, so its integral has the same bound with y in place of x. The whole-space difference identity (8.8), which is valid for L^2 functions with either side possibly infinite by Tonelli and Plancherel, now shows $h \in H^\beta$. The L^2 part is controlled since $d \le 1$. The proof never crosses the axis on a short segment, and it does not require a globally defined weak gradient on the axis itself. □

Proposition 8.27 (A homogeneous scalar polyhedral shift)**.** *On any fixed bounded Lipschitz polyhedron there is $\beta > 1/2$ such that every $u \in H^1(\Omega)$ satisfying $-\Delta u = f \in L^2(\Omega)$ and either homogeneous Dirichlet or homogeneous Neumann condition obeys*

$$\nabla u \in H^\beta(\Omega)^3, \qquad \|\nabla u\|_{H^\beta} \le C(\|f\|_2 + \|u\|_{H^1}).$$

Proof. Choose $\beta = 1/2 + \delta < 1$ using Lemma 8.20. We specify the localization and scaling needed to combine the model estimates. Near each vertex the domain is exactly a straight cone. Multiply u by a radial cutoff supported in that conical neighborhood and equal to one on a smaller one. This preserves either homogeneous condition: the radial direction is tangent to every face of a cone, so the cutoff has zero normal derivative on them. Its equation has right side $\chi f - 2\nabla\chi \cdot \nabla u - u\Delta\chi$, bounded by $C(\|f\|_2 + \|u\|_{H^1})$. Lemma 8.22 gives a vertex constant c in the Neumann case and controls

$$\|r^{-1-\beta}(u-c)\|_2 + \|r^{-\beta}\nabla u\|_2 \le C(\|f\|_2 + \|u\|_{H^1}) \tag{8.27}$$

in the smaller vertex neighborhood; for Dirichlet take $c = 0$. Here the radial and angular first-derivative terms in (8.24) combine to the ordinary gradient.

Divide that smaller neighborhood into dyadic annuli $A_j = \{2^{-j-1} < r < 2^{-j}\} \cap \Omega$. The reference enlarged annulus $\{1/4 < r < 2\} \cap C_S$ has a finite cover by interior balls, face half balls, and dihedral cylinders about its edge segments. Choose cutoffs equal to one on a smaller cover and supported inside these patches. For Neumann conditions they can be chosen with zero normal derivative on all faces they meet: in a face patch use a cutoff even in the normal coordinate; in an edge patch use cutoffs depending on distance to the edge and on its longitudinal coordinate. Such an edge patch meets only its two adjacent faces. Both cutoff factors have zero normal derivative on those faces. Radial annulus cutoffs also have this property. Finite sums and division by a positive sum preserve it if a partition is desired.

Rescale these fixed patches and cutoffs to each A_j, with scale $R_j = 2^{-j}$. Apply the dihedral estimate to each localized $u - c$ in an edge cylinder, and the flat-face or interior H^2 estimate in the other patches. The flat-face estimate is Lemma 8.5 with constant coefficients; interior balls follow from the whole-space Fourier estimate and a cutoff. The scaled right-hand side is bounded by

$$C\left(R_j^{1-\beta}\|f\|_{L^2(A_j')} + R_j^{-\beta}\|\nabla u\|_{L^2(A_j')} + R_j^{-1-\beta}\|u-c\|_{L^2(A_j')}\right),$$

where A_j' is a fixed enlargement of A_j. To verify every power, an L^2 norm scales by $R_j^{3/2}$, a first derivative by R_j^{-1}, a second derivative by R_j^{-2}, and the edge weight by its displayed power. The derivatives of a rescaled cutoff cost R_j^{-1} and R_j^{-2}; these are exactly the last two terms. The left side controls on A_j the quantities $d^{-\beta}\nabla u$ and $d^{1-\beta}D^2u$: in an edge patch d is comparable to the transverse distance, and outside these patches it is comparable to R_j. These comparisons involve only the fixed reference cover.

Square and sum over j. The enlarged annuli have bounded overlap, r is comparable to R_j on each, and (8.27) bounds the last two terms. The first is bounded by $\|f\|_2$ because $1 - \beta > 0$. This proves both weighted estimates in all vertex neighborhoods. Away from the vertices a finite family of fixed dihedral cylinders covers the remaining edge segments; the same dihedral estimate applies, and its cutoff terms are bounded by the unweighted H^1 norm. The remaining face and interior pieces use the ordinary local H^2 estimates. We have therefore proved globally

$$\|d^{-\beta}\nabla u\|_2 + \|d^{1-\beta}D^2u\|_2 \le C(\|f\|_2 + \|u\|_{H^1}).$$

Apply Lemma 8.25 to each component of ∇u to obtain the result. □

Remark 8.28 (Why a further boundary-data argument is required). The homogeneous scalar shift does not by itself prove the Maxwell embedding. The boundary data arising in the two regular decompositions are inhomogeneous. The H^2 extension above and the variational Neumann shift below supply those two distinct steps. In particular no assertion that a positive angular gap automatically handles arbitrary boundary data is used.

8.9 The variational Neumann shift and the full polyhedral result

Lemma 8.29 (Radial estimates for first-order forcing). *Let the parameters β, ν_* be as in Lemma 8.21. For $-v'' + \nu^2 v = -h' + b$, $v(0) = 0$, one has, uniformly for $\nu \geq \nu_* > \beta$,*

$$\|e^{\beta t} v'\|_2 + \nu \|e^{\beta t} v\|_2 \leq C\big(\|e^{\beta t} h\|_2 + \nu^{-1} \|e^{\beta t} b\|_2\big). \tag{8.28}$$

For fixed $0 < \nu < \beta$, the same bound, with an allowed parameter-dependent constant, holds after subtracting $Ae^{-\nu t}$, where

$$A = \int_0^\infty \cosh(\nu s) h(s)\, ds + \nu^{-1} \int_0^\infty \sinh(\nu s) b(s)\, ds.$$

The formula is used for data vanishing near $t = 0$; by continuity the inequality also applies whenever the corresponding weak equation and zero boundary condition are defined.

Proof. For smooth data supported away from zero, integration by parts in the Green representation gives $v(t) = \int \partial_s G_\nu(t,s) h(s)\, ds + \int G_\nu(t,s) b(s)\, ds$. The weighted L^1 bounds of G_ν and its first derivatives were computed in Lemma 8.21. Moreover $\partial_t \partial_s G_\nu$ is a unit delta mass at $t = s$ plus kernels bounded by $C\nu e^{-\nu|t-s|}$ and $C\nu e^{-\nu(t+s)}$. Their weighted operator norms are bounded by $C\nu^2/(\nu^2 - \beta^2)$. These explicit bounds give (8.28); the coefficient of the b term is C/ν because $\nu/(\nu^2 - \beta^2) \leq C/\nu$. For $\beta > \nu$, integrate the first-order forcing by parts in (8.23). This gives the displayed A. Subtracting its decaying exponential leaves the one-sided tail kernels of the proof of Lemma 8.21, and their s derivatives. Their weighted masses are bounded by constants times $(\beta - \nu)^{-1}$ and $(\beta + \nu)^{-1}$; differentiating once in t adds only a bounded delta term and the same exponentials. This proves the asserted first-derivative bound. Weighted L^2 approximation passes all formulas and estimates to the stated data. □

Lemma 8.30 (Inhomogeneous variational cone and edge estimates). *For a conical model with a Lipschitz apex, as supplied by the present polyhedra, or for the dihedral model, replace the scalar equation by*

$$\int \nabla u \cdot \nabla \phi = \int G \cdot \nabla \phi + \int F \phi, \tag{8.29}$$

with $G \in H^1$ and $F \in L^2$, allowing all H^1 tests at the lateral faces. Assume the model functions and data are localized away from their artificial outer boundaries. For $\beta = 1/2 + \delta$ obeying the gaps in the homogeneous estimates, the cone has a constant c such that

$$\|r^{-1-\beta}(u - c)\|_2 + \|r^{-\beta} \nabla u\|_2 \leq C(\|G\|_{H^1} + \|F\|_2). \tag{8.30}$$

On a dihedron,

$$\|r^{-\beta} \nabla u\|_2 + \|r^{1-\beta} D^2 u\|_2 \leq C(\|G\|_{H^1} + \|F\|_2 + \|u\|_{H^1}). \tag{8.31}$$

The constants depend on the fixed model geometry and the positive gaps, not on a subsequent dilation.

Proof. In the cone use $t = -\log r$ and $u = e^{t/2} v$ as before. The transformed weak equation is

$$\int (v_t \phi_t + \nabla_S v \cdot \nabla_S \phi + \tfrac{1}{4} v \phi) = \int (H_r \phi_t + H_S \cdot \nabla_S \phi + H_0 \phi),$$

where, writing the radial and spherical components of G,

$$H_r = -r^{3/2}G_r, \qquad H_S = r^{3/2}G_S, \qquad H_0 = -\tfrac{1}{2}r^{3/2}G_r + r^{5/2}F.$$

The cross term from $(v_t + v/2)(\phi_t + \phi/2)$ integrates to zero; tests vanish at the artificial outer endpoint. In the angular eigenbasis the right side has coefficients $-h_j' + b_j + (H_0)_j$, where $h_j = (H_r, Y_j)$ and $b_j = (H_S, \nabla_S Y_j)$. The vectors $\nabla_S Y_j/\sqrt{\mu_j}$ for positive μ_j are orthonormal, so Bessel's inequality gives $\sum_{\mu_j>0} |b_j|^2/\mu_j \le \|H_S\|_2^2$. The zero mode has $b_0 = 0$. Apply Lemma 8.29 to every positive mode and its constant-subtraction case to the zero mode. Parseval and this Bessel bound show that $e^{\beta t}v_t$ and $e^{\beta t}(1-\Delta_S)^{1/2}v$ are controlled after subtracting the constant mode, by

$$C\big(\|r^{-\beta}G\|_{L^2(C_S)} + \|r^{1-\beta}F\|_{L^2(C_S)}\big).$$

The notation for the angular square root here means precisely the squared coefficient sum with weights $1+\mu_j$, not an additional spectral theorem. The scalar constant is bounded by the same input using the convergent exponential integrals in the first-order Green formula. Conversion back to r gives (8.30). Indeed $r^{-\beta}G$ is bounded by $C\|G\|_{H^1}$ when $\beta < 1$: choose $6/(3-2\beta) < p < 6$, use the H^1 extension and Lemma 8.10 in dimension three, and apply Holder to $r^{-2\beta} \in L^{p/(p-2)}$ on the bounded cone. The F weight is bounded.

For an edge, the change $t = -\log r$ needs no factor in u. Its transformed form is

$$\int (u_t\phi_t + u_\theta\phi_\theta + r^2 u_z\phi_z) = \int (-rG_r\phi_t + rG_\theta\phi_\theta + r^2G_z\phi_z + r^2F\phi).$$

Expand in angular cosine modes and transform in z. For the positive angular modes test with the weighted positive-mode part $e^{2\beta t}u_+$, using bounded weight truncations as in (8.26). The left side controls $\|Q_t\|_2^2 + \|Q_\theta\|_2^2 + \||\kappa| rQ\|_2^2 + \|Q\|_2^2$, because $\alpha^2 - \beta^2 \ge (1-\beta^2/\alpha_1^2)\alpha^2$. On the right, the radial term pairs $e^{\beta t}rG_r$ with $Q_t + \beta Q$; the angular term pairs $e^{\beta t}rG_\theta$ with Q_θ; the longitudinal term pairs $e^{\beta t}rG_z$ with $i\kappa rQ$; and the final term pairs $e^{\beta t}r^2F$ with Q. Taking real parts for the complex Fourier coefficients, Cauchy–Schwarz and Young's inequality absorb all four test norms. Consequently

$$\|r^{-1-\beta}u_+\|_2 + \|r^{-\beta}\nabla u_+\|_2 \le C(\|r^{-\beta}G\|_2 + \|r^{1-\beta}F\|_2).$$

We detail the zero angular mode because treating it as a positive mode would be incorrect. Let bars denote angular averages of u, F, G_r, G_z. Extend $\bar u$ as an axisymmetric function U to the full cylinder and by zero outside its artificial supports. Set $\mathcal{G} = \overline{G_r}e_r + \overline{G_z}e_z$ and $\mathcal{F} = \bar F$, also axisymmetric. Lemma 8.23 gives $U \in H^1(\mathbb{R}^3)$ and $-\Delta U = \mathcal{F} - \operatorname{div}\mathcal{G}$ globally, including the arbitrary-test justification and the absence of any distribution supported on the axis.

Fix $\beta < \beta' < 1$. Angular averaging is L^2 bounded by Jensen's inequality. Away from the axis, the radial and longitudinal derivatives of $\mathcal{G}$ are bounded in L^2 by those of G, while the angular derivative costs at most $|\overline{G_r}|/r$. For every $0 < a < 1$ one has $\|r^{-a}G\|_2 \le C_a\|G\|_{H^1}$ on the model: on each z slice use a bounded H^1 extension from the fixed two-dimensional sector, choose $p > 2/(1-a)$, apply its two-dimensional $H^1 \to L^p$ bound, and use Holder as in the zero-mode proof above. Fubini then integrates the estimate in z. It follows that

$$\|r^{-\beta'}\mathcal{G}\|_2 + \|r^{1-\beta'}\nabla\mathcal{G}\|_2 \le C\|G\|_{H^1}.$$

Lemma 8.26, applied componentwise, therefore gives $\mathcal{G} \in H^{\beta'}(\mathbb{R}^3)^3$,

with this bound; compact artificial supports handle the part away from the axis.

Taking the whole-space Fourier transform of the equation for U gives

$$\widehat{\nabla U} = \frac{i\xi}{|\xi|^2}\widehat{\mathcal{F}} + \frac{\xi(\xi\cdot\widehat{\mathcal{G}})}{|\xi|^2}.$$

The second multiplier is bounded at every frequency. For the first, frequencies $|\xi| \geq 1$ gain one derivative. On $|\xi| < 1$ use the already known L^2 norm of ∇U, bounded by $C\|u\|_{H^1}$ by angular averaging; no low-frequency inverse bound for an arbitrary L^2 forcing is assumed. Therefore

$$\|\nabla U\|_{H^{\beta'}} \leq C(\|G\|_{H^1} + \|F\|_2 + \|u\|_{H^1}).$$

Finally choose $2/(1-\beta) < p < 2/(1-\beta')$. The two-dimensional $H^{\beta'} \to L^p$ estimate and Holder with $r^{-2\beta}$ show, slice by slice, $\|r^{-\beta}\nabla U\|_2 \leq C\|\nabla U\|_{H^{\beta'}(\mathbb{R}^3)}$. The slice norms are controlled because tangential Fourier weights $(1+|\xi'|^2)^{\beta'}$ are bounded by the full three-dimensional weights. This proves the required zero-mode weighted-gradient estimate.

To obtain the weighted Hessian of the full edge solution, cover every annulus $r \asymp \rho$ by balls and flat half balls of radius comparable to ρ; the axial intervals are also of length comparable to ρ. The cover and its fixed enlargements have bounded overlap. Subtract the mean of u on each larger ball or half ball and rescale Lemma 8.5. Poincare's inequality absorbs the zeroth-order term and gives locally

$$\|D^2 u\|_{L^2(\text{smaller patch})} \leq C\big(\rho^{-1}\|\nabla u\|_2 + \|F\|_2 + \|\nabla G\|_2 + \rho^{-1}\|G\|_2\big).$$

Multiply by $\rho^{1-\beta}$, square, and sum over patches and dyadic annuli. The weighted gradient was just bounded, $r^{-\beta}G$ is bounded by $\|G\|_{H^1}$, and the weights on F and ∇G are bounded. This proves (8.31). □

Theorem 8.31 (Polyhedral variational Neumann shift)**.** *On a fixed bounded Lipschitz polyhedron there is $\beta > 1/2$ such that every $u \in H^1(\Omega)$ satisfying (8.29) for all $\phi \in H^1(\Omega)$, with $G \in H^1(\Omega)^3$ and $F \in L^2(\Omega)$, obeys*

$$\|\nabla u\|_{H^\beta(\Omega)} \leq C(\|u\|_{H^1} + \|G\|_{H^1} + \|F\|_2).$$

Proof. Use the vertex and edge localization of Proposition 8.27, choosing cutoffs with zero normal derivative on all faces they meet. For a cutoff χ, the localized equation has data

$$G_\chi = \chi G, \qquad F_\chi = \chi F + G\cdot\nabla\chi - 2\nabla u\cdot\nabla\chi - u\Delta\chi.$$

This identity follows by using $\chi\phi$ as a test and integrating the extra $u\nabla\chi\cdot\nabla\phi$ term once; its boundary term is zero because $\partial_n\chi = 0$. The data have the asserted H^1 and L^2 regularities. The conical part of Lemma 8.30 controls the vertex-weighted gradient and $u - c$. On each dyadic annulus the inhomogeneous edge and flat-face estimates give, in addition to the three terms displayed in the homogeneous proof,

$$C\big(R_j^{1-\beta}\|\nabla G\|_{L^2(A_j')} + R_j^{-\beta}\|G\|_{L^2(A_j')}\big).$$

These powers follow either from the weak rescaling $G \mapsto R_j G$, $F \mapsto R_j^2 F$, or from the local estimate in the preceding proof. Their squared sum is bounded: the derivative weight is bounded and the vertex weight $r^{-\beta}G$ is controlled by the three-dimensional Holder estimate already proved. The corresponding edge weight away from vertices uses the two-dimensional slice estimate. All other terms are controlled exactly as in the homogeneous annulus argument. Hence

$$\|d^{-\beta}\nabla u\|_2 + \|d^{1-\beta}D^2 u\|_2 \leq C(\|u\|_{H^1} + \|G\|_{H^1} + \|F\|_2).$$

Lemma 8.25 finishes the proof. □

Theorem 8.32 (The full polyhedral embedding; original Proposition 3.7)**.** *For every bounded Lipschitz polyhedron Ω there is a number $s > 1/2$ such that*

$$X_T \hookrightarrow H^s(\Omega)^3, \qquad X_N \hookrightarrow H^s(\Omega)^3$$

continuously.

Proof. For $v \in X_N$ use the global decomposition (8.3). On every complementary piece D_a of the enclosing ball, $\nabla q = -w \in H^1(D_a)^3$, so $q \in H^2(D_a)$ with a bound by $\|v\|_X$. The H^2 extension of Lemma 8.19, multiplied by the fixed interface cutoffs and summed over a, gives $Q \in H^2(\Omega)$ with the same trace as q and $\|Q\|_{H^2} \le C\|v\|_X$. Thus $r = q - Q \in H^1_0(\Omega)$ and $\Delta r = \operatorname{div} v - \Delta Q \in L^2$. Proposition 8.27 gives $\nabla r \in H^s$ for some $s > 1/2$. Since $v = w + \nabla Q + \nabla r$ and the first two terms are H^1, this proves the X_N assertion.

For $v \in X_T$ use the local decomposition (8.2). The local pieces can be chosen polyhedral: near a boundary graph take a small rectangular horizontal base, a flat upper cap strictly above the graph, and vertical sides. The graph is piecewise affine with finitely many pieces; this capped graph prism is a simply connected Lipschitz polyhedron. Interior pieces may be cubes. Choose cutoffs χ_j supported away from their artificial boundaries and summing to one on $\overline{\Omega}$ as before. On such a piece write $v = w + \nabla q_j$ and put $u_j = \chi_j q_j$. Testing the normal-trace identity with $\chi_j \phi$ shows that u_j satisfies the variational equation on the entire local piece, with

$$G_j = -\chi_j w + q_j \nabla \chi_j, \qquad F_j = -w \cdot \nabla \chi_j - \nabla q_j \cdot \nabla \chi_j - \chi_j \operatorname{div} v.$$

Here $G_j \in H^1$, $F_j \in L^2$, and both norms, together with $\|u_j\|_{H^1}$, are bounded by $C\|v\|_X$. Theorem 8.31 therefore gives $\nabla u_j \in H^{s_j}$ with $s_j > 1/2$. These gradients vanish in a neighborhood of the artificial boundaries, so extending by zero past those boundaries preserves their fractional norms of orders below one: the cross-boundary double integrals are bounded by their L^2 norms and the fixed support margin, exactly as in Lemma 8.16. Finally,

$$v = w + \sum_j \nabla u_j - \sum_j q_j \nabla \chi_j.$$

The last terms, extended past their artificial boundaries, are H^1. Choose s smaller than one and than the finitely many s_j and the Dirichlet exponent already obtained. This proves both embeddings with a common $s > 1/2$ and the required norm bounds. □

9 Three orthogonal scalar–vector decompositions

Use the spaces Θ, Θ_0, the broken gradient ∇_b, the fluxes C_j, F_i, and the normalized harmonic bases $h_j^T = \nabla_b q_j^T$ and $h_i^N = \nabla q_i^N$ constructed above. Constants in scalar potentials are fixed by zero mean when the scalar space permits constants. The maps R_T and R_N denote the unique potentials in the two main theorems.

Theorem 9.1 (The three decompositions in Section 3.5)**.** *Every $u \in L^2(\Omega)^3$ has each of the following three uniquely normalized decompositions.*

(i) There are $q_N \in H^1(\Omega)/\mathbb{R}$ and $\psi_T \in X_T$ such that

$$u = \nabla q_N + \operatorname{curl} \psi_T, \quad \operatorname{div} \psi_T = 0, \quad C_j(\psi_T) = 0, \quad (\operatorname{curl} \psi_T) \cdot n = 0.$$

(ii) There are $\widetilde{q}_N \in \Theta/\mathbb{R}$ and $\psi_N \in X_N$ such that

$$u = \nabla_b \widetilde{q}_N + \operatorname{curl} \psi_N, \quad \operatorname{div} \psi_N = 0, \quad F_i(\psi_N) = 0 \quad (0 \le i \le I).$$

(iii) There are $q_D \in \Theta_0$ and $\widehat{\psi}_T \in X_T$ such that

$$u = \nabla q_D + \operatorname{curl} \widehat{\psi}_T, \quad \operatorname{div} \widehat{\psi}_T = 0, \quad C_j(\widehat{\psi}_T) = 0.$$

In each decomposition the two summands are orthogonal in L^2. Every scalar map is bounded into its indicated H^1 space, and every vector-potential map is bounded into X.

Proof. Solve respectively the three Riesz problems

$$\begin{aligned}(\nabla q_N, \nabla r) &= (u, \nabla r) && (r \in H^1(\Omega)),\\ (\nabla_b \widetilde{q}_N, \nabla_b r) &= (u, \nabla_b r) && (r \in \Theta),\\ (\nabla q_D, \nabla r) &= (u, \nabla r) && (r \in \Theta_0).\end{aligned}$$

The gradient norms make these quotient or anchored scalar spaces Hilbert spaces, as proved earlier. Riesz therefore gives unique normalized solutions, whose gradient norms are at most $\|u\|_2$. The first residual is divergence-free with zero normal trace, so it lies in the data space of R_T; apply R_T to it. The second residual is orthogonal to $Z = \nabla_b\Theta$, hence lies in the data space of R_N; apply R_N. The third residual is orthogonal to $Z_0 = \nabla\Theta_0$, hence lies in the data space of R_T; apply R_T. The annihilator identifications give all stated conditions. Orthogonality follows by taking the scalar solution itself as test function. Poincaré and the bounds for R_T, R_N give the norm estimates. The scalar orthogonal projections and the normalized potential maps are unique, which gives uniqueness of each decomposition. □

Proposition 9.2 (The finite-dimensional correction formulas)**.** *Let $q_D^0 \in H_0^1(\Omega)$ solve $(\nabla q_D^0, \nabla r) = (u, \nabla r)$ for $r \in H_0^1(\Omega)$. Up to the chosen additive normalization in the first formula,*

$$\widetilde{q}_N = q_N + \sum_{j=1}^{J} C_j(u - \nabla q_N) q_j^T, \tag{9.1}$$

$$q_D = q_D^0 + \sum_{i=1}^{I} F_i(u - \nabla q_D^0) q_i^N. \tag{9.2}$$

All fluxes in these formulas exist: the first residual is divergence-free with zero normal trace, and the second is divergence-free.

Proof. Set $v = u - \nabla q_N$. The field $h = \sum_j C_j(v) h_j^T$ is in K_T, and $v - h$ has zero divergence, zero normal trace, and zero cut fluxes. It is therefore orthogonal to Z. Since $\nabla q_N + h \in Z$, it is the orthogonal projection of u onto Z, proving (9.1). For the second identity set $v = u - \nabla q_D^0$. The field $v - \sum_i F_i(v) h_i^N$ is divergence-free and all its inner boundary fluxes vanish; its outer flux also vanishes by total flux zero. It is orthogonal to Z_0. The removed gradient belongs to Z_0, because $q_D^0 \in \Theta_0$ and $q_i^N \in \Theta_0$. Projection uniqueness proves (9.2). □

10 A full Dirichlet vector potential on a $C^{1,1}$ domain

This section assumes $\partial\Omega$ is $C^{1,1}$. We use the already proved $C^{1,1}$ Maxwell estimate (Theorem 8.7) $\|a\|_{H^1} \le C\|a\|_X$ for $a \in X_N$. No fourth-order regularity theorem is needed below: the scalar fourth-order problem is solved directly in an H^2 energy space.

10.1 The boundary lift and the clamped scalar space

Lemma 10.1 (An H^2 lift of the normal derivative)**.** *For every $g \in H^{1/2}(\partial\Omega)$ there is $Lg \in H^2(\Omega)$ with trace zero and normal derivative trace g, and*

$$\|Lg\|_{H^2(\Omega)} \le C\|g\|_{H^{1/2}(\partial\Omega)}.$$

The map L can be chosen linear.

Proof. We first give the flat construction. For $g \in H^{1/2}(\mathbb{R}^2)$, put $a(\xi) = (1 + |\xi|^2)^{1/2}$ and define on $t > 0$

$$\widehat{S}(\xi, t) = te^{-a(\xi)t}\widehat{g}(\xi),$$

where the Fourier transform is in the two tangential variables. The one-dimensional integrals $\int_0^\infty t^m e^{-2at}\,dt = m!/(2a)^{m+1}$ show, for every derivative of order at most two, that its squared L^2 norm is at most a constant times $\int a(\xi)|\widehat{g}(\xi)|^2\,d\xi$. For example two tangential derivatives contribute at most $a^4 \int t^2 e^{-2at}\,dt \le Ca$; two normal derivatives contribute $\int(-2a+a^2t)^2e^{-2at}\,dt = Ca$. The mixed derivatives satisfy the same bound. Thus $S \in H^2$ with the asserted bound. Its trace is zero and its normal coordinate derivative has trace g, since the multiplier $(1 - at)e^{-at}$ tends to one and is uniformly bounded; dominated convergence gives the trace limit in $H^{1/2}$. The assertion first follows for smooth g and then for all g by the Fourier bound and completion.

For localization, multiplication by a bounded Lipschitz scalar function is bounded on $H^{1/2}(\mathbb{R}^2)$. One direct verification uses the fractional seminorm and the inequality

$$|b(x)g(x) - b(y)g(y)|^2 \le 2\|b\|_\infty^2|g(x) - g(y)|^2 + 2|b(x) - b(y)|^2|g(y)|^2.$$

In the second term use $|b(x)-b(y)| \le \min(L|x-y|, 2\|b\|_\infty)$; the resulting integral against $|x-y|^{-3}$ is finite in dimension two. The fractional-seminorm and Fourier definitions agree, up to a fixed constant: expand the square of $g(x + h) - g(x)$ by Plancherel and change variables $h \mapsto |\xi|^{-1}h$ in its radial integral.

In a $C^{1,1}$ boundary chart write $\Omega = \{(y, x_3) : x_3 > \varphi(y)\}$ and $F(y, t) = (y, \varphi(y)+t)$. Composition with F and F^{-1} is bounded on H^2. For smooth functions this follows from the chain rule: first derivatives of F and second weak derivatives of F are bounded, its Jacobian is one, and the second derivative formula has only a second derivative of the function times two bounded first derivatives of F, plus a first derivative times a bounded second derivative of F. Approximation by smooth functions and weak convergence prove the same formula and estimate for H^2 functions. Since the desired trace is zero, the tangential derivative of the trace is zero. On the boundary the transformed gradient is therefore $(-\nabla\varphi, 1)\partial_t S$, so the exterior normal derivative equals $-\sqrt{1 + |\nabla\varphi|^2}\,\partial_t S$. Apply the flat construction to $-g \circ F(\cdot, 0)/\sqrt{1 + |\nabla\varphi|^2}$, with local cutoffs. The multiplier is Lipschitz, so the preceding bound applies. Choose a finite partition of unity on the boundary, subordinate to these charts. Each localized datum is supported a positive distance δ from the chart edge and can be extended by zero in the tangential plane: the additional fractional-seminorm cross integral is bounded by $C\delta^{-1}\|g\|_2^2$. Multiply the resulting S by a cutoff supported in the chart, equal to one near that datum's support at $t = 0$. This preserves zero value trace and the required normal derivative trace. Transport these finitely many functions to Ω and sum them. The partition sums to one on the boundary, so the sum has normal derivative g. All operations are linear and bounded, which proves the lemma. □

Define

$$\Theta_{00} = \{q \in H^2(\Omega) : q|_{\Gamma_0} = 0, q|_{\Gamma_i} = c_i(q) \text{ constant}, \ \partial_n q = 0 \text{ on } \partial\Omega\}.$$

The constants c_i use the same boundary convention as Θ_0. Let $H_0^2(\Omega)$ be the closure of $C_c^\infty(\Omega)$ in H^2.

Lemma 10.2 (Clamped traces and coercivity)**.** *On a $C^{1,1}$ domain,*

$$H_0^2(\Omega) = \{q \in H^2(\Omega) : q|_{\partial\Omega} = 0, \ \partial_n q = 0\}.$$

For $q \in \Theta_{00}$ one has $\nabla q \in H_0^1(\Omega)^3$ and

$$\|D^2 q\|_2^2 = \|\Delta q\|_2^2, \qquad \|q\|_{H^2} + \sum_{i=1}^{I} |c_i(q)| \le C\|\Delta q\|_2. \tag{10.1}$$

Consequently Θ_{00} and $H_0^2(\Omega)$ are Hilbert spaces for $(q, r) \mapsto (\Delta q, \Delta r)$.

Proof. In a flattened chart the tangential derivatives of an H^2 function's trace are the traces of its tangential derivatives. To verify this without assuming boundary smooth approximation, integrate the tangential weak-derivative identity on almost every slice $t > 0$ against a smooth compactly supported tangential test function. Let t tend to zero along slices where the H^1 trace limits hold, both for the function and for its first derivatives. Their L^2 limits give exactly the boundary weak-derivative identity. Constant value trace and zero normal derivative therefore give zero trace of the entire gradient. Thus $\nabla q \in H^1_0$. For q with zero value trace as well, extend it by zero across the boundary. The value and first derivative traces match zero, so applying the H^1 gluing lemma to q and then to each first derivative shows that its zero extension is H^2. Use a finite smooth partition supported in boundary graph charts, with one additional interior part. Translate each boundary part a small distance in its inward vertical direction. Its support is now a positive distance from the graph, since the graph is Lipschitz; the chart can first be shrunk to leave room at its other sides. Mollify in the ambient physical coordinates with a smaller radius. The resulting functions are in $C^\infty_c(\Omega)$ and converge in H^2: translations and convolution converge in L^2 for the zero extension and every one of its weak derivatives of order at most two. The interior part is treated by ordinary convolution. This proves the difficult inclusion in the displayed characterization; the converse follows from trace continuity.

For any $w \in H^1_0(\Omega)^3$, extension by zero and the Fourier identity $|\xi|^2|\widehat{w}|^2 = |\xi \times \widehat{w}|^2 + |\xi \cdot \widehat{w}|^2$ give

$$\|\nabla w\|_2^2 = \|\operatorname{curl} w\|_2^2 + \|\operatorname{div} w\|_2^2.$$

Apply this with $w = \nabla q$. Its curl is zero, proving the first part of (10.1). Poincaré applied to each component of $\nabla q \in H^1_0$, followed by the anchored Poincaré inequality for $q|_{\Gamma_0} = 0$, controls the remaining H^2 terms. Trace continuity controls each $c_i(q)$. Both spaces are closed under their defining trace conditions, so norm equivalence with H^2 proves completeness. □

10.2 Fluxes of an L^2 harmonic scalar

One must not assume that an L^2 harmonic function has an L^2 gradient up to a Lipschitz or $C^{1,1}$ boundary. The following transposition definition supplies exactly the component fluxes needed for the third potential theorem. Choose smooth η_i equal to one near Γ_i and zero near all other boundary components; in particular $\partial_n \eta_i = 0$. For $b \in L^2(\Omega)$ with $\Delta b = 0$ distributionally, set

$$N_i(b) = -\int_\Omega b \Delta \eta_i. \tag{10.2}$$

The expression denoted $\langle \partial_n b, 1 \rangle_{\Gamma_i}$ below means this quantity.

Lemma 10.3 (Transposition flux identity)**.** *The quantity (10.2) is independent of the cutoff, is bounded by $C_i\|b\|_2$, and satisfies $\sum_{i=0}^{I} N_i(b) = 0$. If $b \in H^1$ is harmonic, it agrees with the usual weak normal flux of ∇b. For $q \in \Theta_{00}$ with $\Delta^2 q = 0$,*

$$(\Delta q, \Delta r) = -\sum_{i=1}^{I} c_i(r) N_i(\Delta q) \qquad (r \in \Theta_{00}). \tag{10.3}$$

Proof. Two admissible cutoffs differ by a function in H^2_0, because both the value and normal derivative traces of their difference are zero. Approximate that difference in H^2 by compactly supported smooth functions. The distributional identity $\Delta b = 0$ gives $(b, \Delta r) = 0$ for each approximant and hence for their limit. This proves independence; boundedness follows from Cauchy–Schwarz. The sum of the cutoffs and the constant function one have the same value and normal derivative traces, so the same argument gives total flux zero. If $b \in H^1$, integration by parts against a cutoff whose gradient is compactly supported gives $-(b, \Delta\eta_i) = (\nabla b, \nabla \eta_i)$, the usual flux. Finally decompose $r = r_0 + \sum_{i=1}^{I} c_i(r)\eta_i$ with $r_0 \in H^2_0$ and apply the same density argument to $b = \Delta q$. □

Proposition 10.4 (The kernel in Proposition 3.21)**.** *Let*

$$K_0 = \{w \in H_0^1(\Omega)^3 : \operatorname{curl} w = 0, \Delta \operatorname{div} w = 0\}.$$

For each $1 \le i \le I$ *there is a unique* $q_i^0 \in \Theta_{00}$ *such that*

$$(\Delta q_i^0, \Delta r) = -c_i(r) \qquad (r \in \Theta_{00}). \tag{10.4}$$

It satisfies $\Delta^2 q_i^0 = 0$ *and* $N_k(\Delta q_i^0) = \delta_{ik}$ *for* $1 \le k \le I$*, with* $N_0(\Delta q_i^0) = -1$*. The gradients* ∇q_i^0 *form a basis of* K_0*; thus* $\dim K_0 = I$*.*

Proof. The coercive Hilbert structure in Lemma 10.2 and continuity of c_i give (10.4) by Riesz. Testing with compactly supported smooth r proves $\Delta^2 q_i^0 = 0$. Testing with η_k and using (10.2) gives the specified fluxes; total flux gives the outer value. The gradients belong to K_0 by Lemma 10.2, and the flux identities prove independence.

If $w \in K_0$, the already proved minimal-curl kernel identification gives $w = \nabla q$ with $q \in \Theta_0$. Since $w \in H^1$, all second weak derivatives of q are in L^2; hence $q \in H^2$. Its gradient has zero trace, so $q \in \Theta_{00}$. Also $\Delta^2 q = \Delta \operatorname{div} w = 0$. Subtract from w the combination $\sum_i N_i(\operatorname{div} w)\nabla q_i^0$. The remainder is $\nabla q'$ for a biharmonic $q' \in \Theta_{00}$ with all inner $N_i(\Delta q')$ zero. Taking $r = q'$ in (10.3) gives $\|\Delta q'\|_2^2 = 0$. Coercivity then gives $q' = 0$. This proves spanning. □

Theorem 10.5 (The full Dirichlet potential, Theorem 3.20)**.** *On a bounded connected* $C^{1,1}$ *domain with the stated cut system, a field* $u \in H(\operatorname{div}; \Omega)$ *satisfies*

$$\operatorname{div} u = 0, \qquad u \cdot n = 0, \qquad C_j(u) = 0 \ (1 \le j \le J)$$

if and only if there is a unique $\psi \in H_0^1(\Omega)^3$ *such that*

$$\operatorname{curl} \psi = u, \qquad \Delta \operatorname{div} \psi = 0, \qquad N_i(\operatorname{div} \psi) = 0 \ (0 \le i \le I).$$

The potential depends linearly on u *and satisfies* $\|\psi\|_{H^1} \le C\|u\|_2$*.*

Proof. Necessity follows from the necessity part of the normal-potential theorem: $H_0^1 \subset \mathcal{D}(C_{\min})$, and its curl is orthogonal to Z, which is precisely the displayed compatibility. For sufficiency take $a = R_N u$. It has zero tangential trace and zero divergence; the $C^{1,1}$ Maxwell estimate gives $\|a\|_{H^1} \le C\|u\|_2$. Since the unit normal is Lipschitz in boundary charts, multiplication by it preserves $H^{1/2}$, so $g = a \cdot n \in H^{1/2}(\partial\Omega)$. Let $s_0 = Lg$ be the lift of Lemma 10.1. Riesz on H_0^2 gives a unique z such that

$$(\Delta z, \Delta r) = -(\Delta s_0, \Delta r) \qquad (r \in H_0^2).$$

Thus $s = s_0 + z$ has trace zero, normal derivative $a \cdot n$, and $\Delta^2 s = 0$. Its norm is bounded by $C\|a\|_{H^1}$. The entire trace of ∇s equals that of a: both tangential components vanish and their normal components agree. Consequently $a - \nabla s \in H_0^1$. Define

$$\psi = a - \nabla s + \sum_{i=1}^{I} N_i(\Delta s)\nabla q_i^0.$$

The added gradients belong to H_0^1 and have zero curl, so $\operatorname{curl} \psi = u$. Its divergence is $-\Delta s + \sum_i N_i(\Delta s)\Delta q_i^0$, a harmonic L^2 function. The normalized flux identities of Proposition 10.4 make each inner N_i of this divergence zero; total flux gives the outer one. The bounds for s and the finite-dimensional correction give $\|\psi\|_{H^1} \le C\|u\|_2$. Two solutions differ by an element of K_0 with all these fluxes zero, which Proposition 10.4 forces to vanish. Linearity follows either from the construction or from uniqueness. □

11 Stokes velocity and the tangential potential formulation

Here Ω is again only bounded, connected, and Lipschitz, with an admissible cut system. Fix $\mu > 0$ and $f \in L^2(\Omega)^3$. Put

$$V = \{v \in H_0^1(\Omega)^3 : \operatorname{div} v = 0\}.$$

Pressure recovery is developed separately at the end and in the following chapter; the variational arguments here do not assume pressure existence.

Lemma 11.1 (Zero full trace and the energy identity)**.** *If $v \in H(\operatorname{curl}) \cap H(\operatorname{div})$ has both zero tangential and zero normal traces, then $v \in H_0^1(\Omega)^3$ and*

$$\|\nabla v\|_2^2 = \|\operatorname{curl} v\|_2^2 + \|\operatorname{div} v\|_2^2.$$

Conversely every H_0^1 field has these zero traces. For $v, w \in V$, $(\nabla v, \nabla w) = (\operatorname{curl} v, \operatorname{curl} w)$ and $\|v\|_{H^1} \le C\|\operatorname{curl} v\|_2$.

Proof. The two zero-trace Green identities say that the zero extension $\widetilde{v}$ has curl and divergence equal to the zero extensions of the corresponding derivatives. Fourier transformation gives

$$\int |\xi|^2 |\widehat{\widetilde{v}}(\xi)|^2 \, d\xi = \|\operatorname{curl} \widetilde{v}\|_2^2 + \|\operatorname{div} \widetilde{v}\|_2^2 < \infty.$$

Thus $\widetilde{v} \in H^1(\mathbb{R}^3)$. Its trace from the exterior is zero, hence its trace from Ω is zero, and the H^1 zero-trace characterization gives $v \in H_0^1$. The identity follows from the same Fourier equality. The converse follows by approximation with compactly supported smooth fields. Polarization gives the bilinear identity on V, and Poincaré gives the asserted norm bound. □

Proposition 11.2 (Existence and uniqueness of the variational velocity)**.** *There is a unique $u \in V$ such that*

$$\mu(\nabla u, \nabla v) = (f, v) \qquad (v \in V), \tag{11.1}$$

and $\|u\|_{H^1} \le C\mu^{-1}\|f\|_2$. Any pair $(u, p) \in H_0^1(\Omega)^3 \times L^2(\Omega)/\mathbb{R}$ solving $-\mu\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ distributionally has this velocity.

Proof. V is a closed subspace of H_0^1 because divergence is continuous into L^2. The gradient inner product is complete and coercive by Poincaré, and the functional $v \mapsto (f, v)$ is bounded. Riesz, with the factor μ, gives existence, uniqueness, and the bound. For a distributional solution with $p \in L^2$, testing first with compactly supported smooth vector fields and then taking H_0^1 limits gives $\mu(\nabla u, \nabla v) - (p, \operatorname{div} v) = (f, v)$. Its restriction to V is (11.1). □

Define the graph space

$$\mathcal{W} = \{\phi \in X : \Delta\phi \in L^2(\Omega)^3,\ \operatorname{curl}\operatorname{curl}\phi \in L^2(\Omega)^3\}$$

with squared norm

$$\|\phi\|_{\mathcal{W}}^2 = \|\phi\|_2^2 + \|\operatorname{curl}\phi\|_2^2 + \|\operatorname{div}\phi\|_2^2 + \|\Delta\phi\|_2^2 + \|\operatorname{curl}\operatorname{curl}\phi\|_2^2.$$

The identity $\nabla \operatorname{div}\phi = \Delta\phi + \operatorname{curl}\operatorname{curl}\phi$ holds distributionally, so $\operatorname{div}\phi \in H^1$ for every $\phi \in \mathcal{W}$. Set

$$W_T = \{\phi \in \mathcal{W} \cap X_T : C_j(\phi) = 0 \text{ for all } j,\ \operatorname{curl}\phi \text{ has zero full trace}\}.$$

The full trace condition is meaningful since $\operatorname{curl}\phi$ belongs to $H(\operatorname{curl}) \cap H(\operatorname{div})$; its divergence is zero. Lemma 11.1 identifies it with $\operatorname{curl}\phi \in V$.

Lemma 11.3 (Lemma 4.16, with the orthogonality proved)**.** *W_T is a Hilbert space for its graph norm, and*

$$\|\phi\|_{\mathcal{W}} \le C\|\Delta\phi\|_2 \qquad (\phi \in W_T). \tag{11.2}$$

In particular $\|\Delta\phi\|_2$ is an equivalent norm on W_T.

Proof. Graph convergence preserves the distributional derivative equations, the normal trace of ϕ, its cut fluxes, and both traces of $\operatorname{curl}\phi$. These observations prove completeness. Write $s = \operatorname{div}\phi$ and $c = \operatorname{curl}\operatorname{curl}\phi$. The normal trace of ϕ is zero, so $\int s = 0$ and $\|s\|_2 \le C\|\nabla s\|_2$. Since $v = \operatorname{curl}\phi \in V$, its curl c has zero normal trace: the adjoint identity $C_{\min} = C^*_{\max}$, tested with $\nabla r \in H(\operatorname{curl})$, whose curl is zero, gives $(c, \nabla r) = (v, \operatorname{curl}\nabla r) = 0$ for every $r \in H^1$. Hence $(c, \nabla s) = 0$, and

$$\|\Delta\phi\|_2^2 = \|\nabla s - c\|_2^2 = \|\nabla s\|_2^2 + \|c\|_2^2. \tag{11.3}$$

Lemma 11.1 gives $\|v\|_2 \le C\|c\|_2$. The previously proved X_T norm inequality, with zero cut fluxes, gives $\|\phi\|_2 \le C(\|v\|_2 + \|s\|_2)$. All five graph terms are therefore controlled by $\|\Delta\phi\|_2$. The reverse inequality is part of the graph norm definition. □

Theorem 11.4 (Theorems 4.15 and 4.17, and the velocity part of 4.18)**.** *There is a unique $\psi \in W_T$ such that*

$$\mu(\Delta\psi, \Delta\phi) = (f, \operatorname{curl}\phi) \qquad (\phi \in W_T). \tag{11.4}$$

If u is the unique variational velocity of Proposition 11.2, then $\psi = R_T u$. Consequently $\operatorname{div}\psi = 0$ *and $u = \operatorname{curl}\psi$. In particular the normalized potential of any primitive-variable Stokes solution solves* (11.4)*.*

Proof. The right side is bounded on W_T by $C\|f\|_2\|\Delta\phi\|_2$, and Lemma 11.3 makes the left side a coercive inner product. Riesz gives existence and uniqueness.

Take now $u \in V$ solving (11.1) and put $\psi = R_T u$. The compatibility holds because u has zero normal trace. The potential has zero divergence, and $\Delta\psi = -\operatorname{curl} u \in L^2$, $\operatorname{curl}\operatorname{curl}\psi = \operatorname{curl} u \in L^2$; thus $\psi \in W_T$. For $\phi \in W_T$, let $v = \operatorname{curl}\phi \in V$ and $s = \operatorname{div}\phi \in H^1$. Because $u \in H^1_0$, its curl has zero normal trace, so $(\operatorname{curl} u, \nabla s) = 0$. Therefore

$$\begin{aligned}(\Delta\psi, \Delta\phi) &= (-\operatorname{curl} u, \nabla s - \operatorname{curl} v)\\ &= (\operatorname{curl} u, \operatorname{curl} v) = (\nabla u, \nabla v).\end{aligned}$$

Equation (11.1) now proves that ψ solves (11.4); uniqueness identifies it with the Riesz solution and proves the remaining assertions. □

12 The normal Stokes formulation and its harmonic correction

Define

$$Z_N = \{\phi \in \mathcal{W} \cap X_N : F_i(\phi) = 0 \ (0 \le i \le I)\},$$

and

$$W_N = \{\phi \in Z_N : \text{there is } h \in K_T \text{ such that } (\operatorname{curl}\phi + h) \times n = 0\}.$$

This is the source's space: it would be incorrect to replace its last condition by $\operatorname{curl}\phi \times n = 0$. The normal trace of $\operatorname{curl}\phi$ is already zero because $\phi \in X_N$, and every $h \in K_T$ has zero normal trace. Thus the field in parentheses, when it exists, belongs to V.

Lemma 12.1 (Existence, uniqueness, and control of the harmonic correction)**.** *For $\phi \in W_N$, the harmonic field in its definition is unique; denote it by $H\phi$. The maps H and $T\phi = \operatorname{curl}\phi + H\phi$ are linear, and*

$$\|H\phi\|_2 + \|\operatorname{curl}\phi\|_2 + \|T\phi\|_{H^1} \le C\|\operatorname{curl}\operatorname{curl}\phi\|_2. \tag{12.1}$$

Every $u \in V$ is $T\psi$ for a divergence-free $\psi \in W_N$. More precisely, if

$$h = \sum_{j=1}^{J} C_j(u) h_j^T, \qquad \psi = R_N(u-h),$$

then $H\psi = h$ and $T\psi = u$.

Proof. If two corrections exist, their difference lies in K_T and has zero tangential trace as well as zero normal trace. By Lemma 11.1 it is an H_0^1 field of zero curl and zero divergence, so its gradient and then the field itself are zero. Uniqueness also proves linearity. Let $v = \operatorname{curl}\phi$ and $h = H\phi$. Since $\phi \in \mathcal{D}(C_{\min})$, one has $v \in Z^\perp$; in particular $(v,h) = 0$ because $K_T \subset Z$. The field $U = v + h$ belongs to V and has curl $\operatorname{curl} v$ because h is curl-free. Consequently

$$\|v\|_2^2 + \|h\|_2^2 = \|U\|_2^2, \qquad \|U\|_{H^1} \le C\|\operatorname{curl} v\|_2,$$

which gives (12.1).

For $u \in V$, subtract the displayed harmonic field h. The residual $u - h$ has zero divergence, zero normal trace, and zero cut fluxes, so R_N is defined. Its output ψ has zero divergence, zero tangential trace, and zero boundary fluxes. Its second-order graph derivatives are $\operatorname{curl}\operatorname{curl}\psi = \operatorname{curl} u$ and $\Delta\psi = -\operatorname{curl} u$. Finally $\operatorname{curl}\psi + h = u \in V$, so $\psi \in W_N$ and uniqueness of the correction gives $H\psi = h$. □

Lemma 12.2 (Lemma 4.20, including completeness of its space)**.** *W_N is a Hilbert space for the graph norm, and*

$$\|\phi\|_{\mathcal{W}} \le C\|\Delta\phi\|_2 \qquad (\phi \in W_N).$$

Thus $\|\Delta\phi\|_2$ is an equivalent norm on W_N.

Proof. A graph-Cauchy sequence in W_N has a limit in Z_N, since the distributional equations, tangential trace, and boundary fluxes pass to limits. Estimate (12.1), applied to differences, makes its harmonic corrections Cauchy in K_T. This finite-dimensional space is closed, so those corrections converge to $h \in K_T$. The corrected velocities converge in H^1 by the same estimate and have zero trace; their limit is $\operatorname{curl}\phi + h \in V$. Thus the graph limit belongs to W_N, proving completeness.

Set $s = \operatorname{div}\phi$, $c = \operatorname{curl}\operatorname{curl}\phi$. All boundary fluxes of ϕ vanish, so $\int s = 0$. The corrected velocity $U = T\phi$ belongs to V, and $c = \operatorname{curl} U$ has zero normal trace. Therefore $(c, \nabla s) = 0$ and the exact identity (11.3) holds here too. Poincaré controls s by ∇s, and (12.1) controls $\operatorname{curl}\phi$ by c. The X_N norm inequality, with zero boundary fluxes, controls ϕ by its curl and divergence. These bounds control all graph terms by $\|\Delta\phi\|_2$, as required. □

Theorem 12.3 (Theorems 4.19 and 4.21, and the velocity part of 4.22)**.** *There is a unique $\psi \in W_N$ satisfying*

$$\mu(\Delta\psi, \Delta\phi) = (f, \operatorname{curl}\phi + H\phi) \qquad (\phi \in W_N). \tag{12.2}$$

It is divergence-free, and $u = \operatorname{curl}\psi + H\psi$ is the unique variational Stokes velocity. More precisely, for that velocity u it is the potential $R_N(u - \sum_j C_j(u)h_j^T)$. In particular the corresponding normalized potential of any primitive-variable Stokes solution solves (12.2).

Proof. Lemmas 12.1 and 12.2 bound the right side by $C\|f\|_2\|\Delta\phi\|_2$ and make the left side coercive. Riesz gives existence and uniqueness. For the variational velocity u, construct the divergence-free $\psi \in W_N$ of Lemma 12.1. Then $\Delta\psi = -\operatorname{curl} u$. For $\phi \in W_N$ put $v = T\phi \in V$ and $s = \operatorname{div}\phi$. Since $H\phi$ is curl-free, $\Delta\phi = \nabla s - \operatorname{curl} v$. The same zero-normal orthogonality and energy identity used in the previous theorem give $(\Delta\psi, \Delta\phi) = (\nabla u, \nabla v)$. The variational velocity equation proves (12.2), and uniqueness proves all assertions. □

13 Reduction of pressure recovery to a divergence inverse

The preceding sections prove the two potential formulations, their well-posedness, their equivalence to the divergence-free velocity formulation, and the zero-divergence gauges. To obtain the pressure in $L^2(\Omega)/\mathbb{R}$, we first give a short Hilbert-space reduction. The following chapter then proves the required bounded divergence inverse explicitly, including its singular-kernel estimate.

We use the closed Hilbert subspace

$$L^2_0(\Omega) = \{g \in L^2(\Omega) : \int_\Omega g = 0\}.$$

It is closed because integration is a continuous functional on $L^2(\Omega)$ when Ω is bounded. Mean-zero representatives identify $L^2(\Omega)/\mathbb{R}$ with this subspace.

Proposition 13.1 (Pressure recovery from a quantitative divergence inverse)**.** *Suppose there is a bounded linear map*

$$B : L^2_0(\Omega) \longrightarrow H^1_0(\Omega)^3, \qquad \operatorname{div} Bg = g, \qquad \|Bg\|_{H^1} \le C_B\|g\|_2.$$

Then the variational velocity u has a unique pressure $p \in L^2_0(\Omega)$ satisfying $-\mu\Delta u + \nabla p = f$ distributionally, and

$$\|p\|_2 \le C_B(\mu\|\nabla u\|_2 + C\|f\|_2).$$

Consequently the full pressure assertions of Corollaries 4.18 and 4.22 follow from this additional bounded-inverse statement.

Proof. Define the continuous functional on $H^1_0(\Omega)^3$

$$F(v) = \mu(\nabla u, \nabla v) - (f, v).$$

It vanishes on V. The functional $g \mapsto F(Bg)$ is bounded on L^2_0, so Riesz gives a unique $p \in L^2_0$ with $(p, g) = F(Bg)$ and the displayed norm bound. For arbitrary $v \in H^1_0$, its divergence has zero mean, and $v - B\operatorname{div} v \in V$. Hence $F(v) = (p, \operatorname{div} v)$, which is precisely the weak primitive-variable Stokes equation. If two pressures satisfy it, their difference has zero distributional gradient and is constant on connected Ω; mean zero makes it zero. □

Remark 13.2 (Separation of the pressure module). The bounded divergence inverse used in this reduction is proved in Theorem 14.4. Combining that theorem with the present proposition yields the full pressure conclusions of Corollaries 4.18 and 4.22, as recorded in Corollary 14.5. Keeping the quantitative divergence construction separate identifies the exact analytic dependency.

14 A bounded divergence inverse and the L^2 Stokes pressure

We prove the quantitative divergence statement needed for pressure recovery. The proof includes the singular-kernel estimate, so no Calderón–Zygmund or Bogovskiĭ theorem is imported.

Lemma 14.1 (A mean-zero angular kernel is L^2 bounded)**.** *Let $a \in C^1(S^2)$ with $\int_{S^2} a = 0$ and let χ be a bounded radial cutoff of compact support. The operators*

$$T_\varepsilon f(x) = \int_{|z|>\varepsilon} \chi(|z|) \frac{a(z/|z|)}{|z|^3} f(x-z)\, dz$$

have $L^2(\mathbb{R}^3)$ norms bounded by $C\|\chi\|_\infty \|a\|_{C^1(S^2)}$, independently of ε. As $\varepsilon \downarrow 0$ they converge strongly on L^2. The same conclusion holds for a smooth angular function $a(x,\theta)$ of compact support in x, with zero spherical mean for each x; the bound then depends on finitely many x derivatives and one angular derivative of a.

Proof. For the translation-invariant case, the Fourier multiplier is

$$m_\varepsilon(\xi) = \int_\varepsilon^\infty \chi(r) A(r|\xi|, \xi/|\xi|)\, \frac{dr}{r}, \qquad A(t,\omega) = \int_{S^2} a(\theta) e^{-it\omega\cdot\theta}\, d\theta.$$

At $\xi = 0$ the multiplier is zero. For $0 < t \le 1$, zero spherical mean and $|e^{-is} - 1| \le |s|$ give $|A(t,\omega)| \le Ct\|a\|_\infty$. For $t \ge 1$, rotate the north pole to ω and use latitude $z = \omega\cdot\theta \in [-1,1]$ and longitude α. The surface measure is $d\alpha\, dz$, so

$$A(t,\omega) = \int_{-1}^1 e^{-itz} b_\omega(z)\, dz, \qquad b_\omega(z) = \int_0^{2\pi} a(\theta(z,\alpha))\, d\alpha.$$

The latitude parametrization has $|\partial_z \theta| = (1-z^2)^{-1/2}$. Thus b_ω is absolutely continuous, has bounded endpoint values, and

$$|b_\omega(1)| + |b_\omega(-1)| + \int_{-1}^1 |b_\omega'(z)|\, dz \le C\|a\|_{C^1(S^2)}.$$

One-dimensional integration by parts gives $|A(t,\omega)| \le Ct^{-1}\|a\|_{C^1}$, uniformly in ω. Split the radial integral at $r|\xi| = 1$. The two bounds integrate respectively as $\int_0^{1/|\xi|} |\xi|\, dr = 1$ and $\int_{1/|\xi|}^\infty (|\xi| r^2)^{-1}\, dr = 1$. They prove a uniform multiplier bound. For fixed ξ the omitted small-radius part tends to zero by the first bound, so the multipliers converge pointwise. Plancherel and dominated convergence give both the norm bound and strong convergence on L^2.

For x dependence, choose a cube Q with the x support strictly inside it and expand periodically in that variable:

$$a(x,\theta) = \sum_{k\in\mathbb{Z}^3} e^{2\pi i k\cdot x/L} a_k(\theta).$$

Here harmless changes of origin and side length are understood. Integrating by parts with $(1-\Delta_x)^2$ gives

$$\|a_k\|_{C^1(S^2)} \le C(1+|k|^2)^{-2} \max_{|\beta|\le 4} \|\partial_x^\beta a\|_{C^1(S^2);L^\infty_x}.$$

The sequence on the right is summable over $\mathbb{Z}^3$. For clarity, equality with this absolutely convergent Fourier series uses only the following elementary uniqueness argument. On a circle the Fejér kernel $F_n(t) = n^{-1}|\sum_{j=0}^{n-1} e^{ijt}|^2$ is nonnegative, has integral 2π, and is at most C_δ/n when the distance of t from a multiple of 2π is at least δ. Its normalized convolution therefore converges uniformly to every continuous periodic function, by uniform continuity and splitting the integral into the δ neighborhood and its complement. The product of three such kernels has the same approximation property on the three-dimensional torus. A continuous periodic function with all Fourier coefficients zero has all these convolutions zero and is consequently zero. The absolutely convergent series above has the same coefficients as a, so their difference is zero; the coefficient bounds also give uniform convergence of its first angular derivatives. Each a_k has spherical mean zero. The translation-invariant result therefore applies to its convolution operator,

and multiplication by $e^{2\pi ik\cdot x/L}$ preserves L^2 norm. To make the global operator identity explicit, let $T_{\varepsilon,k}$ be the translation-invariant truncated operator with angular coefficient a_k. Then

$$T_\varepsilon f = \mathbf{1}_Q \sum_{k\in\mathbb{Z}^3} e^{2\pi ik\cdot x/L} T_{\varepsilon,k} f.$$

For each positive truncation the identity follows by the uniformly convergent coefficient series and the integrability of the truncated kernel. Its operator-norm series is bounded by a fixed summable sequence, independently of ε. Multiplication by $\mathbf{1}_Q$ is an L^2 contraction. For strong convergence first make the Fourier tail uniformly small using that summable bound; then pass to the limit in the remaining finite sum, using the translation-invariant result. No derivative of $\mathbf{1}_Q$ is taken in this argument.

This proves the variable-coefficient statement. □

Lemma 14.2 (A divergence solver on a star-shaped set)**.** *Let $D \subset \mathbb{R}^3$ be bounded and open, star-shaped with respect to every point of a ball $B \Subset D$. There is a linear map from the mean-zero functions of $L^2(D)$ to $H^1(\mathbb{R}^3)^3$, denoted B_D, with*

$$\operatorname{div} B_D g = \widetilde{g}, \quad B_D g = 0 \text{ almost everywhere outside } D, \quad \|B_D g\|_{H^1(\mathbb{R}^3)} \le C_D \|g\|_{L^2(D)}.$$

Here $\widetilde{g}$ is the zero extension.

Proof. Choose $\rho \in C_c^\infty(B)$ with integral one. For a function g extended by zero, define

$$B_D g(x) = \int_D g(y)(x-y) \int_1^\infty \rho\big(y + t(x-y)\big) t^2 \, dt \, dy. \tag{14.1}$$

We prove its support, divergence, and norm properties separately. If an integrand is nonzero, $a = y + t(x-y) \in B$ for some $t \ge 1$, and $x = (1-1/t)y + (1/t)a$ lies on a segment contained in D. Thus the formula is zero outside D.

For a fixed y, denote its vector kernel by $K_y(x)$. For $\varphi \in C_c^\infty(\mathbb{R}^3)$, change variables $a = y + t(x-y)$ to obtain

$$\begin{aligned} -\int K_y(x)\cdot\nabla\varphi(x)\,dx &= -\int \rho(a) \int_1^\infty (a-y)t^{-2}\cdot\nabla\varphi(y+(a-y)/t)\,dt\,da \\ &= \int \rho(a) \int_1^\infty \frac{d}{dt}\varphi(y+(a-y)/t)\,dt\,da \\ &= \varphi(y) - \int \rho(a)\varphi(a)\,da. \end{aligned}$$

The integrals are absolutely convergent: a, y stay in bounded sets and $\int_1^\infty t^{-2}\,dt < \infty$. Integrating against $g(y)$ gives $\operatorname{div} B_D g = \widetilde{g} - \rho \int_D g$. For mean-zero g this is the required equation.

It remains to prove, rather than assume, the H^1 bound. Write $z = x - y = r\theta$ and replace t by $1+s$ in the kernel. With

$$A_m(x,\theta) = \int_0^\infty \rho(x+\tau\theta)\tau^m \, d\tau \qquad (m = 0,1,2),$$

the ith component becomes exactly

$$K_i(x,z) = \theta_i\big(r^{-2}A_2(x,\theta) + 2r^{-1}A_1(x,\theta) + A_0(x,\theta)\big). \tag{14.2}$$

For x in a fixed bounded set all these coefficients are smooth in x, θ with bounded derivatives: the support of ρ bounds τ uniformly, so differentiation under a fixed finite integral is legitimate. Insert a smooth x cutoff equal to one on $\overline{D}$, and a radial z cutoff equal to one for $|z| \le \operatorname{diam} D$.

This does not change (14.1): outside D the original integrand is already zero, and inside D all contributing $x-y$ have the indicated length. We may therefore work with compactly supported coefficients and kernels. The bound $|K(x,z)| \le C|z|^{-2}$ on a bounded z ball proves the L^2 bound by the elementary two-sided integral estimate

$$\sup_x \int |K(x,x-y)|\,dy < \infty, \qquad \sup_y \int |K(x,x-y)|\,dx < \infty.$$

Indeed Cauchy–Schwarz with weight $|K|$, followed by Fubini, gives

$$\|Kf\|_2^2 \le (\sup_x \int |K|)(\sup_y \int |K|)\|f\|_2^2.$$

When differentiating the integral, both x and $z = x-y$ are differentiated. All explicit x derivatives and all derivatives of the last two terms in (14.2) are bounded by $C|z|^{-2}$ and satisfy the same integral estimate. Cutoff derivatives are also integrable. Only a z derivative of the leading term $a_i(x,\theta)r^{-2}$ needs special treatment. Away from $z=0$ it is $r^{-3}a_{ij}(x,\theta)$, with smooth coefficients. Its spherical mean is zero. To check this, integrate that derivative over an annulus: its two boundary integrals are both $\int_{S^2} a_i(x,\theta)\theta_j\,d\theta$ and cancel, while its volume integral is the logarithm of the radius ratio times $\int_{S^2} a_{ij}(x,\theta)\,d\theta$. Distributionally the derivative is the principal-value kernel just described plus

$$d_{ij}(x)\delta_0, \qquad d_{ij}(x) = \int_{S^2} a_i(x,\theta)\theta_j\,d\theta.$$

This identity follows by integration by parts outside a ball of radius ε; on its boundary replace the test function by its value at zero, and bound the resulting error by $C\varepsilon$. The coefficient d_{ij} is bounded, so its integral operator is bounded multiplication. Lemma 14.1 bounds the principal-value term. We have therefore bounded every first weak derivative by $C_D\|g\|_2$. The derivative identities hold first for smooth compactly supported g; their distributional integration-by-parts proof and the uniform L^2 bounds pass to arbitrary $L^2(D)$ functions by density. If mean-zero smooth approximants are desired, choose $g_n \in C_c^\infty(D)$ converging to g in $L^2(D)$ and replace them by $g_n - (\int_D g_n)\rho$. These still have compact support in D, have exactly zero mean, and converge to a mean-zero g, since $|\int_D (g_n - g)| \le |D|^{1/2}\|g_n - g\|_2$. The established norm and distributional identities therefore pass to the limit without any compatibility assumption being lost.

This proves the global H^1 bound and completes the lemma. □

Lemma 14.3 (A finite star-shaped cover and transfer of means)**.** *Every bounded connected Lipschitz domain Ω has finitely many open subsets $D_1,\dots,D_m \subset \Omega$, each star-shaped with respect to an interior ball, and a bounded linear decomposition*

$$g = \sum_{k=1}^m g_k, \qquad g_k \in L^2(D_k), \quad \int_{D_k} g_k = 0, \qquad \sum_k \|g_k\|_2 \le C\|g\|_2$$

for every $g \in L^2_0(\Omega)$, with each g_k extended by zero in the sum.

Proof. Here are the geometric and algebraic details. In a Lipschitz boundary chart, translate the graph so that its central height is zero and let its Lipschitz constant be L. After choosing a sufficiently small horizontal disk of radius r, choose a constant height H still inside the chart with $H > \sup\varphi + 3Lr + 2r$. The set

$$D = \{(y,t) : |y| < r,\ \varphi(y) < t < H\}$$

is star-shaped with respect to a sufficiently small ball centered near $(0, H - r/2)$. Indeed the horizontal disk and the upper half-space $t < H$ are convex. For a segment from $(y, t) \in D$ to (z, s) in that ball, the Lipschitz inequality gives

$$\varphi((1-\lambda)y + \lambda z) \le \varphi(y) + \lambda L|z - y| < (1-\lambda)t + \lambda s$$

for $0 < \lambda \le 1$, since $s > \sup \varphi + L|z - y|$. A smaller cylinder gives a neighborhood of the boundary point covered by this D. Such sets, together with interior balls, yield a finite cover of Ω: use compactness of the boundary, and then compactness of the remaining interior part. Choose ambient open sets O'_k and O_k with $\overline{O'_k} \subset O_k$ such that the O'_k cover $\overline{\Omega}$ and $O_k \cap \Omega \subset D_k$. For a graph piece, take their horizontal supports strictly inside $|y| < r$ and their upper supports strictly below the cap $t = H$; smaller boundary neighborhoods still cover the relevant boundary patch. For an interior piece use two concentric balls contained in D_k. Choose nonnegative smooth ambient bumps supported in O_k and positive on $\overline{O'_k}$. Their sum is positive on a neighborhood N of $\overline{\Omega}$. Divide by this sum and multiply by a smooth cutoff supported in N and equal to one near $\overline{\Omega}$. The resulting χ_k are smooth, sum to one on Ω, and have relative support inside D_k. In particular $\chi_k g$, regarded as a function on D_k, has exactly the support required below.

The intersection graph of the sets D_k is connected: otherwise the unions belonging to two disjoint graph components would separate the connected open set Ω. Choose a rooted spanning tree. For every nonroot vertex k, pick a smooth function ρ_k of integral one supported in a ball inside $D_k \cap D_{\mathrm{parent}(k)}$. Set $f_k = \chi_k g$, $m_k = \int f_k$, and let M_k be the sum of m_ℓ over the subtree rooted at k. For a nonroot vertex define

$$g_k = f_k + \sum_{\ell \text{ child of } k} M_\ell \rho_\ell - M_k \rho_k;$$

for the root omit the last term. Each g_k is supported in D_k. Its mean is $m_k + \sum_{\ell \text{ child}} M_\ell - M_k = 0$; for the root the same conclusion uses $\int g = 0$. All transfer terms cancel in the sum, so $\sum g_k = g$. Finally $|m_k| \le \|\chi_k\|_2 \|g\|_2$ and only finitely many fixed transfer functions occur. These inequalities prove the stated bound and linearity. □

Theorem 14.4 (Quantitative divergence inverse on a Lipschitz domain)**.** *There is a bounded linear map*

$$B : L^2_0(\Omega) \longrightarrow H^1_0(\Omega)^3, \qquad \operatorname{div} Bg = g, \qquad \|Bg\|_{H^1} \le C_\Omega \|g\|_2.$$

Proof. Decompose g by Lemma 14.3 and set $Bg = (\sum_k B_{D_k} g_k)|_\Omega$. The sum is a global H^1 field, vanishes outside Ω, and has divergence equal to the zero extension of g. Its boundary trace is zero by matching interior and exterior traces, so its restriction belongs to $H^1_0(\Omega)^3$. The finite sum of the estimates in Lemmas 14.2 and 14.3 gives the bound. Every construction was linear. □

Corollary 14.5 (The full pressure conclusions of Corollaries 4.18 and 4.22)**.** *The variational Stokes velocity has a unique mean-zero pressure $p \in L^2_0(\Omega)$ with $-\mu\Delta u + \nabla p = f$ distributionally. In the tangential potential formulation, $u = \operatorname{curl} \psi$; in the normal potential formulation, $u = \operatorname{curl} \psi + H\psi$. Thus both potential formulations are equivalent to the complete primitive-variable Stokes problem, including its L^2 pressure.*

Proof. Apply the bounded divergence inverse of Theorem 14.4 in Proposition 13.1. The two velocity identifications and divergence-free gauges were proved in the corresponding potential theorems. □

15 The cylindrical potential of Remark 3.23

Let $\Omega = S \times (a,b)$, where $S \subset \mathbb{R}^2$ is bounded, connected, and Lipschitz. Assume the chosen cuts have the product form $\Sigma_j = \sigma_j \times (a,b)$. Write $\ell = b-a$ and $\overline{w}(x) = \ell^{-1}\int_a^b w(x,z)\,dz$ for vertical averaging. There is a minor trace issue in the source's displayed conditions: an arbitrary L^2 horizontal potential need not have a normal trace on a vertical cut. Its integrated flux is interpreted below through the averaged field, whose divergence and normal trace are defined. This agrees with the displayed classical integrals whenever the field is smooth.

Theorem 15.1 (A precise weak form of Remark 3.23)**.** *For every divergence-free $u \in L^2(\Omega)^3$, there is a unique $\psi = (w_1, w_2, 0) \in H(\mathrm{curl};\Omega)$ satisfying*

$$\operatorname{curl}\psi = u, \qquad \operatorname{div}_2 \overline{w} = 0, \qquad \overline{w}\cdot n_S = 0, \qquad C_j((\overline{w},0)) = 0 \quad (1 \le j \le J).$$

The last flux is that of the z-independent three-dimensional field $(\overline{w},0)$ through Σ_j. The map $u \mapsto \psi$ is linear and bounded from L^2 into $H(\mathrm{curl})$. Conversely, existence of such a potential implies $\operatorname{div} u = 0$.

Proof. Define the zero-mean vertical primitive

$$Pf(z) = \int_a^z f(t)\,dt - \frac{1}{\ell}\int_a^b\int_a^s f(t)\,dt\,ds.$$

It satisfies $\partial_z Pf = f$, $\overline{Pf} = 0$, and $\|Pf\|_{L^2(a,b)} \le C\ell\|f\|_{L^2(a,b)}$ by Cauchy–Schwarz. Apply it on almost every vertical line and set $w^0 = (Pu_2, -Pu_1)$. The first two components of $\operatorname{curl}(w^0,0)$ are u_1, u_2. The divergence-free identity gives

$$\partial_z\big(\operatorname{curl}_2 w^0 - u_3\big) = 0$$

in distributions. A distribution with zero derivative in an interval is independent of that variable: any scalar test function of integral zero is the derivative of a compactly supported smooth function on the interval, so the distribution depends only on its integral. Averaging the last identity and using $\overline{w^0} = 0$ therefore yields

$$\operatorname{curl}_2 w^0 = u_3 - \overline{u_3}.$$

In particular this distribution is in L^2, a fact not assumed in the construction of w^0.

Solve the two-dimensional variational Dirichlet problem

$$(\nabla p, \nabla r)_S = (\overline{u_3}, r)_S \qquad (r \in H_0^1(S))$$

for $p \in H_0^1(S)$. Riesz and Poincaré give a unique solution and $\|\nabla p\|_2 \le C\|\overline{u_3}\|_2$. Put $A = (\partial_2 p, -\partial_1 p)$. Then $\operatorname{div}_2 A = 0$ and $\operatorname{curl}_2 A = -\Delta_2 p = \overline{u_3}$. Approximate p in $H_0^1(S)$ by compactly supported smooth functions. Their rotated gradients have zero normal trace, and convergence in $H(\mathrm{div};S)$ proves $A\cdot n_S = 0$. For each approximant the flux through a cut curve is the integral of a tangential derivative of that compactly supported scalar function, which is zero by the one-dimensional fundamental theorem of calculus. Equivalently the product-field flux through Σ_j is zero. Continuity of the scalar cut flux in the zero-normal $H(\mathrm{div})$ space passes this conclusion to $(A,0)$.

Set $w = w^0 + A$. Its average is A, so it has all the required average conditions and cut fluxes; its three curl components are exactly u. The estimates for P and p, together with $\|\overline{u_3}\|_{L^2(S)} \le \ell^{-1/2}\|u_3\|_{L^2(\Omega)}$, give the stated norm bound.

For uniqueness the difference of two potentials has zero curl. Its first two curl equations say that its horizontal components are independent of z, hence the difference is $(B(x),0)$. The average

conditions imply that this field has zero divergence and zero normal trace on the cylinder; it has zero cut fluxes as well. Thus it is a curl-free field satisfying the unique tangential-potential gauge, so it is zero by the already proved kernel characterization. Linearity follows from the linear construction or uniqueness. The converse is the distributional identity $\operatorname{div}\operatorname{curl} = 0$. □

16 The finite-element constructions: definitions, exactness, and proof obligations

This section proves the concrete finite-element results in Section 4.1 of the source paper. Polynomial exactness and moment unisolvence are proved explicitly. Weak face and edge moments are constructed before they are used on rough fields. A stable scalar projector supplies the boundary-constrained discrete estimate, and a Fourier low/high split supplies the full fractional approximation range. Neither a black-box commuting projection nor a polyhedral fractional shift theorem is needed for the finite-element conclusions.

16.1 Meshes, cuts, and the actual polynomial spaces

Assume now that Ω is a bounded Lipschitz polyhedron. A mesh $\mathcal{T}_h$ is a finite collection of closed nondegenerate tetrahedra whose union is $\overline{\Omega}$, whose interiors are disjoint, and whose distinct members intersect, if at all, in a common full face, edge, or vertex. Write $h_K = \operatorname{diam} K$ and $h = \max_K h_K$. Shape regularity means

$$h_K/\rho_K \le \gamma \qquad (K \in \mathcal{T}_h),$$

where ρ_K is the inradius and the number γ is independent of the mesh. No lower bound $h_K \ge ch$ is imposed: a shape-regular family need not be quasi-uniform. Source Hypothesis 4.1 requires, for each mesh, an admissible cutting system whose closed cuts are unions of mesh faces. The cuts may depend on the mesh. All arguments that use uniform constants below are phrased through intrinsic normalized potentials so that a changing presentation of the cuts does not introduce an unproved uniform cut-domain constant.

Let $P_m(K)$ be the restrictions of polynomials of degree at most m, and let $\mathcal{H}_m$ be the homogeneous polynomials of degree m. Set $P_m = \{0\}$ if $m < 0$. Fix an integer $k \ge 1$ and put

$$N_k(K) = P_{k-1}(K)^3 \oplus \{b \in \mathcal{H}_k^3 : x \cdot b(x) = 0\}. \tag{16.1}$$

This is the first Nédélec family with the indexing used in the source; its lowest-order member is $N_1(K)$, not a complete vector P_1 space. Changing the coordinate origin changes only lower-degree terms and does not change this space. More explicitly, translating a homogeneous degree-k term changes it by a polynomial of degree at most $k-1$; the highest-degree part retains $x \cdot b = 0$.

The auxiliary Raviart–Thomas space, with $r = k - 1$, is

$$RT_r(K) = P_r(K)^3 \oplus x\mathcal{H}_r. \tag{16.2}$$

It is used here to identify curls and normal moments. Its divergence-free subspace is precisely $\{z \in P_r(K)^3 : \operatorname{div} z = 0\}$; this fact is proved below. In particular one must not replace the source's piecewise P_{k-1} curl space by the entire Raviart–Thomas space.

Define the assembled spaces

$$\begin{aligned}
Y_h &= \{v \in H(\mathrm{curl}, \Omega) : v|_K \in N_k(K) \text{ for every } K\},\\
Y_h^0 &= Y_h \cap H_0(\mathrm{curl}, \Omega),\\
\widetilde{\Theta}_h &= \{q \in H^1(D) : q|_K \in P_k(K) \text{ for every } K\},\\
\Theta_h &= \widetilde{\Theta}_h \cap \Theta,\\
\Theta_h^0 &= \{q \in H^1(\Omega) : q|_K \in P_k(K), \quad q|_{\Gamma_0} = 0, \quad q|_{\Gamma_i} \text{ constant } (i \geq 1)\}.
\end{aligned}$$

Here D is the cut domain for this mesh; element interiors are contained in D, so restrictions are unambiguous. The superscript 0 on Θ_h^0 denotes the boundary-constant space, and does not mean that the function vanishes on every boundary component.

For piecewise polynomial fields, membership in $H(\mathrm{curl})$ is equivalent to equality of the two tangential traces on every interior face. Indeed, integration by parts element by element expresses the distributional curl as the elementwise curl plus the face-supported tangential jumps. An L^2 distribution cannot contain a nonzero measure on a face, since a face has three-dimensional measure zero. Testing locally inside a face proves that each jump vanishes. Conversely, when the jumps vanish the elementwise integration identities sum without interior boundary terms and give an L^2 curl. The same argument gives the corresponding characterization of $H(\mathrm{div})$ by matching normal traces. The boundary version identifies zero tangential trace with vanishing boundary-face tangential polynomials.

16.2 Local polynomial exactness and unisolvent moments

Lemma 16.1 (Polynomial kernel and polynomial curl image)**.** *For every tetrahedron K,*

$$\ker(\mathrm{curl}\,|_{N_k(K)}) = \nabla P_k(K), \qquad \mathrm{curl}\, N_k(K) = \{z \in P_{k-1}(K)^3 : \mathrm{div}\, z = 0\}.$$

Proof. A curl-free polynomial field a has the polynomial primitive

$$q(x) = \int_0^1 a(tx) \cdot x \, dt.$$

The differentiation calculation used in the scalar Poincaré lemma gives $\nabla q = a$; all identities are polynomial identities on $\mathbb{R}^3$, so the fact that the origin need not lie in K causes no problem. If $a \in N_k$, its degree is at most k, so q has degree at most $k+1$. Let q_{k+1} be its homogeneous degree-$(k+1)$ part. The degree-k part of a is ∇q_{k+1} and satisfies $x \cdot \nabla q_{k+1} = 0$. Euler's identity gives $(k+1)q_{k+1} = 0$, hence $q \in P_k$. The reverse inclusion follows because $\nabla P_k \subset P_{k-1}^3 \subset N_k$.

The curl of a member of N_k has degree at most $k-1$ and zero divergence. Conversely decompose a divergence-free polynomial z into homogeneous parts z_m, $0 \leq m \leq k-1$. Their divergences have different homogeneous degrees, so every z_m is divergence-free. The vector identity

$$\mathrm{curl}(x \times z_m) = x\, \mathrm{div}\, z_m - 2z_m - (x \cdot \nabla) z_m = -(m+2) z_m$$

shows that $a_m = -x \times z_m/(m+2)$ has curl z_m. For $m < k-1$ it belongs to P_{k-1}^3; for $m = k-1$ it is homogeneous of degree k and orthogonal to x. Thus $\sum_m a_m \in N_k$ has curl z. □

Orient each face and edge once. The Nédélec moments are

$$\int_E (v \cdot t_E) p, \qquad p \in P_{k-1}(E), \tag{16.3}$$

$$\int_F (v \times n_F) \cdot p, \qquad p \in P_{k-2}(F)^2, \tag{16.4}$$

$$\int_K v \cdot p, \qquad p \in P_{k-3}(K)^3. \tag{16.5}$$

Here $P_m(F)^2$ denotes tangential vector polynomials on the plane of F, rather than an additional choice of physical two-dimensional coordinates. The Raviart–Thomas moments are

$$\int_F (z \cdot n_F) p \quad (p \in P_r(F)), \qquad \int_K z \cdot p \quad (p \in P_{r-1}(K)^3). \tag{16.6}$$

Lemma 16.2 (Unisolvence of the two moment systems)**.** *The moments* (16.6) *determine a unique member of* $RT_r(K)$*, and the moments* (16.3)–(16.5) *determine a unique member of* $N_k(K)$*.*

Proof. First consider $z \in RT_r(K)$ with all its moments zero. Its normal trace on each face has degree at most r: the factor $x \cdot n_F$ is constant on that face. The face moments therefore give $z \cdot n_F = 0$. For $q \in P_r(K)$, integration by parts gives

$$\int_K (\operatorname{div} z) q = -\int_K z \cdot \nabla q = 0.$$

Since $\operatorname{div} z \in P_r$, its divergence is zero. Writing $z = a + xb$ with $a \in P_r^3$ and $b \in \mathcal{H}_r$, the highest-degree part of its divergence is $(r+3)b$. Hence $b = 0$ and $z \in P_r^3$. This also proves the asserted description of the divergence-free subspace of RT_r.

Use an affine transformation to the reference tetrahedron $\{x_i \geq 0,\ x_1 + x_2 + x_3 \leq 1\}$ and transform the vector field contravariantly. This transformation preserves the moment conditions, the Raviart–Thomas space, and zero normal traces, as is checked by the chain rule and the affine face Jacobian. On the three coordinate faces, $z_i = 0$ implies $z_i = x_i p_i$, with $p_i \in P_{r-1}$. Its cell moment against (p_1, p_2, p_3) gives

$$0 = \int_K \sum_{i=1}^{3} x_i p_i^2,$$

so each p_i is zero. If $r = 0$, z is constant and the same three normal conditions give $z = 0$ directly. Finally

$$\dim RT_r = 3\binom{r+3}{3} + \binom{r+2}{2} = 4\binom{r+2}{2} + 3\binom{r+2}{3},$$

which equals the number of scalar moments. Injectivity of the square moment map therefore proves existence and uniqueness for arbitrary moment values.

Now let $v \in N_k$ have all its moments zero, and set $z = \operatorname{curl} v$. For a face polynomial $p \in P_{k-1}(F)$, the two-dimensional Stokes identity, obtained by integration by parts on the planar triangle, expresses $\int_F z \cdot n_F p$ through edge moments of degree at most $k-1$ and face moments against $\nabla_F p$, of degree at most $k-2$. It is therefore zero. For $p \in P_{k-2}(K)^3$, the curl Green identity expresses $\int_K z \cdot p$ through a cell moment against $\operatorname{curl} p \in P_{k-3}^3$ and face moments of degree at most $k-2$; it too is zero. Thus $z \in RT_{k-1}$ has all its Raviart–Thomas moments zero, and the preceding paragraph gives $z = 0$. By Lemma 16.1, $v = \nabla q$ with $q \in P_k$.

The edge moments now imply that the tangential derivative of q is zero on every edge. The connected edge graph of a tetrahedron makes these edge constants all equal. Subtract that common constant, so q vanishes on all edges. On a face, integration by parts against the tangential polynomials in (16.4) gives

$$\int_F qp = 0 \qquad (p \in P_{k-3}(F)).$$

Indeed the divergence of tangential P_{k-2}^2 is all of P_{k-3}, as follows by integrating a polynomial in one planar coordinate. Since q vanishes on the three edges, its face restriction is the product of the three face barycentric coordinates times some $p \in P_{k-3}$. Use this p as test: positivity of the product in the face interior gives $q|_F = 0$. When $k < 3$, divisibility already forces this restriction to be zero. Thus q vanishes on all four faces and is their barycentric-coordinate product times

$p \in P_{k-4}(K)$. The cell moments and integration by parts give $\int_K qp = 0$ for all such polynomials; testing with this last factor gives $q = 0$. The cases $k < 4$ again follow directly from degree.

For the dimension count, the map $\mathcal{H}^3_k \to \mathcal{H}_{k+1}$, $b \mapsto x \cdot b$, is onto: $b = \nabla q/(k+1)$ maps to q by Euler's identity. Hence

$$\dim N_k = 3\binom{k+2}{3} + 3\binom{k+2}{2} - \binom{k+3}{2} = \frac{k(k+2)(k+3)}{2}.$$

There are $6k$ edge moments, $4k(k-1)$ face moments, and $k(k-1)(k-2)/2$ cell moments, with zero-sized lists at low order. Their sum is the same dimension. The square moment map is therefore bijective, completing the proof. □

For smooth fields denote the resulting interpolants by I^N_K and I^{RT}_K. Their definitions and the integration identities just used give the exact commuting identities

$$\operatorname{curl} I^N_K v = I^{RT}_K \operatorname{curl} v, \qquad \operatorname{div} I^{RT}_K z = \Pi_r \operatorname{div} z, \tag{16.7}$$

where Π_r is the $L^2(K)$ projection onto $P_r(K)$. For the second identity, test the divergence difference against every $q \in P_r$, integrate by parts, and use precisely the face and cell moments (16.6). For the first identity, equality of all Raviart–Thomas moments follows from the same Stokes and curl Green calculations as in the preceding proof, with the differences of moments in place of zero moments. Unisolvence then gives equality of the fields. These calculations prove commutation for smooth inputs; they do not, without a continuity theorem, define edge traces of arbitrary $H(\operatorname{curl})$ fields.

On any face, the tangential trace of $N_k(K)$ is the corresponding two-dimensional first-family space. In that plane a curl-free member is a gradient of P_k: integrate it along rays for a polynomial primitive and eliminate its degree-$(k+1)$ homogeneous part by the same Euler identity as in Lemma 16.1. Its edge and face moments are unisolvent by the same proof: first use planar Stokes to kill its scalar curl, then use its polynomial scalar primitive, the edge constants, and the triangular bubble. Consequently matching edge and face moments give matching tangential traces. The smooth interpolants assemble into $I^N_h v \in Y_h$, and smooth zero-tangential input gives an element of Y^0_h. The analogous conclusion for normal traces follows immediately because those traces are polynomials of degree at most r determined by the face moments.

16.3 Discrete kernels and mixed solvability, without a uniformity claim

Proposition 16.3 (The two concrete discrete kernels)**.** *The following equalities hold on every admissible mesh:*

$$\ker(\operatorname{curl}|_{Y_h}) = \nabla_b \Theta_h, \qquad \ker(\operatorname{curl}|_{Y^0_h}) = \nabla \Theta^0_h.$$

Thus the first equality proves source Lemma 4.3, and every curl-free field in either space is piecewise of degree at most $k-1$, as in source Remark 4.4.

Proof. Let $v_h \in Y_h$ have zero curl. The proved continuous maximal-kernel theorem gives $v_h = \nabla_b q$ with $q \in \Theta$. On each element Lemma 16.1 gives a polynomial $p_K \in P_k(K)$ with the same gradient. The function $q - p_K$ has zero gradient on the connected element interior, and is constant there. Hence $q|_K \in P_k(K)$ and $q \in \Theta_h$. Conversely a member of Θ_h has a globally curl-free broken gradient by the continuous constant-jump theorem, and its elementwise gradients belong to $P^3_{k-1} \subset N_k$.

For $v_h \in Y^0_h$ use the proved minimal-kernel theorem instead: $v_h = \nabla q$, $q \in \Theta_0$. The same local argument makes q piecewise P_k, hence a member of Θ^0_h. Conversely the continuous boundary-constant theorem puts $\nabla\Theta^0_h$ in the minimal kernel. The degree statement follows from taking gradients of degree-k scalar polynomials. □

The following notation handles both formulations without identifying their distinct boundary conditions. In case T let

$$X_h = Y_h, \quad Q_h = \Theta_h/\mathbb{R}, \quad Gq = \nabla_b q;$$

in case N let

$$X_h = Y_h^0, \quad Q_h = \Theta_h^0, \quad Gq = \nabla q.$$

In both cases G is injective on Q_h, its image is the curl kernel, and $\|q\|_{Q_h} := \|Gq\|_2$ is a norm. Define

$$V_h = \{v_h \in X_h : (v_h, Gq_h) = 0 \text{ for every } q_h \in Q_h\}.$$

For $u \in L^2(\Omega)^3$ the source's mixed systems are precisely

$$\begin{aligned} (\operatorname{curl} w_h, \operatorname{curl} v_h) + (v_h, G\theta_h) &= (u, \operatorname{curl} v_h) \quad (v_h \in X_h), \\ (w_h, Gq_h) &= 0 \quad (q_h \in Q_h). \end{aligned} \tag{16.8}$$

Theorem 16.4 (Mixed uniqueness and existence on each mesh)**.** *For every $u \in L^2(\Omega)^3$, system (16.8) has exactly one solution (w_h, θ_h). Its multiplier is zero in Q_h. These are the complete assertions of source Propositions 4.5 and 4.11.*

Proof. The finite-dimensional orthogonal decomposition is $X_h = GQ_h \oplus V_h$. On V_h the bilinear form $(v, w) \mapsto (\operatorname{curl} v, \operatorname{curl} w)$ is an inner product: its zero vectors lie in both GQ_h and its orthogonal complement. Finite-dimensional Riesz representation gives a unique $w_h \in V_h$ representing $v_h \mapsto (u, \operatorname{curl} v_h)$. Take $\theta_h = 0$. The first equation holds on V_h by construction, and on GQ_h because all curls there vanish. The second equation is the definition of V_h, so this is a solution.

In any solution take $v_h = G\theta_h$ in the first equation. Its curl is zero and the resulting identity is $\|G\theta_h\|_2^2 = 0$. Injectivity gives $\theta_h = 0$. The second equation puts w_h in V_h, where the representing vector is unique. This proves the claim. All norms on this fixed finite-dimensional space are equivalent, but that statement has not supplied a bound independent of h. □

The discrete inf-sup constant in the gradient norm is at least one: for $q_h \neq 0$, choose $v_h = Gq_h$ and use its zero curl to obtain

$$\sup_{v_h \neq 0} \frac{(v_h, Gq_h)}{\|v_h\|_{H(\operatorname{curl})}} \geq \|Gq_h\|_2. \tag{16.9}$$

For a quotient $H^1(D)$ norm, equivalence follows from the cut-domain Poincaré inequality for that mesh. Uniformity of such equivalence under mesh-dependent cut choices is not asserted or required by (16.9).

Proposition 16.5 (Exact curl projection and its best error)**.** *Put $W_h = \operatorname{curl} X_h$. The solution just constructed obeys*

$$\operatorname{curl} w_h = P_{W_h} u, \qquad \|u - \operatorname{curl} w_h\|_2 = \min_{z_h \in W_h} \|u - z_h\|_2.$$

In particular $\|\operatorname{curl} w_h\|_2 \leq \|u\|_2$ and $\operatorname{div} \operatorname{curl} w_h = 0$ distributionally.

Proof. The first mixed equation with zero multiplier says exactly $(u - \operatorname{curl} w_h, z_h) = 0$ for every $z_h \in W_h$. The space W_h is finite-dimensional and closed. For any $z_h \in W_h$, the orthogonal decomposition

$$\|u - z_h\|_2^2 = \|u - \operatorname{curl} w_h\|_2^2 + \|\operatorname{curl} w_h - z_h\|_2^2$$

proves both the projection statement and the exact best-error identity. The contraction follows by taking $z_h = 0$; the divergence statement is the distributional identity $\operatorname{div} \operatorname{curl} = 0$. □

16.4 Weak edge moments and uniform bounds

We construct the weak interpolation operator and its scaled estimates before using it on continuous potentials.

Weak edge interpolation: source Lemma 4.7. For a tetrahedron K and $p > 2$, the source asserts continuity of I_K^N on

$$\mathcal{A}_p(K) = \{v \in L^p(K)^3 : \operatorname{curl} v \in L^p(K)^3,\ v \times n \in L^p(\partial K)^2\}. \tag{16.10}$$

The norm must include all three displayed quantities. The face and cell moments are already bounded by Hölder's inequality; the edge moments are not ordinary restrictions of L^p functions. One must construct their unique compatible extensions, prove equality of the definitions from the two incident faces, and extend the commuting and boundary-preservation identities. The smooth proof of (16.7) does not establish these points by itself.

The exponent $p > 2$ has a concrete role. With $p' = p/(p-1) < 2$, one can lift a polynomial on one edge of a triangle to $W^{1,p'}$ on that triangle, setting the trace to zero on its other edges. At a vertex the resulting derivative can behave as $1/r$, which is integrable to the p'th power in two dimensions exactly when $p' < 2$. One then lifts this face function, extended by zero on the other faces, into $W^{1,p'}(K)$. The source applies Stokes and the divergence identity for $\operatorname{curl} v$ to express the edge moment as

$$\int_K \operatorname{curl} v \cdot \nabla\Psi + \int_F (v \times n_F) \cdot \nabla_F \psi, \tag{16.11}$$

with orientation signs chosen to match the selected edge direction. Hölder gives a bound once these lifts and the weak identities are proved. We now supply that compatibility argument for the explicit polynomial moments; it does not assume that every $H(\operatorname{curl})$ field has an ordinary edge restriction.

Lemma 16.6 (Weak edge moments, including their compatibility)**.** *For $p > 2$ the canonical Nédélec interpolant extends linearly and continuously to $\mathcal{A}_p(K)$, with bound*

$$\|I_K^N v\|_2 \le C_{K,k,p}\big(\|v\|_{L^p(K)} + \|\operatorname{curl} v\|_{L^p(K)} + \|v \times n\|_{L^p(\partial K)}\big).$$

Its weak edge moments computed from the two incident faces agree after orientation. It commutes with curl into the weak Raviart–Thomas interpolant of Lemma 16.11. For assembly, assume additionally that $v \in H(\operatorname{curl}, \Omega)$ and that $v|_K \in \mathcal{A}_p(K)$ for every cell, with a fixed $2 < p < \infty$. Then the local interpolants assemble in Y_h. If, in addition, $v \in H_0(\operatorname{curl}, \Omega)$, they assemble in Y_h^0. Elementwise membership in $\mathcal{A}_p(K)$ alone is not an assembly hypothesis: it does not impose equality of the two tangential traces at an interior face. This proves the assertion of source Lemma 4.7, including the compatibilities used after that lemma.

Proof. Write $a = p' < 2$. The scalar $W^{1,a}$ facts used below are proved in Lemma 16.10. On a reference triangular face, let $E = \{\lambda_3 = 0\}$ and let μ be its prescribed polynomial. A concrete lift is

$$\psi = \mu\left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right) \frac{\lambda_1}{\lambda_1 + \lambda_3} \frac{\lambda_2}{\lambda_2 + \lambda_3}.$$

It has trace μ on E and zero on the other two edges. It is bounded, smooth away from vertices, and its derivative is bounded by a constant times the sum of the reciprocal distances to vertices. At the opposite vertex the product of the two fractions vanishes quadratically and cancels the denominator in the differentiated argument of μ. Thus $\psi \in W^{1,a}(F)$ because $\int_0^1 r^{1-a}\, dr < \infty$.

Extend it into K, with zero traces on the other faces, by the formula (16.13) with q replaced by this ψ. Differentiating the finite rational expression bounds its derivative by a finite sum of

$C(\lambda_i + \lambda_j)^{-1}$. The same two-dimensional transverse integral proves membership in $W^{1,a}(K)$. Denote this extension by Ψ. Define the edge moment by (16.11). Its two integrals are bounded by Hölder. Changing Ψ without changing its face traces does not change the result, since $\operatorname{div}\operatorname{curl} v = 0$ and their difference belongs to $W_0^{1,a}(K)$.

We need the following restricted weak Green identity. If g is continuous across the edges of the boundary, is smooth on each open face away from vertices, is bounded, and has face derivatives bounded by $C\sum_P |x-P|^{-1}$, then it has a $W^{1,a}(K)$ lift Φ such that

$$\int_K \operatorname{curl} v \cdot \nabla\Phi + \sum_F \int_F (v \times n_F) \cdot \nabla_F g = 0. \tag{16.12}$$

Here is an approximation proof for exactly this class of data. Extend the data from each face by barycentric projection onto it and blend these extensions using

$$w_i = \frac{\lambda_i^{-1}}{\sum_j \lambda_j^{-1}}$$

in the interior. The limit on face i is its prescribed datum. Near an edge, derivatives of the weights are $O(d^{-1})$; the two corresponding face extensions differ by $O(d)$ times their local Lipschitz bound, because their values agree on the edge. Subtracting that common value before differentiating the weights cancels the edge singularity. Only vertex bounds $C\sum_P |x-P|^{-1}$ remain. In three dimensions they are L^a for $a < 3$, so the blended lift Φ is in $W^{1,a}(K)$.

Multiply Φ by a smooth cutoff vanishing within distance ε of each vertex and equal to one outside distance 2ε, and denote the result by Φ_ε; let g_ε be its boundary trace. The resulting function is Lipschitz on the closed tetrahedron. More precisely, the ath powers of the bulk gradient-error norms are $O(\varepsilon^{3-a})$, while the ath powers of the face tangential-gradient-error norms are $O(\varepsilon^{2-a})$. These estimates follow by integrating r^{-a} with the bulk volume factor $r^2\,dr$ or the face area factor $r\,dr$, and by estimating the differentiated cutoff by C/ε. The zeroth-order errors tend to zero as well. Thus $\Phi_\varepsilon \to \Phi$ in $W^{1,a}(K)$ and $\nabla_F g_\varepsilon \to \nabla_F g$ in $L^a(F)$ on each face.

Here is the additional boundary-derivative property of this specific lift that is needed for mollification. Fix a point x_0 in the relative interior of face i, and let τ be any vector tangent to that face. In a neighborhood of x_0, all λ_j with $j \neq i$ are bounded below, and

$$w_i = \frac{1}{1 + \lambda_i \sum_{j\neq i} \lambda_j^{-1}}, \qquad w_j = \frac{\lambda_i/\lambda_j}{1 + \lambda_i \sum_{\ell\neq i} \lambda_\ell^{-1}} \quad (j \neq i).$$

Since $\partial_\tau \lambda_i = 0$, these formulas give $w_i \to 1$, $w_j \to 0$, and $\partial_\tau w_i, \partial_\tau w_j = O(\lambda_i)$. The barycentric projection onto face i tends to the identity on that face. Projections onto the other faces remain away from their vertices, so the assumed face first-derivative bounds are uniform there, including when the projected points approach an open edge. Differentiating the finite blend therefore shows

$$\partial_\tau \Phi_\varepsilon(x) \longrightarrow \partial_\tau g_\varepsilon(x_0) \quad \text{as } x \to x_0 \text{ from inside } K.$$

This is stronger than merely having a Lipschitz extension.

Use the finite graph-reflection construction of Lemma 16.10 to extend Φ_ε to an ambient Lipschitz function. Near a relative face-interior point, the graph of that boundary is affine, and each active reflection fixes its plane pointwise. Its derivative therefore maps τ to τ. The same tangential derivative limit holds on the reflected side. Multiplication by the smooth partition functions preserves the limits; their sum gives $\partial_\tau g_\varepsilon$ because the partition equals one near the boundary. Thus the extension has this tangential limit from both sides of the face.

Mollify that particular extension. The mollifications converge in bulk $W^{1,a}$, and the two-sided tangential derivative limit implies that their tangential derivatives converge at every relative

face-interior point where the displayed limits are taken. For fixed ε their Lipschitz constants are uniformly bounded by that of the extension. Dominated convergence therefore gives convergence in face L^a; the edges have face measure zero. This argument does not assert the corresponding boundary-gradient convergence for an arbitrary Lipschitz extension.

The ordinary curl Green identity applied to the gradient of each smooth mollification says that its two integrals in (16.12) sum to zero. First pass through the mollifications and then let $\varepsilon \downarrow 0$, using Hölder and the convergence just proved. This proves (16.12). The identity is independent of the chosen lift because the curl is divergence-free.

For two faces incident to E, take their lifts of the same μ and put zero on the other faces. The resulting boundary function is continuous across every edge: its traces agree on E and are zero on the other edges. It satisfies the class just considered. Equation (16.12) says that the two proposed edge moments add to zero in the induced boundary orientations. They therefore give the same moment for a fixed edge orientation. Applying that identity to the difference of two face lifts with identical edge traces also shows independence of the face lift within this class. For smooth fields the formula reduces to planar Stokes, hence agrees with the original edge integral.

The bounded face, cell, and now edge moments determine $I_K^N v$ by unisolvence and give the asserted continuity. To verify commutation, for any $q \in P_{k-1}(F)$ sum the three edge lift formulas on a face. The sum of the face lifts minus q has zero trace on its edges, so (16.12) gives precisely the weak Stokes identity with test q. This identifies the face Raviart–Thomas moments of $\operatorname{curl} I_K^N v$ with those of $\operatorname{curl} v$. Their cell moments agree by the ordinary weak curl Green formula with polynomial vector tests. Raviart–Thomas unisolvence proves commutation.

Across a shared mesh face, the two tangential traces of the input agree as distributions and hence as the given L^p functions. Their face moments agree. The two volume lifts of the same face function glue to a $W^{1,a}$ function supported on the two cells; its outer traces vanish. Testing the global divergence-free field $\operatorname{curl} v$ proves that the corresponding normal moments agree with opposite outward signs. The edge formula therefore agrees from the two cells as well. Repeating through the face-connected star of an edge gives globally consistent edge moments. The face-connectedness follows by intersecting the conforming tetrahedra with a small transverse disk at an interior point of the edge: the disk in the Lipschitz domain is a connected full disk or sector, partitioned into their successive angular sectors. Face unisolvence proves conformity.

If v has zero tangential trace, the continuous curl theorem gives zero normal trace for $\operatorname{curl} v$. The zero-extension divergence identity extends its zero-normal Green identity to $W^{1,a}(\Omega)$ tests, as in the proof of Lemma 16.11. Thus the volume term of a boundary-face edge formula is zero, and the face term is zero because $v \times n = 0$. All boundary face and edge moments vanish. Face unisolvence now gives the required zero tangential trace of the interpolant. □

Remark 16.7 (Why assembly needs a global graph-space hypothesis). Let K_+ and K_- have common face in $\{x_3 = 0\}$ and lie on opposite sides. Put $v = e_1$ on K_+ and $v = 0$ on K_-. Every local field is smooth and belongs to every $\mathcal{A}_p$, and its interpolant is itself because constants are reproduced. The tangential jump across the common face is nonzero. Testing the piecewise field distributionally therefore gives a nonzero surface term in its curl, so it is not in the global $H(\operatorname{curl})$ space. Elementwise regularity alone cannot imply global conformity. All uses of Lemma 16.6 below supply its global hypothesis.

A concrete lift for weak face moments. There is a useful explicit part of the corresponding Raviart–Thomas construction. On the reference tetrahedron let $\lambda_0, \dots, \lambda_3$ be its barycentric coordinates and let $F = \{\lambda_0 = 0\}$. For a polynomial q on F put, in the interior,

$$\Psi_q(x) = q\left(\frac{\lambda_1}{1-\lambda_0}, \frac{\lambda_2}{1-\lambda_0}, \frac{\lambda_3}{1-\lambda_0}\right) \prod_{i=1}^{3} \frac{\lambda_i}{\lambda_i + \lambda_0}. \tag{16.13}$$

This function is bounded, equals q on the relative interior of F, and has zero trace on the relative interiors of the other faces. Differentiating its rational factors gives

$$|\nabla \Psi_q| \le C_q \left(1 + \sum_{i=1}^{3} (\lambda_0 + \lambda_i)^{-1}\right).$$

To check the only apparent extra singularity, near the opposite vertex $1 - \lambda_0 = s$ is small, while the product of the three factors is $O(s^3)$; differentiating the polynomial in normalized coordinates costs only $O(s^{-1})$ and is harmless. Each displayed singular term is transverse to an edge, and its p'th power has finite integral for $p' < 2$: in two transverse nonnegative coordinates its integral is bounded by a constant times $\int_0^1 r^{1-p'}\, dr$. Thus $\Psi_q \in W^{1,p'}(K)$ for every $1 < p' < 2$.

If $z \in L^p(K)^3$ is divergence-free and $p > 2$, this supplies the bounded candidate face moment $\int_K z \cdot \nabla \Psi_q$. Independence of the lift follows from density of compactly supported smooth functions in the zero-trace $W^{1,p'}$ space, by applying the distributional divergence identity. A full use of this argument therefore also needs the $W^{1,p'}$ versions of trace, gluing, and zero-trace density; the earlier H^1 proofs cannot be cited verbatim without checking their exponents. They can be adapted by replacing Cauchy–Schwarz with Hölder. Moreover, compatibility of these face moments with the global normal trace must be proved, including summation over all faces forming a cut.

The required scaled stability estimate. Let $s \in (1/2, 1]$. For fields $v \in H^s(K)^3$ whose curl belongs to $P_{k-1}(K)^3$, the canonical weak edge interpolant must satisfy

$$\|I_K^N v\|_{L^2(K)} \le C\big(\|v\|_{L^2(K)} + h_K^s |v|_{H^s(K)} + h_K \|\operatorname{curl} v\|_{L^2(K)}\big), \tag{16.14}$$

with a constant depending only on k, s, and shape regularity. The fractional seminorm here is the Slobodeckij seminorm when $s < 1$. This is the precise estimate needed in the source's proof; an unscaled continuity assertion on each physical element is not enough. The route through (16.10) also requires $H^s(K)$ face traces in $L^p(F)$ for some $p > 2$, a consequence of the positive margin $s - 1/2$, not merely of $s > 0$.

For clarity, the affine scaling uses the covariant pullback $\widehat{v} = B_K^T (v \circ F_K)$ and the exact formula

$$(I_K^N v) \circ F_K = B_K^{-T} I_{\widehat{K}}^N \widehat{v}, \qquad \operatorname{curl} \widehat{v} = (\det B_K) B_K^{-1} (\operatorname{curl} v) \circ F_K.$$

The formula follows first for smooth fields from the chain rule and matching transformed moments, and then requires the asserted weak continuity for rough fields. On a fixed polynomial space the inverse bound follows from equivalence of finite-dimensional norms, and its physical scaling is

$$\|z_h\|_{L^p(K)} \le C h_K^{3/p - 3/2} \|z_h\|_{L^2(K)}.$$

Shape regularity controls $\|B_K\| \le C h_K$, $\|B_K^{-1}\| \le C h_K^{-1}$, and $c h_K^3 \le |\det B_K| \le C h_K^3$. These factors explain the last term in (16.14); they must not be replaced by quasi-uniform mesh assumptions.

Lemma 16.8 (Scaled stability of the weak edge interpolant)**.** *Estimate* (16.14) *holds for every* $s \in (1/2, 1]$*, on a shape-regular family, when* $\operatorname{curl} v \in P_{k-1}(K)^3$*.*

Proof. On the reference cell choose $p > 2$ close enough to 2 that the H^s norm controls both the volume L^p norm and the L^p norms of the face traces. The fractional extension, plane trace, and dyadic embedding lemmas in the regularity section prove this because $s - 1/2 > 0$; for $s = 1$ use the proved H^1 extension and the same plane trace estimate. Lemma 16.6 therefore gives

$$\|I_{\widehat{K}}^N \widehat{v}\|_2 \le C\big(\|\widehat{v}\|_{H^s(\widehat{K})} + \|\operatorname{curl} \widehat{v}\|_{L^p(\widehat{K})}\big).$$

Under the covariant transformation the output L^2 norm gains a factor at most $Ch_K^{1/2}$. The transformed H^s norm is bounded by $Ch_K^{-1/2}(\|v\|_2 + h_K^s|v|_{H^s(K)})$, directly from the double-integral seminorm (or first derivatives if $s = 1$). The transformed curl term is bounded by $Ch_K^{2-3/p}\|\operatorname{curl} v\|_{L^p(K)}$. The polynomial inverse estimate above turns its final coefficient into $h_K^{1/2+2-3/p+3/p-3/2} = h_K$. These bounds prove exactly (16.14). □

Theorem 16.9 (Uniform coercivity for Y_h; original Proposition 4.6)**.** *In the case $X_h = Y_h$, $\|w_h\|_2 \le C\|\operatorname{curl} w_h\|_2$ for every $w_h \in V_h$, with C independent of the shape-regular mesh. No polyhedral fractional regularity theorem is needed here.*

Proof. For $f_h = \operatorname{curl} w_h$, the bounded regular-potential theorem supplies $a \in H^1(\Omega)^3$ with $\operatorname{curl} a = f_h$ and $\|a\|_{H^1} \le C\|f_h\|_2$. Its hypotheses hold because every curl has zero component flux. The weak interpolant is defined for a: its curl is piecewise polynomial, and its face traces have an L^p gain with $p > 2$. Commutation gives $\operatorname{curl} I_h^N a = f_h$. Thus $w_h - I_h^N a \in \nabla_b \Theta_h$, and orthogonality gives $\|w_h\|_2 \le \|I_h^N a\|_2$. Sum Lemma 16.8 with $s = 1$ over the mesh and use the bound on a to obtain the result. This uses the unconstrained regular potential, not the possibly less regular boundary-normalized potential. □

16.5 A concrete divergence-preserving approximation at every positive order

The following construction closes the Raviart–Thomas approximation bridge without assuming a fractional Bramble–Hilbert theorem. It uses the fractional embedding and reflection estimates proved in the regularity section. The $H^\sigma(\Omega)$ norm is the restriction norm: take the infimum of the $H^\sigma(\mathbb{R}^3)$ norms of extensions.

Lemma 16.10 (The $W^{1,a}$ scalar facts used for moments)**.** *For $1 < a < \infty$, scalar extension, trace, zero-trace density, and interface gluing on a bounded Lipschitz set hold in $W^{1,a}$ in the same senses used earlier for H^1. In particular two $W^{1,a}$ functions glue across a flat face when their traces agree, and the kernel of the trace is the closure of compactly supported smooth functions in $W^{1,a}$.*

Proof. Here are the changes in the already given scalar proof, to specify the exponents. On an interval, absolute continuity and Hölder give

$$|f(0)|^a \le 2^{a-1}\left(h^{-1}\int_0^h |f|^a + h^{a-1}\int_0^h |f'|^a\right).$$

Integrating transverse variables proves the flat L^a trace bound. The coordinate-line characterization follows from Fubini and the one-dimensional weak derivative argument. Reflection has norm bounded by $2^{1/a}$, since its function and first-derivative ath powers integrate to twice those on the half box. In the vertical integration identity the interface terms cancel precisely when the two traces agree. The graph composition rule has the same bounded coefficients as before, and its change of volume is still the elementary vertical translation. Finite localization therefore gives extension with fixed bounded support. Convolution converges in L^a both for the function and each weak derivative. If the trace is zero, extend by zero, translate the localized pieces inward by a positive distance, and mollify at less than half that distance, exactly as in the scalar zero-trace proof. The supports are then compact inside the domain and the convergence is in $W^{1,a}$. Conversely trace continuity makes every limit of such functions have zero trace. These arguments prove all the assertions, without a Hilbert-space representation theorem or a fractional trace space. □

Lemma 16.11 (Weak normal moments and a solenoidal interpolant)**.** *Fix $p > 2$ and let $p' = p/(p-1)$. For every divergence-free $z \in L^p(K)^3$ the Raviart–Thomas moments have a well-defined extension, yielding $J_K z \in P_{k-1}(K)^3$ with $\operatorname{div} J_K z = 0$. This operator reproduces every divergence-free member of $P_{k-1}(K)^3$ and satisfies*

$$\|J_K z\|_2 \le C h_K^{3/2-3/p} \|z\|_{L^p(K)}. \tag{16.15}$$

For a globally divergence-free $z \in L^p(\Omega)^3$ the local fields assemble into $J_h z \in H(\operatorname{div}, \Omega)$. They preserve every boundary-component flux. If z has zero normal trace, they preserve that condition and the total flux through each mesh-aligned cut.

Proof. For a polynomial q on a face F define its outward face moment by $M_F(z, q) = \int_K z \cdot \nabla \Psi_q$, using (16.13). Any two $W^{1,p'}$ lifts with the same face traces differ by $W_0^{1,p'}(K)$. The distributional divergence equation extends to that space by Lemma 16.10 and Hölder. Hence the value is independent of the lift. For smooth fields it is the ordinary face moment: first approximate the lift in $W^{1,p'}$ by ambient smooth functions, use the smooth divergence formula, and pass to the trace limit. Together with the ordinary cell moments these values determine $J_K z \in RT_{k-1}$ by Lemma 16.2.

For $q \in P_{k-1}(K)$, the sum of the separate lifts of its face restrictions differs from q by a zero-trace $W^{1,p'}$ function. Thus $\sum_F M_F(z, q) = \int_K z \cdot \nabla q$. Integrating the polynomial $\operatorname{div} J_K z$ against q and using its moments proves that integral is zero. Since the divergence is itself in P_{k-1}, it vanishes. The polynomial Raviart–Thomas argument then puts $J_K z$ in P_{k-1}^3. Polynomial reproduction follows from the equality with classical moments and unisolvence.

On the reference cell all moment functionals are bounded on L^p solenoidal fields by the finite collection of $W^{1,p'}$ lifts and Hölder's inequality. The inverse moment map is a map between finite-dimensional spaces and is bounded; this gives $\|J_{\widehat{K}} \widehat{z}\|_2 \le C \|\widehat{z}\|_p$. The contravariant transformation is $\widehat{z} = (\det B_K) B_K^{-1} (z \circ F_K)$. Its L^p norm is at most $C h_K^{2-3/p} \|z\|_p$, whereas transforming the output back multiplies its L^2 norm by at most $C h_K^{-1/2}$. The moments transform identically, by the affine volume and oriented face Jacobians. This proves (16.15).

For two adjacent cells, glue their face lifts with the same trace q on their common face and zero traces elsewhere. Lemma 16.10 makes the result a zero-trace $W^{1,p'}$ function on the two-cell patch. The global divergence identity says that the two outward face moments add to zero. Consequently the two normal polynomials of $J_h z$ agree with opposite outward orientations. This proves global conformity.

For a boundary component, sum the cell lifts of the constant 1 on its boundary faces, setting their other face traces to zero. They glue to a $W^{1,p'}(\Omega)$ function χ having trace one on that component and zero on the others. Subtracting the smooth cutoff η_i gives a zero-trace function, so

$$\sum_{F \subset \Gamma_i} M_F(z, 1) = \int_\Omega z \cdot \nabla \chi = \int_\Omega z \cdot \nabla \eta_i = F_i(z).$$

This proves preservation of component fluxes.

If $z \cdot n = 0$, its zero extension is divergence-free and in $L^p(\mathbb{R}^3)^3$. Consequently its zero-normal Green identity extends to every $W^{1,p'}(\Omega)$ test: extend the test to an ambient compactly supported $W^{1,p'}$ function, approximate smoothly, and use the divergence-free zero extension. A boundary-face lift, set equal to zero on all other cells, is such a global test, because its traces vanish on the interior faces of its cell. Testing it proves that its boundary-face moment is zero. Thus every boundary normal polynomial of $J_h z$ vanishes.

Finally make a broken lift for a cut by adding, on its minus side, the face lifts of 1 on its constituent mesh faces, and putting zero on their plus sides. It is $W^{1,p'}$ on the cut domain, has

jump one on that cut and zero on the other cuts. Subtract the earlier H^1 unit-jump lift. The difference has no jumps and glues to $W^{1,p'}(\Omega)$. The zero-normal Green identity extends to all such tests by the preceding paragraph. It follows that the sum of its oriented face moments equals $C_j(z)$. The same summation for $J_h z$ proves cut-flux preservation. □

Lemma 16.12 (Averaged Taylor polynomials with a distributional definition)**.** *For every nonnegative integer m there is a linear operator $Q_{m,K} : L^2(K) \to P_m(K)$ reproducing $P_m(K)$, with*

$$\|Q_{m,K} f\|_{L^p(K)} \le C h_K^{3/p-3/2} \|f\|_2, \qquad \|f - Q_{m,K} f\|_{L^\infty(K)} \le C h_K^{m+1} \|D^{m+1} f\|_{L^\infty(K)} \tag{16.16}$$

for smooth f in the second estimate. Applied componentwise, $Q_{k-1,K}$ takes distributionally divergence-free fields to divergence-free polynomials.

Proof. Fix once and for all a nonnegative $\theta \in C_c^\infty(B(0,1))$ with integral one. Choose an interior ball $B(a_K, r_K) \subset K$ with $c_\gamma h_K \le r_K \le C_\gamma h_K$, and set

$$\theta_K(y) = r_K^{-3} \theta((y - a_K)/r_K).$$

Shape regularity supplies these balls. This choice gives, for all multiindices needed below,

$$\|D^\alpha \theta_K\|_\infty \le C_{\alpha,\gamma} h_K^{-3-|\alpha|}.$$

In particular Leibniz's rule shows that each summand of $D_y^\alpha\{\theta_K(y)(x-y)^\alpha\}$ is bounded by $C h_K^{-3}$ for $x \in K$: a derivative of order $|\beta|$ on θ_K costs $h_K^{-3-|\beta|}$, while the remaining polynomial factor has degree $|\beta|$ and size at most $C h_K^{|\beta|}$. The constants are uniform over the mesh. For smooth f take the Taylor polynomial of degree m about y and average it against $\theta_K(y)$. Equivalently, after integration by parts, define for every L^2 input

$$Q_{m,K} f(x) = \sum_{|\alpha| \le m} \frac{(-1)^{|\alpha|}}{\alpha!} \int_K f(y) D_y^\alpha\{\theta_K(y)(x-y)^\alpha\}\, dy.$$

The preceding scaled construction bounds every kernel by $C h_K^{-3}$ on $K \times \operatorname{supp} \theta_K$. Cauchy–Schwarz gives the L^∞ bound for Qf, and then the first estimate in (16.16). Taylor's formula on the segment from y to x, which stays in the convex tetrahedron, gives polynomial reproduction and the second estimate. For smooth functions direct differentiation gives $\partial_i Q_{m,K} f = Q_{m-1,K} \partial_i f$ when $m \ge 1$. The integrated formula makes both sides meaningful for distributions and preserves this identity by integration by parts against the compactly supported kernels. Summing its components proves the divergence assertion; for $m = 0$ the output is constant. □

Theorem 16.13 (Concrete solenoidal approximation for all $0 < \sigma \le k$)**.** *For a shape-regular family and every $0 < \sigma \le k$, the interpolant of Lemma 16.11 is defined for each $u \in H^\sigma(\Omega)^3$ with* $\operatorname{div} u = 0$ *and satisfies*

$$\|u - J_h u\|_2 \le C_\sigma h^\sigma \|u\|_{H^\sigma(\Omega)}. \tag{16.17}$$

It has all the conformity, divergence, boundary, and total-flux properties of that lemma.

Proof. Choose $0 < t < \min(1, \sigma)$ and then $p > 2$ sufficiently close to 2 that $3(1/2 - 1/p) < t$. The intrinsic fractional extension and embedding of Lemma 8.16, followed by scaling as in (8.17), give

$$h_K^{3/2-3/p} \|f\|_{L^p(K)} \le C\big(\|f\|_{L^2(K)} + h_K^t |f|_{H^t(K)}\big). \tag{16.18}$$

The global fractional embedding also puts u in $L^p(\Omega)$, so $J_h u$ is defined. Choose an extension $U \in H^\sigma(\mathbb{R}^3)$ of norm at most twice the restriction norm. It need not be divergence-free outside Ω.

It suffices to consider $h \le 1$; larger h are bounded by $\operatorname{diam}\Omega$ and can be absorbed into the constant. Fix a smooth Fourier cutoff χ, equal to one on the unit ball and zero outside the ball of radius two, and decompose $U = L + H$ by $\widehat{L}(\xi) = \chi(h\xi)\widehat{U}(\xi)$. Plancherel and the support of $1 - \chi(h\xi)$ give

$$\|H\|_2 + h^t \|H\|_{H^t} \le Ch^\sigma \|U\|_{H^\sigma}, \qquad \|D^k L\|_2 \le Ch^{\sigma-k}\|U\|_{H^\sigma}. \tag{16.19}$$

Write $Q_K = Q_{k-1,K}$ and $e_K = u - Q_K u$. It is divergence-free. Reproduction and Lemma 16.11 therefore give

$$\|u - J_K u\|_2 \le Ch_K^{3/2-3/p}\|e_K\|_{L^p(K)}.$$

For the high-frequency contribution $H - Q_K H$, use (16.18) and the first Taylor bound. Squaring and summing gives at most

$$C\big(\|H\|_2^2 + h^{2t}|H|^2_{H^t(\mathbb{R}^3)}\big) \le Ch^{2\sigma}\|U\|^2_{H^\sigma}.$$

The sum of local fractional seminorms is at most the global one.

For the low-frequency contribution the second Taylor bound gives

$$h_K^{3/2-3/p}\|L - Q_K L\|_{L^p(K)} \le Ch_K^{k+3/2}\sup_K |D^k L|.$$

The following explicit estimate controls those suprema uniformly even on a graded mesh:

$$\sum_K |K| \sup_K |f|^2 \le C\|f\|^2_{L^2(\mathbb{R}^3)} \quad \text{if } \operatorname{supp}\widehat{f} \subset \{|\xi| \le 2/h\}. \tag{16.20}$$

To prove it choose a smooth cutoff equal to one on that Fourier ball, with inverse transform $\kappa_h(x) = h^{-3}\kappa(x/h)$. Then $f = \kappa_h * f$, and repeated integration by parts in its smooth compactly supported Fourier integral gives $|\kappa(x)| \le C_M(1 + |x|)^{-M}$. For $x, y \in K$ one has $|x - y| \le h$, so

$$|f(y)| \le C_M \int h^{-3}(1 + |x - z|/h)^{-M}|f(z)|\,dz.$$

Take the supremum over $y \in K$, square, and integrate x over K. Summing and using the L^1 norm of this convolution kernel proves (16.20) by Young's inequality. Apply it to each derivative of L. Shape regularity makes $|K|$ comparable to h_K^3, so the squared low-frequency sum is at most $Ch^{2k}\|D^k L\|_2^2$. Equation (16.19) bounds this by $Ch^{2\sigma}\|U\|^2_{H^\sigma}$. Adding the two contributions and taking square roots proves (16.17). The conservation assertions are those already proved in Lemma 16.11. □

16.6 The concrete curl image and the full fractional error range

Write

$$\begin{aligned} Z_h &= \{z_h \in H(\operatorname{div}, \Omega) : z_h|_K \in P_{k-1}(K)^3\}, \\ Z_h^* &= \{z_h \in Z_h : \operatorname{div} z_h = 0, F_i(z_h) = 0 \text{ for every } i\}, \\ Z_h^{0,*} &= \{z_h \in Z_h : \operatorname{div} z_h = 0, z_h \cdot n = 0,\ C_j(z_h) = 0 \text{ for every } j\}. \end{aligned}$$

The proved continuous necessity and the local polynomial lemma give

$$\operatorname{curl} Y_h \subset Z_h^*, \qquad \operatorname{curl} Y_h^0 \subset Z_h^{0,*}. \tag{16.21}$$

We next prove surjectivity, including the boundary-constrained case.

Lemma 16.14 (Discrete scalar lifts with prescribed jumps and boundary values)**.** *There exist $r_{h,j} \in \Theta_h$, already piecewise affine, with $[r_{h,j}]_\ell = \delta_{j\ell}$. If $q \in \Theta$ has a polynomial trace in $P_k(F)$ on every boundary mesh face, then there is $q_h \in \Theta_h$ with the same boundary trace and the same constant jumps as q.*

Proof. Use the unit-jump lift r_j constructed earlier from finitely many ambient smooth cutoffs times local side indicators. On each local side it is smooth up to the boundary, so it has well-defined values at the corresponding copies of the mesh vertices. Interpolate those values affinely on every tetrahedron. Noncut faces have matching vertex values and therefore matching affine traces. Across a cut face the differences at all three vertices are one for cut j and zero for the other cuts; its affine jump is that constant. Disjoint cut closures ensure that a vertex is not on two different cuts. This constructs $r_{h,j}$; no uniform bound on their norms is asserted or used.

Let $c_j = [q]_j$ and set $q_0 = q - \sum_j c_j r_{h,j}$. Its jumps vanish, so it glues to $H^1(\Omega)$, and its boundary traces remain piecewise P_k. They agree polynomially along every common boundary edge. Here is a justification at the trace threshold: flattening a boundary chart and the proved plane trace theorem put the trace in local $H^{1/2}$ on its two-dimensional graph. A nonzero jump between two face polynomials along an open edge interval makes its Slobodeckij integral infinite. Indeed, in coordinates along and across the edge, restrict the two transverse distances to $r, r' > 0$ and the difference of the along-edge coordinates to magnitude at most $r + r'$. A jump bounded away from zero gives a lower bound proportional to $\int_0^\varepsilon \int_0^\varepsilon (r + r')^{-2}\, dr\, dr'$, which diverges. Thus the two polynomial traces agree on the whole edge, by continuity. Values also agree at the vertices because the face star of a boundary vertex is connected and its successive faces meet along edges.

Extend these boundary polynomials to a continuous piecewise P_k function by scalar Lagrange degrees of freedom, assigning the prescribed value at every boundary barycentric-lattice node and zero at every remaining node. To recall unisolvence without importing a finite-element theorem: a polynomial vanishing at the degree-k barycentric lattice vanishes on every face by dimension induction. It is then the product of all barycentric coordinates times a polynomial of degree $k - d - 1$. At the interior nodes this latter polynomial vanishes on the affine image of its own barycentric lattice; degree induction makes it zero. If $k = d + 1$, the remaining factor is constant and the unique interior lattice node makes it zero; the affine lattice rescaling is used only when $k - d - 1 > 0$. In dimension one the assertion is the root bound for a polynomial. This proves both the nodal unisolvence and matching of neighboring face polynomials, hence global H^1 conformity of the extension $q_{0,h}$. Finally $q_h = q_{0,h} + \sum_j c_j r_{h,j}$ has precisely the required trace and jumps. □

Theorem 16.15 (Both discrete curl images)**.** *The inclusions* (16.21) *are equalities:*

$$\operatorname{curl} Y_h = Z_h^*, \qquad \operatorname{curl} Y_h^0 = Z_h^{0,*}.$$

Proof. For $z_h \in Z_h^*$ the bounded regular-potential theorem gives $a \in H^1(\Omega)^3$ with $\operatorname{curl} a = z_h$. The weak edge interpolant is defined for this a, and commutation and polynomial reproduction give $\operatorname{curl} I_h^N a = z_h$. This proves the first surjectivity assertion. Notice that no ordinary edge restriction of the H^1 field has been assumed.

Now let $z_h \in Z_h^{0,*}$. The first assertion supplies $a_h \in Y_h$ with this curl. The continuous normal-potential theorem supplies $w \in H_0(\operatorname{curl}, \Omega)$ with the same curl. The continuous maximal-kernel theorem then gives $a_h - w = \nabla_b q$ for some $q \in \Theta$. On a boundary mesh face F, its trace has surface gradient equal to the tangential polynomial $a_h|_F$, because the tangential trace of w is zero. The equality between the tangential derivative of the scalar trace and the trace of its gradient follows by local H^1 approximation on the flat patch and distributional integration against smooth compactly supported face tests.

That face polynomial has zero surface curl: its scalar surface curl is $z_h \cdot n_F = 0$. The two-dimensional polynomial exactness argument therefore makes it the gradient of a scalar $P_k(F)$ polynomial. Subtracting this polynomial from the trace of q leaves a distribution on the connected face with zero tangential gradient, hence a constant. Thus the trace of q itself belongs to $P_k(F)$ on every boundary face. Lemma 16.14 gives $q_h \in \Theta_h$ with the same boundary trace and jumps. The field $a_h - \nabla_b q_h$ lies in Y_h, has curl z_h, and has zero tangential trace on every boundary face. It therefore belongs to Y_h^0, proving the second surjectivity assertion without a polyhedral shift theorem. □

Lemma 16.16 (A stable scalar projector for the boundary-polynomial subspace)**.** *Let*

$$S_h = \{q \in \Theta : q|_F \in P_k(F) \text{ on every boundary mesh face } F\}.$$

There is a linear map $A_h : S_h \to \Theta_h$ preserving these boundary traces and all constant jumps, reproducing Θ_h, and satisfying

$$\|\nabla_b A_h q\|_2 \le C\|\nabla_b q\|_2. \tag{16.22}$$

The constant depends only on k and mesh shape regularity, not on the chosen mesh-aligned cut presentation. This assertion is deliberately made on S_h; no pointwise boundary values are assigned to general H^1 traces.

Proof. Use the barycentric-lattice nodes from Lemma 16.14, retaining their two copies at a cut. On a face F containing a node a, let $\pi_{a,F} \in P_k(F)$ be the $L^2(F)$ dual polynomial to evaluation at that node:

$$\int_F \pi_{a,F} p = p(a) \quad (p \in P_k(F)).$$

It exists by the positive-definite polynomial Gram matrix, and scaling gives $\|\pi_{a,F}\|_{L^2(F)} \le C h_F^{-1}$. There is similarly a cell dual with L^2 norm at most $C h_K^{-3/2}$. Both functionals take the value one on the constant function one.

At a physical boundary node choose an incident boundary face and use its dual average of the trace of q. This is its actual polynomial node value, independent of which incident boundary face is selected, after removing the constant jump as in the preceding scalar-lift proof. For a pair of cut-node copies with no incident physical boundary face, choose a common cut face and use its two dual trace averages. Their difference is $[q]_j$, because the trace difference is that constant and the dual integrates constants correctly. If a cut-node pair meets the physical boundary, use an incident boundary face for one copy and assign the other by adding or subtracting $[q]_j$. When both copies meet physical boundary faces this agrees with both prescribed polynomial values, by the proof of Lemma 16.14. This rule also covers a tangential cut contact where only one local copy has an incident physical boundary face. All remaining nodes use a dual cell average in one incident tetrahedron. Make these choices once for the mesh. The resulting nodal coefficients define $A_h q$; they preserve the boundary polynomials and jumps, and reproduce every member of Θ_h by nodal unisolvence.

We give the local stability argument because global cut-domain Poincaré constants are not suitable for mesh-dependent cuts. The star of a node consists of uniformly boundedly many tetrahedra of mutually comparable diameters. To verify these familiar geometric facts, normalize each tetrahedron by its diameter: shape regularity bounds the singular values of its affine map above and below, hence bounds all solid and dihedral angles below. The total solid angle at a vertex is at most 4π, so its valence is bounded. Two cells sharing a face have comparable diameters, because each edge of a shape-regular cell has length comparable to its diameter. The face adjacency graph of a vertex, edge, or face star is connected: a small local link is a disk or sphere cut into successive tetrahedral sectors, since the original domain is Lipschitz. A chain has length bounded by the valence, giving the asserted comparability throughout the star.

For a node a, let its star be the union of the tetrahedra containing it, with its relative interior understood for Sobolev functions. Choose $\varepsilon_a > 0$ smaller than the distance from a to every face of these tetrahedra which does not contain a, and smaller than its distance to every nonincident cell. These finitely many distances are positive. On that ball, the star coincides with the corresponding local part of Ω. Every tetrahedron in the star contains the segment from a to each of its points. An internal face of the star contains a, because the two incident cells both contain a and their intersection is exactly that common face. Thus all internal faces, and in particular all cut faces internal to the star, are radial faces through a.

If no cut face is internal to the star, set $g = q$; there is no jump to remove. Otherwise its cut contains a, and disjoint cut closures exclude every other cut from the internal faces. Along a ray from a the cut-side label is constant, because crossing an internal radial face is impossible unless the ray lies in that face. Radial restriction to $B(a, \varepsilon_a)$ therefore identifies the components of the cut star with the two local branches stipulated by the admissible cut geometry, including at a cut-rim node. Write $J = [q]_j = q^- - q^+$ for that cut's fixed jump. Choose the branch shifts explicitly by

$$(g^-, g^+) = \begin{cases} (q^-, q^+ + J), & K \text{ is on the minus side,} \\ (q^- - J, q^+), & K \text{ is on the plus side.} \end{cases}$$

In either case $g = q$ on the specified cell K. The traces then agree across every internal face. Elementwise integration by parts shows that the resulting function g is in H^1 of the uncut star, with gradient exactly the broken gradient of q; no additional edge or vertex distribution is introduced, since the first-derivative boundary terms are only face terms and cancel pairwise.

The coefficient on the copy of a belonging to K is the chosen face or cell dual average of this g. For a cut-rim node whose other branch supplies the physical boundary face, this is exactly what the rule of transferring its value by the jump constant does. Thus that exceptional nodal rule uses the same uncut-star estimate, without a Poincare constant on either possibly thin cut branch.

Let c_K be the mean of q on K. On the star one has

$$\|g - c_K\|_{L^2(\mathrm{star}(a))} \le C h_K \|\nabla g\|_{L^2(\mathrm{star}(a))}. \tag{16.23}$$

For completeness, the cell Poincaré inequality follows by affine scaling from the reference tetrahedron. Across a common face F of two cells, their means c_1, c_2 satisfy

$$|c_1 - c_2|\,|F|^{1/2} \le \|g - c_1\|_{L^2(F)} + \|g - c_2\|_{L^2(F)} \le C h_K^{1/2} (\|\nabla g\|_{L^2(K_1)} + \|\nabla g\|_{L^2(K_2)}).$$

The last inequality is the scaled H^1 trace inequality followed by cell Poincaré. Propagate this estimate along a bounded face chain from K and sum the cell Poincaré inequalities. This proves (16.23) with a uniform constant.

The face-dual estimate, scaled trace, and (16.23) now give for the chosen coefficient α_a

$$|\alpha_a - c_K| \le C h_K^{-1/2} \|\nabla_b q\|_{L^2(\mathrm{star}(a))}.$$

The cell-dual case follows from its L^2 bound and (16.23). A local nodal shape function has gradient norm at most $C h_K^{1/2}$, by affine scaling. Subtract c_K from all the nodal coefficients on K; this does not change the gradient since the shape functions sum to one. Summing their finitely many estimates bounds $\|\nabla A_h q\|_{L^2(K)}$ by the broken gradient norm over its node stars. Each cell occurs in boundedly many such stars, again by bounded valence. Squaring and summing proves (16.22). □

Theorem 16.17 (Uniform coercivity for Y_h^0; original Proposition 4.12)**.** *In the case $X_h = Y_h^0$, $\|w_h\|_2 \le C \|\operatorname{curl} w_h\|_2$ for every $w_h \in V_h$, uniformly on the shape-regular family.*

Proof. Set $f_h = \operatorname{curl} w_h$. As in Theorem 16.9, a bounded regular H^1 potential and its weak edge interpolant give $a_h \in Y_h$ with $\operatorname{curl} a_h = f_h$ and $\|a_h\|_2 \le C\|f_h\|_2$. Let $w = R_N f_h$ be the continuous boundary-normalized potential; the already proved main theorem gives $\|w\|_2 \le C\|f_h\|_2$ and zero tangential trace. The maximal-kernel theorem gives $a_h - w = \nabla_b q$ with $q \in \Theta$. Exactly as in the proof of Theorem 16.15, its boundary traces are piecewise P_k, so $q \in S_h$. No bound for the mean of q, or for a cut-domain Poincaré constant, is required.

The field $z_h = a_h - \nabla_b A_h q$ belongs to Y_h^0 and has curl f_h, because A_h preserves the boundary trace and jumps. The stable scalar estimate gives

$$\|z_h\|_2 \le \|a_h\|_2 + C\|\nabla_b q\|_2 \le \|a_h\|_2 + C\|a_h - w\|_2 \le C\|f_h\|_2.$$

Finally $w_h - z_h$ lies in the discrete minimal curl kernel $\nabla\Theta_h^0$, orthogonal to w_h. Thus $\|w_h\|_2^2 = (w_h, z_h)$, and cancellation gives the desired bound. □

The approximation theorem just proved supplies, for every $0 < \sigma \le k$, the following concrete properties on the indicated continuous data space:

$$\begin{aligned} &J_h u \in RT_{k-1}(\mathcal{T}_h), \qquad \operatorname{div} J_h u = 0, \\ &\|u - J_h u\|_2 \le C_\sigma h^\sigma \|u\|_{H^\sigma(\Omega)}, \end{aligned} \tag{16.24}$$

It preserves all boundary-component fluxes. In the second case it additionally preserves zero exterior normal trace and zero cut fluxes. Since a divergence-free RT_{k-1} field is elementwise P_{k-1}, these requirements put $J_h u$ in Z_h^* or $Z_h^{0,*}$.

In particular the proof has not used an ordinary L^2 face restriction when $0 < \sigma \le 1/2$. It uses the rational $W^{1,p'}$ lifts, weak divergence, and the strictly positive L^p gain supplied by $\sigma > 0$. Its low/high Fourier split and averaged Taylor operator supply the full fractional interval; the conservation argument identifies the sum over cut faces with the earlier volume definition of the total cut flux.

Theorem 16.18 (Optimal curl errors; original Theorems 4.8 and 4.13)**.** *Let $u \in H^\sigma(\Omega)^3$, $0 < \sigma \le k$, satisfy the appropriate continuous compatibility conditions. Then the actual mixed finite-element solution satisfies*

$$\|u - \operatorname{curl} w_h\|_2 \le C_\sigma h^\sigma \|u\|_{H^\sigma(\Omega)}.$$

Proof. Theorems 16.13 and 16.15 put $J_h u$ in the actual discrete curl image W_h. Proposition 16.5 therefore gives

$$\|u - \operatorname{curl} w_h\|_2 \le \|u - J_h u\|_2 \le C_\sigma h^\sigma \|u\|_{H^\sigma(\Omega)}.$$

Notice that this particular curl-error conclusion uses the concrete image and approximation theorem, not the uniform norm bound for the potential itself. This observation simplifies the dependency order, and permits their proof without first proving the boundary-normalized discrete coercivity estimate. □

16.7 Continuous mixed formulations and the potential estimate

Proposition 16.19 (The continuous mixed formulations for arbitrary data)**.** *The continuous systems (4.4) and (4.12) are well posed for every $u \in L^2(\Omega)^3$, with zero multiplier. In the first case their potential is $R_T P_E u$; in the second it is $R_N P_{Z^\perp} u$. Here $E = \operatorname{ran} C_{\max}$ and $Z^\perp = \operatorname{ran} C_{\min}$ are the proved closed continuous data spaces. This includes the arbitrary-data assertions of Remarks 4.2 and 4.10.*

Proof. In each case use its proved normalized inverse on the indicated orthogonal projection of u. The resulting w is orthogonal to the corresponding continuous curl kernel, which is exactly the second mixed equation. Its curl is the indicated projection, so $(\operatorname{curl} w, \operatorname{curl} v) = (u, \operatorname{curl} v)$ for every permitted test v. Take the multiplier equal to zero. Conversely testing the first equation with the gradient of the multiplier makes that gradient zero, hence makes the multiplier zero in its scalar quotient or boundary-constant space. The homogeneous difference then has zero curl and is orthogonal to the same kernel, so vanishes. The bounded inverse and the contractive projection give continuity. □

Lemma 16.20 (Potential best approximation in the graph norm)**.** *For either boundary case, if w is the continuous normalized potential of compatible data and w_h the mixed solution, then*

$$\|w - w_h\|_{H(\operatorname{curl})} \le C \inf_{v_h \in X_h} \|w - v_h\|_{H(\operatorname{curl})}.$$

Proof. For any $v_h \in X_h$ subtract its L^2 orthogonal projection g_h onto the discrete kernel and put $z_h = v_h - g_h \in V_h$. Since w is orthogonal to that kernel, $\|g_h\|_2 \le \|v_h - w\|_2$ and $\|w - z_h\|_{H(\operatorname{curl})} \le 2\|w - v_h\|_{H(\operatorname{curl})}$. Curl Galerkin orthogonality and Cauchy–Schwarz give

$$\|\operatorname{curl}(w_h - z_h)\|_2 \le \|\operatorname{curl}(w - z_h)\|_2.$$

Apply the appropriate uniform discrete coercivity theorem to $w_h - z_h \in V_h$ and then the triangle inequality. Taking the infimum proves the assertion. □

Lemma 16.21 (Edge interpolation for a smoother potential)**.** *If $1 \le \sigma \le k$ and $w \in H^{\sigma+1}(\Omega)^3$, then*

$$\|w - I_h^N w\|_{H(\operatorname{curl})} \le C h^\sigma \|w\|_{H^{\sigma+1}(\Omega)}.$$

If w has zero tangential trace, its interpolant belongs to Y_h^0.

Proof. The last assertion is the proved weak boundary preservation. Commutation and Theorem 16.13 already give the curl part, because $\operatorname{curl} w \in H^\sigma$ is divergence-free. We detail the L^2 part to avoid importing a general fractional interpolation-error theorem.

Fix $0 < t < 1/2$. The weak edge bound on the reference cell, the H^1 face trace bound, and the small fractional L^p gain for the curl give, after scaling,

$$\|I_K^N v\|_2 \le C\big(\|v\|_2 + h_K\|\nabla v\|_2 + h_K^{1+t}|\nabla v|_{H^t(K)}\big). \tag{16.25}$$

Indeed select $p > 2$ sufficiently close to 2, apply (16.18) to the components of the curl, and use $|\operatorname{curl} v|_{H^t} \le C|\nabla v|_{H^t}$. This bound is valid for $v \in H^{1+t}$ and requires no polynomial curl.

Extend w to $W \in H^{\sigma+1}(\mathbb{R}^3)^3$ with norm at most twice the restriction norm, and split $W = L + H$ at Fourier frequency $1/h$ as above. For $h \le 1$,

$$\|H\|_2 + h\|\nabla H\|_2 + h^{1+t}\|\nabla H\|_{H^t} \le C h^{\sigma+1}\|W\|_{H^{\sigma+1}}.$$

Sum (16.25) for $H|_K$; it gives the required high-frequency bound for $H - I_h^N H$.

For L use $Q_K = Q_{k-1,K}$ componentwise. Its image is contained in N_k, so $I_K^N Q_K L = Q_K L$. For $k \ge 2$, Taylor's formula and the derivative commutation of Q_K give on K

$$\|D^j(L - Q_K L)\|_\infty \le C h_K^{k-j} \sup_K |D^k L|, \qquad j = 0, 1, 2.$$

For $j = 2 = k$ the estimate just says that the second derivative of the affine Taylor polynomial is zero. The intrinsic fractional seminorm is bounded by the Lipschitz bound of the first derivative:

$$|\nabla(L - Q_K L)|_{H^t(K)} \le C h_K^{5/2-t} \sup_K |D^2(L - Q_K L)|.$$

This follows by using the pointwise difference bound $|x-y|\sup|D^2(L-Q_KL)|$ in its double integral. Substitution in (16.25) therefore bounds the cell error by $Ch_K^{k+3/2}\sup_K|D^kL|$. The bandlimited estimate (16.20) and $\|D^kL\|_2 \le Ch^{\sigma-k}\|W\|_{H^\sigma}$ give a global bound $Ch^\sigma\|W\|_{H^{\sigma+1}}$.

If $k=1$, necessarily $\sigma=1$. The constant $Q_{0,K}L$ gives the same $j=0,1$ bounds with $\sup|DL|$, while the fractional term uses $D^2(L-Q_{0,K}L)=D^2L$. The corresponding two global terms are bounded by $C(h\|DL\|_2+h^2\|D^2L\|_2)$, again by (16.20). This is at most $Ch\|W\|_{H^2}$. As before $h>1$ is absorbed into a domain-dependent constant. The L^2 and curl estimates together prove the lemma. □

Corollary 16.22 (The smoother-potential estimate of Remark 4.9)**.** *If the tangential potential w of the source's Theorem 3.12 belongs to $H^{\sigma+1}(\Omega)^3$, $1\le\sigma\le k$, then its mixed approximation satisfies*

$$\|w-w_h\|_{H(\mathrm{curl})} \le Ch^\sigma\|w\|_{H^{\sigma+1}(\Omega)}.$$

The analogous implication holds for the normal potential when it has that additional regularity.

Proof. Insert $I_h^N w$ into Lemma 16.20 and use Lemma 16.21. □

Correspondence with the original results. The discrete kernel theorem proves Lemma 4.3 and Remark 4.4. The mixed-solvability theorem proves Propositions 4.5 and 4.11; Theorems 16.9 and 16.17 prove Propositions 4.6 and 4.12. Lemma 16.6 proves Lemma 4.7 with the weak compatibility details needed for assembly. The concrete curl-image and fractional approximation theorems give Theorems 4.8 and 4.13 for every $0<\sigma\le k$. The normal discrete bound uses the stable scalar projector rather than the source's polyhedral regularity route; all its constants are uniform on a shape-regular family, including graded meshes and mesh-dependent admissible cuts.

17 The source's additional mixed-boundary assertions

For completeness this section records precisely the unnumbered assertions (3.39)–(3.42) in the source. They are *false under their stated hypotheses*: Propositions 18.3 and 18.4 give smooth solid-torus counterexamples to existence in the first case and uniqueness in the second. They are recorded as source assertions, not as established theorems of this manuscript. A corrected result would require appropriate relative-topology conditions in addition to a development of partial traces and their component fluxes.

The source partitions the boundary into disjoint relatively open Lipschitz parts Γ_N,Γ_T with

$$\partial\Omega=\overline{\Gamma_N}\cup\overline{\Gamma_T},\qquad \Gamma_N\cap\Gamma_T=\varnothing.$$

Write Γ_ℓ^N and Γ_ℓ^T for their connected components, and $\partial\Sigma=\bigcup_j\partial\Sigma_j$ for the union of cut rims. The notation F_G below denotes the component flux on a relatively open boundary piece G, whose weak realization is part of the additional partial-trace development just mentioned.

If $\partial\Sigma\subset\Gamma_T$, the source asserts that

$$\operatorname{div}u=0,\quad u\cdot n=0\text{ on }\Gamma_N,\quad F_{\Gamma_\ell^T}(u)=0\text{ for every }\ell$$

is equivalent to the existence of a unique $\psi\in X$ satisfying

$$\operatorname{curl}\psi=u,\quad \operatorname{div}\psi=0,\quad \psi\times n=0\text{ on }\Gamma_N,\quad \psi\cdot n=0\text{ on }\Gamma_T,$$

$$F_{\Gamma_\ell^N}(\psi)=0\text{ for every }\ell,\qquad C_j(\psi)=0\text{ for every }j.$$

If instead $\partial\Sigma \subset \Gamma_N$, the source adds the conditions $C_j(u) = 0$ to the data and removes the conditions $C_j(\psi) = 0$ from the potential; all the other displayed conditions remain the same. In the first case the potential's zero normal condition occurs near the cut rims, and in the second case the data's zero normal condition occurs there. This explains why the respective cut fluxes are the ones written, but does not itself prove their partial-boundary trace construction or the asserted existence and uniqueness.

18 Corrections and qualifications needed for the source statements

18.1 Remark 3.22 lacks a necessary regularity hypothesis

The printed Remark 3.22 asserts that for every positive integer m, on a $C^{m,1}$ domain, every $u \in H(\mathrm{div})$ satisfying (3.43) has a potential in H_0^m. Taken literally, this is false for every $m \geq 2$. The following counterexample uses a smooth ball, so boundary regularity and the topology do not account for the failure.

Proposition 18.1 (Counterexample to the literal higher-order remark)**.** *In a smooth ball $\Omega \subset \mathbb{R}^3$ there is a compactly supported $u \in H(\mathrm{div};\Omega)$ with $\operatorname{div} u = 0$ and $u \cdot n = 0$ that is not the curl of any $H^2(\Omega)^3$ field. In particular it has no H_0^m potential for any $m \geq 2$.*

Proof. Choose $\chi \in C_c^\infty(\Omega)$ equal to one on a small box about the origin, and put

$$A = (0, 0, \chi(x)|x_1|), \qquad u = \operatorname{curl} A.$$

The scalar $\chi|x_1|$ belongs to H_0^1, so $u \in L^2$ and has compact support. Commutation of weak derivatives gives $\operatorname{div} u = 0$, hence $u \in H(\mathrm{div})$. Its normal trace is zero. For example, approximate A in H^1 by compactly supported smooth vector fields; their curls converge to u in the $H(\mathrm{div})$ graph norm and have zero normal trace. A ball has no cuts, so all conditions (3.43) are satisfied.

On the box where $\chi = 1$, $u = (0, -\operatorname{sign}(x_1), 0)$. Consequently $\partial_1 u_2 = -2\delta_{\{x_1=0\}}$ there in the distributional sense. This distribution is not represented by an L^2 function: take test functions $\varphi_\varepsilon(x) = \eta(x_1/\varepsilon)\zeta(x_2, x_3)$ supported in the box, with $\eta(0) = 1$ and $\int \zeta \neq 0$. Their L^2 norms tend to zero like $\varepsilon^{1/2}$, whereas their pairing with the plane measure is the fixed nonzero number $\int \zeta$. Cauchy–Schwarz rules out an L^2 representative. Thus $u \notin H^1(\Omega)^3$. If $u = \operatorname{curl} \psi$ with $\psi \in H^2$, every component of u would be an H^1 first derivative, a contradiction. Since $H^m \subset H^2$ for integers $m \geq 2$, the final assertion follows. □

Any corrected higher-order assertion must at least assume the data regularity $u \in H^{m-1}$; the requirement $\psi \in H_0^m$ also imposes corresponding boundary compatibility. Indeed differentiation maps H_0^m into H_0^{m-1} by its definition as a closure of compactly supported smooth functions. No complete corrected theorem is inferred from the source's abbreviated remark here.

Proposition 18.2 (A separate smooth boundary obstruction)**.** *Even imposing arbitrarily high Sobolev regularity on the data does not repair the literal higher-order remark. On the unit ball, the smooth field $u(x) = (-x_2, x_1, 0)$ satisfies (3.43), but it is not the curl of any field in $H_0^2(\Omega)^3$.*

Proof. Its divergence is zero, and $u \cdot n = u \cdot x = 0$ on the sphere; the ball requires no cuts. If $\psi \in H_0^2(\Omega)^3$, choose $\psi_j \in C_c^\infty(\Omega)^3$ converging to it in H^2. Then $\operatorname{curl} \psi_j$ converges to $\operatorname{curl} \psi$ in H^1, so $\operatorname{curl} \psi \in H_0^1(\Omega)^3$ by definition. The displayed u has nonzero boundary trace on an open part of the sphere, for instance near $(1, 0, 0)$ where its value is $(0, 1, 0)$. The trace-kernel characterization therefore excludes $u \in H_0^1$. This proves the contradiction. For $m \geq 2$, $H_0^m \subset H_0^2$, so the same obstruction applies. □

18.2 The harmonic-divergence flux in Proposition 3.21

For $w \in K_0$, one knows only that $b = \operatorname{div} w$ belongs to L^2 and is harmonic. It is not justified to infer $\nabla b \in L^2$ and apply an $H(\mathrm{div})$ vector-potential theorem to that gradient. Our proof instead uses the component flux

$$N_i(b) = -\int_\Omega b\Delta\eta_i,$$

where η_i is smooth, one near Γ_i, and zero near the other boundary components. Lemma 10.3 proves its independence from the cutoff by H_0^2 density, its continuity on the harmonic L^2 space, and its agreement with ordinary normal flux whenever $b \in H^1$. The biharmonic variational identity then proves the kernel theorem without the unproved gradient regularity.

18.3 The velocity in the proof of Theorem 4.19

The displayed reconstruction preceding Theorem 4.19 is $u = \operatorname{curl}\psi + H\psi$. The sentence in its proof that writes $u = \psi + H\psi$ omits the curl. Our normal Stokes formulation consistently uses the former expression, and retains the harmonic correction both in its boundary condition and in its right-hand side.

18.4 Both mixed-boundary assertions need additional topology hypotheses

For the smooth boundary parts used in these examples, zero partial normal trace on a relatively open set $G \subset \partial\Omega$ means that the normal Green functional vanishes on scalar tests whose smooth boundary traces have compact support in G. Zero partial tangential trace is defined in the same way using the curl Green functional and vector tests. Both functionals depend only on the boundary trace of the test: a zero-trace difference belongs to H_0^1 and is annihilated by the relevant weak derivative identity. Thus a test need not itself be supported in a boundary collar. On the smooth boundaries used here, every smooth boundary datum has a smooth collar extension: extend it constantly in the normal coordinate of a local smooth chart, multiply by a collar cutoff, and sum a finite partition of unity. Consequently an H^1 test with such a trace may be replaced by this smooth extension, because their difference has zero full trace.

The conditions on the cut rims in (3.39)–(3.42) do not suffice for the asserted existence and uniqueness. We give two counterexamples on the same smooth solid torus. They use only local partial traces, and in the second example the component flux is defined unambiguously by its variational identity.

Fix $R > a > 0$, write $r = (x_1^2 + x_2^2)^{1/2}$ and $z = x_3$, and let

$$\Omega = \{x : (r - R)^2 + z^2 < a^2\}.$$

Its azimuthal angle θ is defined modulo 2π. Although the angle itself is not globally single-valued, the field

$$v = \nabla\theta = \Big(-\frac{x_2}{r^2}, \frac{x_1}{r^2}, 0\Big)$$

is globally smooth near $\overline{\Omega}$. Direct differentiation gives $\operatorname{curl} v = 0$ and $\operatorname{div} v = 0$. The boundary normal lies in the span of the radial and vertical unit vectors, so $v \cdot n = 0$ on the entire boundary. Take the meridian disk $\Sigma = \{\theta = 0\} \cap \Omega$ as the single cut. Its complement is diffeomorphic to $\{(s, z) : s^2 + z^2 < a^2\} \times (0, 2\pi)$ through

$$(s, z, \theta) \longmapsto ((R + s)\cos\theta, (R + s)\sin\theta, z).$$

Thus it is simply connected and has precisely the pseudo-Lipschitz cut geometry of the source. In particular this is an admissible one-cut system, with $J = 1$. For $0 < \varepsilon < \pi/4$, let B be the open

boundary band $|\theta| < \varepsilon$ and let A be the complementary open boundary annulus $\varepsilon < \theta < 2\pi - \varepsilon$. Both are connected smooth boundary parts, their closures cover the boundary, and the cut rim lies in B.

Proposition 18.3 (Failure of mixed existence in (3.39)–(3.40))**.** *Set $\Gamma_T = B$, $\Gamma_N = A$, and $u = v$. All the data conditions (3.39) and the cut-rim hypothesis $\partial\Sigma \subset \Gamma_T$ hold, but there is no $\psi \in H(\operatorname{curl}; \Omega)$ with $\operatorname{curl}\psi = u$ and $\psi \times n = 0$ on Γ_N.*

Proof. The data are smooth, divergence-free, and normal-zero on the whole boundary. Thus the partial normal condition and the flux on the single connected component Γ_T are zero. Choose a nonnegative, nonzero, smooth periodic function $\chi(\theta)$ supported in a closed subinterval of $(\varepsilon, 2\pi - \varepsilon)$, and put $h = \chi(\theta)v$. The chain rule in any local angle chart gives $\nabla\chi = \chi'(\theta)v$, hence

$$\operatorname{curl} h = \nabla\chi \times v + \chi \operatorname{curl} v = 0.$$

The field h is smooth and its boundary trace is supported strictly inside Γ_N. If the asserted ψ existed, its zero partial tangential trace and the curl Green identity tested with h would give

$$(u, h) = (\operatorname{curl}\psi, h) = (\psi, \operatorname{curl} h) = 0.$$

But $(u, h) = \int_\Omega \chi(\theta)|v|^2 > 0$. This contradiction already prevents existence before any of the additional potential gauges in (3.40) are imposed. □

Proposition 18.4 (Failure of mixed uniqueness in (3.41)–(3.42))**.** *Set $\Gamma_N = B$, $\Gamma_T = A$, and take zero data $u = 0$. The cut-rim hypothesis $\partial\Sigma \subset \Gamma_N$ and all the data conditions (3.41) hold. There are nevertheless at least two potentials satisfying every condition in (3.42).*

Proof. Choose a nonnegative, nonzero, smooth periodic χ supported strictly inside the angle interval of A, and set $b = \chi(\theta)v$. As in the preceding proof, b is curl-free and vanishes near Γ_N. Consider the closed scalar space

$$Q = \{q \in H^1(\Omega) : q|_{\Gamma_N} = 0\}.$$

The gradient norm is equivalent to its H^1 norm. Indeed Poincaré controls $q - q_\Omega$, while the positive area of Γ_N and the trace inequality bound $|q_\Omega|\,|\Gamma_N|^{1/2}$ by $\|q - q_\Omega\|_{H^1}$ because the trace of q on that set is zero. Thus Riesz supplies $q \in Q$ with

$$(\nabla q, \nabla s) = -(b, \nabla s) \qquad (s \in Q).$$

Put $h = b + \nabla q$. It has zero curl. Testing with compactly supported smooth s gives zero divergence, so $h \in X$. Its partial tangential trace is zero on Γ_N: the trace of b is zero there, and a gradient of an H^1 function with zero trace on an open boundary part has zero tangential trace on that part. Here is a direct verification of the gradient assertion for these smooth boundary parts. Let w be a smooth vector test whose boundary trace has compact support in Γ_N. Choose a smooth scalar cutoff β equal to one near that boundary support and whose ambient support meets $\partial\Omega$ only in a compact subset of Γ_N. The field $w - \beta w$ has zero full trace, so the curl Green functional of ∇q annihilates it. Since $\operatorname{curl}\nabla q = 0$,

$$\langle (\nabla q) \times n, \operatorname{tr} w \rangle = \int_\Omega \nabla q \cdot \operatorname{curl}(\beta w) = \int_{\partial\Omega} (\operatorname{tr} q) \operatorname{curl}(\beta w) \cdot n = 0.$$

The scalar Green formula justifies the middle equality; the last boundary integrand vanishes because its smooth vector factor has boundary support in Γ_N, where $\operatorname{tr} q = 0$. This proves the required zero partial tangential trace without a separate partial-trace density theorem.

The variational identity says

$$(h, \nabla s) = 0 \qquad (s \in Q).$$

Together with zero divergence, this is exactly zero normal trace on Γ_T, including independence from the choice of an extension of trace data on Γ_N. Since Γ_N is connected, its sole component flux is evaluated by the constant trace one on Γ_N. Extend that trace by the global constant function one. The flux is $(h, \nabla 1) + (\operatorname{div} h, 1) = 0$. It is independent of the extension because any two extensions differ by an element of Q and the preceding variational identity applies. Hence all the potential conditions in (3.42) hold for h and for zero.

They are distinct. The field v has zero divergence and zero normal trace on the whole boundary, so $(\nabla q, v) = 0$ by the scalar Green identity. Therefore

$$(h, v) = (b, v) = \int_\Omega \chi(\theta)|v|^2 > 0,$$

which proves $h \neq 0$. □

These examples satisfy smooth geometric hypotheses and the stated cut conditions. Consequently a mixed Maxwell estimate or a more detailed regularity proof cannot establish the two printed assertions as they stand. A valid replacement must account for the relative topology of the chosen boundary partition, including the appropriate compatibility functionals and harmonic gauges; a condition only on where the rims of one absolute cut system lie does not provide that information.

References

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